

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

alpha-1-Acid Glycoprotein (AGP_L)

Data File: AGP_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Alpha-1-Acid Glycoprotein (AGP) is synthesized in the liver and structurally belongs to the lipocalin superfamily of secretory proteins, such as retinol-binding protein and alpha-1-microglobulin (Schmid, 1975). AGP is a sensitive acute phase reactant whose concentration can increase by a factor of 3 within 24-48 hours when inflammation occurs (Tietz, 1995). It can also be used to differentiate between acute phase reactions (elevated serum level) and estrogen effects (normal or decreased serum level); whereas, the serum level of other positive reactants, such as ceruloplasmin and haptoglobin, increases during such reactions (Ganrot, 1974). Moderate and isolated increases occur when glomerular filtration is inhibited in the early stages of uremia. The determination is used in the assessment of the activity of acute and recurring inflammations as well as of tumors with cell necrosis (Ganrot, 1974).

Eligible Sample

Examined participants 1-5 years old and 12-49 years old females were eligible.

Description of Laboratory Methodology

The Tina-quant Roche AGP assay is based on the principle of immunological agglutination. Anti- alpha-1-acid glycoprotein antibodies react with antigen in the sample to form an antigen/antibody complex. Following agglutination, this is measured turbidimetrically.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

This was a new component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[alpha-1-Acid Glycoprotein](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences’ QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be

used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBDAGPLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXAGP) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDAGPLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD in ng/mL) for LBXFER:

Variable Name	SAS Label	LLOD
LBXAGP	alpha-1-acid glycoprotein	0.1 g/L

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.
- Ganrot K. Plasma protein pattern in acute infectious disease. *Scand J Clin Lab Invest* 1974; 34:75-81.
- Schmid K. α 1-Acid glycoprotein. In: *The Plasma Proteins*, 2nd ed. Putnam FW, ed. New York: Academic Press 1975: 183-228.
- Tietz NW, ed. *Clinical Guide to laboratory Tests*, 3rd ed. Philadelphia, PA: WB Saunders Company 1995; 66-67.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 5 YEARS
Target:	Females only 12 YEARS - 49 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6213.3002375 to 253478.77765	Range of Values	2051	2051	
0	No blood sample provided	513	2564	
.	Missing	0	2564	

LBXAGP - alpha-1-acid glycoprotein (g/L)

Variable Name: LBXAGP
SAS Label: alpha-1-acid glycoprotein (g/L)
English Text: alpha-1-acid glycoprotein (g/L)
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.261 to 2.76	Range of Values	1950	1950	
.	Missing	614	2564	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Albumin & Creatinine - Urine (ALB_CR_L)

Data File: ALB_CR_L.xpt

First Published: September 2025

Last Revised: NA

Component Description

Albumin is the most abundant plasma protein in healthy individuals. Human serum albumin is synthesized by the liver and serves many important roles in human physiology such as, maintaining oncotic pressure, and transport of various hormones, vitamins, and drugs throughout the body. Kidney elimination of serum albumin may be observed in severe kidney disease. Following the urinary albumin excretion has been shown to be a diagnostic and prognostic marker for kidney and cardiovascular events. Unfortunately, this marker displays variable correlation between diagnostic vendors. The correlation deviations are attributed to calibration and assay differences between platforms.

Creatinine is a breakdown product of creatine phosphate in muscle and is usually produced at a fairly constant rate by the body, depending on muscle mass. Creatinine is excreted by glomerular filtration during normal kidney function. Creatinine may be measured in both serum and urine. Creatinine measurement is useful in the diagnosis and treatment of kidney diseases, in monitoring kidney dialysis, and as a calculation basis for other urinary analytes (e.g. total protein, microalbumin).

Eligible Sample

Examined participants aged 3 years and older were eligible.

Description of Laboratory Methodology

Urinary Albumin

The liquid chromatography tandem mass spectrometry (LC-MS/MS) assay quantifies albumin concentrations in human urine following enzymatic digestion. This measurement procedure utilizes proteolysis with trypsin, targeting a peptide specific to human serum albumin.

The lab method used in August 2021-August 2023 to quantify urinary albumin is different from the fluorescent immunoassay method used in previous cycles. Please see additional information in the Analytic Notes section regarding the data comparison between cycles.

Urinary Creatinine

Creatinine is produced by creatine and creatinine phosphate as a result of muscle metabolic processes. It is then excreted by glomerular filtration during normal kidney function. Creatinine may be measured in both serum and urine. Creatinine measurement is useful in the diagnosis and treatment of kidney diseases, in monitoring kidney dialysis, and as a calculation basis for other urinary analytes (e.g., total protein, microalbumin).

In this enzymatic method creatinine is converted to creatine under the activity of creatininase. Creatine is then acted upon by creatinase to form sarcosine and urea. Sarcosine oxidase

converts sarcosine to glycine and hydrogen peroxide, and the hydrogen peroxide reacts with a chromophore in the presence of peroxidase to produce a colored product that is measured at 546 nm (secondary wavelength = 700 nm). This is an endpoint reaction that agrees well with recognized high-performance liquid chromatography methods, and it has the advantage over Jaffe picric acid-based methods that are susceptible to interferences from non-creatinine chromogens.

There were no changes to the lab method in the NHANES August 2021-August 2023 cycle for urinary creatinine measurement. However, the lab equipment used for the measurement was updated from the Cobas 6000 Analyzer to Cobas 8000. Please refer to the Analytic Notes section for additional information.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Urine Albumin](#) (September 2025)

[Urine Creatinine](#) (September 2025)

Laboratory Quality Assurance and Monitoring

Urine specimens were processed, stored, and shipped to University of Minnesota- Advanced Research Diagnostics Laboratory (ARDL), Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manuals \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to University of Minnesota for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPMs](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Three variables were created in this data file. The variables were created using the following formulas:

URXUMA and URXUMS:

The urine albumin value in $\mu\text{g/mL}$ (URXUMA) was converted to mg/L (URXUMS) by multiplying by 1.00 (rounded 2 decimals).

URXUCR and URXCRS:

The urine creatinine value in mg/dL (URXUCR) was converted to $\mu\text{mol/L}$ (URXCRS) by multiplying by 88.4 (rounded 1 decimal).

URDACT:

The urine albumin/creatinine ratio in mg/g (URDACT) was calculated by dividing URXUMA by URXUCR and multiplying by 100 (rounded 2 decimal places).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., URXUMALC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed URX (ex., URXUMA) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex.,

URXUMALC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD) in ug/ml for URXUMA and in mg/dL for URXUCR:

Variable Name	Analyte description	LLOD
URXUMA	Albumin, Urine	0.02 mg/L
URXUCR	Creatinine, urine (mg/dL)	1.1 mg/dL

Albumin regression equations to compare 2017-March 2020 and August 2021-August 2023 data:

In August 2021-August 2023, urinary albumin was quantified using the liquid chromatography tandem mass spectrometry method instead of the fluorescein immunoassay method used in previous cycles. A method validation (bridging) study was performed to compare results from this method change. Randomly selected urine samples (n=195) from previous NHANES were measured using both methods and the results were used to conduct the analysis. On average, urine albumin values measured with the new method used in August 2021-August 2023 were 6.20% higher than values measured with the old method used in 2017-March 2020 (p = 0.0001). Data from the bridging study indicated the correlation coefficient (r) between the measurements was 0.982. Regression analyses were performed using Analyse-it, v4.30.4. Given that the data showed proportional differences in variability, a weighted Deming regression was chosen to describe the relation between the urine albumin results (mg/L) from the two methods as below:

Forward: $URXUMA_{New\ Method} = 1.381 + 0.8407 * URXUMA_{Old\ Method}$; 95% CI of intercept (1.164 to 1.598) and slope (0.8187 to 0.8627).

Backward: $URXUMA_{Old\ Method} = -1.643 + 1.189 * URXUMA_{New\ Method}$; 95% CI of intercept (-1.930 to -1.355) and slope (1.158 to 1.221).

These regression equations are provided for analytic use and should be applied according to the analytic aims and interests. The backwards equation can be used to ensure comparability between the August 2021-August 2023 cycle and previous years that were measured using the fluorescein immunoassay method (such as in 2017-March 2020). The forward equation may be used if the interest is ensuring comparability with data collected using the liquid chromatography tandem mass spectrometry method in August 2021-August 2023 when comparing or combining the August 2021-August 2023 values with previous years of data. For more detailed information on the albumin data in the previous cycles, please refer to the documentations accompanying these datasets.

No Correction Needed for Urinary Creatinine Results for NHANES August 2021-August 2023

A method validation (bridging) study was performed to compare results from a laboratory instrument change that occurred for the August 2021-August 2023 survey cycle. The Cobas 6000 Analyzer was updated to Cobas 8000. Randomly selected urine samples (n=195) from previous NHANES were measured using both instruments and the results were used to conduct the analysis. A statistically significant differences of 3% was observed between the values from the two instruments, however, the known variability between urine creatinine measurements, as stated in Westgard's Desirable Biological Variation Database is 12.2% (Westgard, 2014) for repeatability or inaccuracy. Data from the bridging study also indicated the correlation coefficient (r) between the measurements was 0.999. Therefore, no adjustment to the urine creatinine value (URXUCR) is recommended for the NHANES August 2021-August 2023 data.

References

- Chavers BM, Simonson J, Michael AF. A solid-phase fluorescent immunoassay for the measurement of human urinary albumin. *Kidney Int.* 1984;25:576–578.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.
- Westgard QC (2014). Desirable Biological Variation Database specifications. Accessed from Desirable Biological Variation Database specifications – Westgard on August 5, 2024.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 3 YEARS - 150 YEARS

URXUMA - Albumin, urine (ug/mL)

Variable Name: URXUMA**SAS Label:** Albumin, urine (ug/mL)**English Text:** Albumin, urine (ug/mL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 7942.47	Range of Values	8153	8153	
.	Missing	340	8493	

URXUMS - Albumin, urine (mg/L)

Variable Name: URXUMS**SAS Label:** Albumin, urine (mg/L)**English Text:** Albumin, urine (mg/L)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 7942.47	Range of Values	8153	8153	
.	Missing	340	8493	

URDUMALC - Albumin, urine comment code

Variable Name: URDUMALC

SAS Label: Albumin, urine comment code

English Text: Albumin, urine comment code

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7665	7665	
1	Below lower detection limit	488	8153	
.	Missing	340	8493	

URXUCR - Creatinine, urine (mg/dL)

Variable Name: URXUCR

SAS Label: Creatinine, urine (mg/dL)

English Text: Creatinine, urine (mg/dL)

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4 to 873	Range of Values	8154	8154	
.	Missing	339	8493	

URXCRS - Creatinine, urine (umol/L)

Variable Name: URXCRS

SAS Label: Creatinine, urine (umol/L)

English Text: Creatinine, urine (umol/L)

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
353.6 to 77173.2	Range of Values	8154	8154	
.	Missing	339	8493	

URDUCRLC - Creatinine, urine comment code

Variable Name: URDUCRLC**SAS Label:** Creatinine, urine comment code**English Text:** Creatinine, urine comment code**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	8154	8154	
1	Below lower detection limit	0	8154	
.	Missing	339	8493	

URDACT - Albumin creatinine ratio (mg/g)

Variable Name: URDACT

SAS Label: Albumin creatinine ratio (mg/g)

English Text: Albumin creatinine ratio (mg/g)

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 14708.28	Range of Values	8153	8153	
.	Missing	340	8493	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Butyrylcholinesterase Activity & Concentration (BCHE_L)

Data File: BCHE_L.xpt

First Published: August 2026

Last Revised: NA

Component Description

Butyrylcholinesterase (BChE) is an enzyme primarily synthesized in the liver and found in high concentrations in blood plasma. BChE can be used as an indicator of reduced cholinesterase (ChE). Due to its higher concentrations in blood plasma, because it is freely circulating in blood, and can be obtained directly from either serum or plasma, BChE activity is often a better indicator for acute ChE inhibition (Bartels et. al., 2000). While BChE is naturally produced, its activity can be affected by environmental exposures. BChE activity measurements can be used to indicate various health and exposure conditions including liver dysfunction, drug sensitivity, and exposure to organophosphate (OP) pesticides and nerve agents (Perez et. al. 2015).

NHANES offers a unique opportunity to determine the baseline for blood BChE biomarkers (butyrylcholinesterase activity and butyrylcholinesterase concentration) that will serve as a clinically significant indicator of cholinesterase inhibition and establish references ranges for the general population.

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

Butyrylcholinesterase Activity

The method described here quantifies BChE activity in human serum/plasma using absorption spectroscopy to monitor the rate of substrate hydrolysis (Ellman et. al., 1961) and (Munro et. al., 1994). Samples are incubated with butyrylthiocholine (BTCh) and dithiobisnitrobenzoic acid (DTNB). During this incubation, active BChE will hydrolyze BTCh into butyrate and thiocholine. The thiol generated by the formation of thiocholine rapidly reacts with DTNB cleaving the disulfide bond to form nitrothiobenzoate (NTB) which presents an absorption band centered at 412 nm. Monitoring the absorption at 412 nm during the incubation provides information on the rate at which NTB is generated which occurs at a 1:1 ratio with the formation of thiocholine.

Butyrylcholinesterase Concentration

The method described here utilizes immunomagnetic separation liquid chromatography-tandem mass spectrometry (LC-MS/MS) to quantify unadducted BChE (Knaack et. al., 2012). Unadducted BChE measurement can quickly identify samples with reduced levels of the endogenous protein. Immunomagnetic beads conjugated to monoclonal BChE antibodies specifically purify both unadducted and OP-adducted BChE protein from serum/plasma. BChE protein bound to the bead is then digested with pepsin to yield a nine amino acid biomarker peptide containing the BChE active site. Unadducted BChE has a unique chemical structure

that can be chromatographically separated from adducted BChE and detected by tandem mass spectrometry.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

This is a new component for the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Butyrylcholinesterase Activity Laboratory Procedure Manual](#) (August 2026)

[Butyrylcholinesterase Concentration Laboratory Procedure Manual](#) (August 2026)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate refrigerated (2-8°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendment mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed.

Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and analytic issues.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection limits

The lowest reportable limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBXBALC) indicates whether the result was below the lowest reportable limit: "0" means that the result was at or above the lowest reportable limit, "1" indicates that the result was below the lowest reportable limit. The other variable prefixed LBX (ex., LBXBA) provides the analytic result for that analyte. For analytes with analytic results below the lowest reportable limit (ex., LBXBALC=1), an imputed fill value was placed in the analyte results field. This value is the lowest reportable limit divided by square root of 2 (LRL/sqrt [2]).

Lowest Reportable Limit (LRL, in ng/mL) for Serum:

VARIABLE NAME	SAS LABEL	LRL
LBXBA	Butyrylcholinesterase Activity (U/mL)	0.05
LBXCH	Butyrylcholinesterase Con (ng/mL)	2

References

- Barterls, C.F.; Xie,W.; Miller-Lindholm, A.K; Schoper, L.; Lockridge, O.; Biochem Pharmacol, 2000, 60, 479-487
- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. Statist. Med. (2008) 27(20):4094-40106.
- Ellman, G. L., Courtney, K. D., Andres Jr, V., & Featherstone, R. M. (1961). A new and rapid colorimetric determination of acetylcholinesterase activity. Biochemical Pharmacology, 7(2), 88-95. DOI: 10.1016/0006-2952(61)90145-9. Worek, F.; Mast, U.; Kiderlen, D.; Diepold, C.; Eyer, P.

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- Munro, N. (1994). Toxicity of the organophosphate chemical warfare agents GA, GB, and VX: implications for public protection. *Environmental Health Perspectives*, 102(1), 18-37. DOI: 10.1289/ehp.9410218
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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

LBXBA - Butyrylcholinesterase Activity (U/mL)

Variable Name: LBXBA

SAS Label: Butyrylcholinesterase Activity (U/mL)

English Text: Butyrylcholinesterase Activity (U/mL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.02 to 8.99	Range of Values	7061	7061	
.	Missing	1007	8068	

LBDBALC - Butyrylcholinesterase Act comt code

Variable Name: LBDBALC

SAS Label: Butyrylcholinesterase Act comt code

English Text: Butyrylcholinesterase Activity (U/mL) Comment Code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7061	7061	
1	Below lower detection limit	0	7061	
.	Missing	1007	8068	

LBXCH - Butyrylcholinesterase Con (ng/mL)

Variable Name: LBXCH

SAS Label: Butyrylcholinesterase Con (ng/mL)

English Text: Butyrylcholinesterase Concentration (ng/mL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15.73 to 194.14	Range of Values	7060	7060	
.	Missing	1008	8068	

LBDCHLC - Butyrylcholinesterase Conc comt code

Variable Name: LBDCHLC

SAS Label: Butyrylcholinesterase Conc comt code

English Text: Butyrylcholinesterase Concentration (ng/mL) Comment Code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7060	7060	
1	Below lower detection limit	0	7060	
.	Missing	1008	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Standard Biochemistry Profile (BIOPRO_L)

Data File: BIOPRO_L.xpt

First Published: September 2025

Last Revised: NA

Component Description

These series of measurements are used in the diagnosis and treatment of certain liver, heart, and kidney diseases; acid-base imbalance in the respiratory and metabolic systems; other diseases involving lipid metabolism; various endocrine disorders; as well as other metabolic or nutritional disorders.

Alanine Aminotransferase (ALT)

Alanine aminotransferase measurements are used in the diagnosis and treatment of certain liver diseases (e.g., viral hepatitis and cirrhosis) and heart diseases. Elevated levels of the transaminases can indicate myocardial infarction, hepatic disease, muscular dystrophy, or organ damage. Serum elevations of ALT activity are rarely observed, except in parenchymal liver disease since ALT is a more liver-specific enzyme than aspartate aminotransferase (AST).

Albumin

Albumin measurements are used in the diagnosis and treatment of diseases involving the liver and/or kidneys and are frequently used to assess nutritional status because plasma levels of albumin are dependent on protein intake.

Alkaline Phosphatase (ALP)

Alkaline phosphatase measurements are used in the diagnosis and treatment of liver, bone, and parathyroid disease.

Aspartate Aminotransferase (AST)

AST measurements are used in the diagnosis and treatment of certain types of liver and heart disease. Elevated levels of the transaminases can signal myocardial infarction, hepatic disease, muscular dystrophy, or organ damage.

Bicarbonate (HCO₃)

Together with pH determination, bicarbonate measurements are used in the diagnosis and treatment of numerous potentially serious disorders associated with acid-base imbalance in the respiratory and metabolic systems.

Blood Urea Nitrogen (BUN)

BUN measurements are used in the diagnosis of certain renal and metabolic diseases. The determination of serum urea nitrogen is the most widely used test for the evaluation of kidney function. The test is frequently requested in conjunction with the serum creatinine test for the

differential diagnosis of prerenal, renal, and post renal uremia. High BUN levels are associated with impaired renal function, increased protein catabolism, nephritis, intestinal obstruction, urinary obstruction, metallic poisoning, cardiac failure, peritonitis, dehydration, malignancy, pneumonia, surgical shock, Addison's disease, and uremia. Low BUN levels are associated with amyloidosis, acute liver disease, pregnancy, and nephrosis. Normal variations are observed according to a person's age and sex, the time of day, and their diet – particularly in their protein intake.

Creatinine

Creatinine measurements are useful in the diagnosis and treatment of renal diseases.

Creatine Phosphokinase (CPK)

Measurements of creatine phosphokinase are used in the diagnosis and treatment of myocardial infarction, skeletal muscle diseases, and diseases of the central nervous system.

Gamma-glutamyl Transaminase (GGT)

GGT measurements are principally used to diagnose and monitor hepatobiliary disease. It is currently the most sensitive enzymatic indicator of liver disease, with normal values rarely found in the presence of hepatic disease. It is also used as a sensitive screening test for occult alcoholism. Elevated levels are found in patients who chronically take drugs, such as phenobarbital and phenytoin.

Globulin

Globulins are a diverse group of proteins that transport various substances in the blood. They are also involved in various defense mechanisms within the body. Measurements of globulin are calculated (**Total protein - Albumin**) and are used to determine the serum globulin concentration.

Glucose

Glucose measurements are used in the diagnosis and treatment of pancreatic islet cell carcinoma and of carbohydrate metabolism disorders, including diabetes mellitus, neonatal hypoglycemia, and idiopathic hypoglycemia.

Iron

Iron (non-heme) measurements are used in the diagnosis and treatment of diseases, such as iron deficiency anemia, chronic renal disease, and hemochromatosis (a disease associated with widespread deposit in the tissues of two iron-containing pigments, hemosiderin and hemofuscin, and characterized by pigmentation of the skin).

Lactate Dehydrogenase (LDH)

LDH measurements are used in the diagnosis and treatment of liver diseases, such as acute viral hepatitis, cirrhosis, and metastatic carcinoma of the liver; cardiac diseases, such as myocardial infarction; and tumors of the lungs or kidneys.

Osmolality

Serum osmolality is a measure of the number of dissolved particles in a solution and is used to evaluate hydration status and detect potential toxins and foreign substances in the blood.

(Osmolality is a calculated value on the chemistry analyzer: $[(1.86 \times \text{Na}) + (\text{GLUC}/18) + (\text{BUN}/2.8) + 9]$).

Phosphorus

There is a reciprocal relationship between serum calcium and inorganic phosphorus. Any increase in the level of inorganic phosphorus causes a decrease in the calcium level by a mechanism not clearly understood. Hyperphosphatemia is associated with vitamin D hypervitaminosis, hypoparathyroidism, and renal failure. Hypophosphatemia is associated with rickets, hyperparathyroidism, and Fanconi syndrome.

Measurements of inorganic phosphorus are used in the diagnosis and treatment of various disorders, including parathyroid gland, kidney diseases, and vitamin D imbalance.

Potassium, Chloride, and Sodium

Hypokalemia (low serum potassium level) is associated with body potassium deficiency, excessive potassium loss caused by prolonged diarrhea or prolonged periods of vomiting and increased secretion of mineralocorticosteroids. Hyperkalemia (increased serum potassium level) is associated with oliguria, anuria, and urinary obstruction.

Low serum chloride values are associated with salt-losing nephritis; Addisonian crisis, prolonged vomiting, and metabolic acidosis caused by excessive production or diminished excretion of acids. High serum chloride values are associated with dehydration and conditions causing decreased renal blood flow, such as congestive heart failure.

Sodium measurements are used in the diagnosis and treatment of diseases involving electrolyte imbalance.

Total Bilirubin

Elevated levels are associated with hemolytic jaundice, paroxysmal hemoglobinuria, pernicious anemia, polycythemia, icterus neonatorum, internal hemorrhage, acute hemolytic anemia, malaria, and septicemia.

Low bilirubin levels are associated with aplastic anemia, and certain types of secondary anemia resulting from toxic therapy for carcinoma and chronic nephritis.

Total Calcium

Calcium measurements are used in the diagnosis and treatment of parathyroid disease, bone diseases, chronic renal disease, and tetany. Urinary calcium measurement is used in the differential diagnosis of hypercalciuria.

Total Cholesterol

An elevated cholesterol level is associated with diabetes, nephrosis, hypothyroidism, biliary obstruction, and those rare cases of idiopathic hypercholesterolemia and hyperlipidemia; low levels are associated with hyperthyroidism, hepatitis, and sometimes severe anemia or infection.

Total Protein

Total protein measurements are used in the diagnosis and treatment of a variety of diseases involving the liver, kidney, or bone marrow, as well as other metabolic or nutritional disorders.

Triglycerides

Triglyceride measurements are used in the diagnosis of diabetes mellitus, nephrosis, liver obstruction, and other diseases involving lipid metabolism and various endocrine disorders and in the treatment of patients with these diseases.

Uric Acid

Uric acid measurements are used in the diagnosis and treatment of numerous renal and metabolic disorders, including renal failure, gout, leukemia, psoriasis, starvation, or other wasting conditions and in the treatment of patients receiving cytotoxic drugs.

Eligible Sample

All examined participants aged 12 years and older were eligible.

Description of Laboratory Methodology

NOTE: Glucose, cholesterol, and triglyceride were analyzed 1) as part of the biochemistry profile in this dataset; and 2) at other institutions, which are considered the reference methods for these measures. The results in this dataset do not replace the reference method data. See the Analytical Notes section below for more detailed information.

All methods were measured on the Roche Cobas 8000 analyzer. See Laboratory Method Files for more detailed information about analyte methodologies, principles, and operating procedures.

Alanine Aminotransferase (ALT)

The method to measure alanine aminotransferase (ALT) catalyzes the reaction of alpha-ketoglutarate with L-alanine to form L-glutamate and pyruvate. Under the action of LDH, pyruvate converts to lactate, and NADH is converted to NAD. The decrease in absorbance of NADH, measured at 340 nm (secondary wavelength is 700 nm), is directly proportional to the serum activity of ALT. It is a kinetic rate reaction.

Albumin

The method to measure albumin concentration utilizes the dye bromocresol purple (BCP). When the dye binds selectively with albumin in a pH range of 5.2-6.8, a color change occurs that is measured at 600 nm. The secondary wavelength is 700 nm. This is a 2-point, endpoint reaction that is specific for albumin.

Aspartate Aminotransferase (AST)

Aspartate aminotransferase (AST) activity is determined by a modification of the method recommended by the International Federation of Clinical Chemistry (IFCC). AST catalyzes the reaction of alpha-ketoglutarate with L-aspartate to form L-glutamate and oxaloacetate. Under the action of malate dehydrogenase (MDH), oxaloacetate converts to malate, and NADH is oxidized to NAD. The decrease in absorbance of NADH, measured at 340 nm (secondary wavelength = 700 nm), is directly proportional to the serum activity of AST. It is a kinetic rate reaction.

Alkaline Phosphatase (ALP)

The method to measure alkaline phosphatase (ALP) utilizes a simple reaction wherein ALP acts upon a substrate (p-nitrophenol phosphate, or PNPP) in the presence of magnesium and zinc

activators to form a colored product (p-nitrophenol) whose appearance is measured at 450 nm. The rate of p-nitrophenol formation is directly related to the amount of alkaline phosphatase in the specimen.

Bicarbonate (HCO_3)

The method to measure bicarbonate (HCO_3) utilizes an enzyme-based reaction.

Phosphoenolpyruvate (PEP) is added to the specimen containing bicarbonate. Under the action of phosphoenolpyruvate carboxylase (PEPC) the PEP accepts the bicarbonate and is converted to oxaloacetate. Then under the action of malate dehydrogenase, and in the presence of an NADH analog, the oxaloacetate is converted to malate, with the NADH analog converting to an NAD analog. The rate of disappearance of NADH analog is measured at 415 nm, and it is directly proportional to the amount of bicarbonate in the specimen.

Blood Urea Nitrogen (BUN)

The method to measure blood urea nitrogen utilizes a coupled enzyme reaction (urease, followed by glutamate dehydrogenase), with measurement of NADH (converting to NAD^+) occurring at 340 nm.

Creatine Phosphokinase (CPK)

The method to measure creatine phosphokinase (CPK) or creatine kinase (CK) utilizes a coupled enzyme reaction. Creatine phosphate and adenosine diphosphate (ADP) are acted upon by CK in the serum specimen. Creatine and ATP are produced from this reaction, and ATP reacts with glucose under the action of hexokinase to produce glucose-6-phosphate and ADP. The glucose-6-phosphate reacts with NADP under the action of glucose-6-phosphate dehydrogenase to produce NADPH and a by-product. The photometrically (340 nm) measured rate of NADPH formation is directly proportional to the CK activity in the specimen.

Creatinine

Creatinine is measured using an enzymatic method in which creatinine is converted to creatine under the activity of creatininase. Creatine is then acted upon by creatinase to form sarcosine and urea. Sarcosine oxidase converts sarcosine to glycine and hydrogen peroxide, and the hydrogen peroxide reacts with a chromophore in the presence of peroxidase to produce a colored product that is measured at 546 nm (secondary wavelength = 700 nm). This is an endpoint reaction that agrees well with recognized HPLC methods, and it has the advantage over Jaffe picric acid-based methods that are susceptible to interferences from non-creatinine chromogens.

Gamma-glutamyl Transaminase (GGT)

The method to measure gamma-glutamyl transaminase (GGT) is a slight modification (slightly different substrate) of the method introduced by Szasz in 1969 (Whitfield JB, et. al., 1972) In the presence of glycylglycine, L-gamma-glutamyl-3-carboxy-4-nitroanilide is converted by GGT to 5-amino-2-nitrobenzoate and L-gamma-glutamyl-glycylglycine. The rate of colored product formation is directly related to the amount of GGT in the specimen, and the rate of its appearance is measured at 415 nm (secondary wavelength 700 nm). This is a kinetic (Rate-A) reaction.

Glucose

The method to measure glucose utilizes an enzymatic method that converts glucose to glucose-6-phosphate (G-6-P) by hexokinase in the presence of ATP, a phosphate donor. Glucose-6-phosphate dehydrogenase then converts the G-6-P to gluconate-6-P in the presence of NADP^+ . As the NADP^+ is reduced to NADPH during this reaction, the resulting increase in

absorbance at 340 nm (secondary wavelength = 700 nm) is measured. This is an endpoint reaction that is specific for glucose.

Iron

The Roche method of iron measurement is a three-step process using the FerroZine reagent: Fe^{3+} is liberated from transferrin by acid/detergent, Fe^{3+} is reduced to Fe^{2+} by ascorbate, and the reduced iron then reacts with the FerroZine reagent to form a colored complex. The intensity of this final product is directly proportional to the iron concentration in the specimen.

Lactate Dehydrogenase (LDH)

The Roche method of LDH measurement is derived from the formulation recommended by the International Federation of Clinical Chemistry (IFCC) and is optimized for performance and stability. In the presence of cofactor NAD^+ , LDH converts L-lactate to pyruvate. NAD^+ is reduced to NADH during this reaction. The initial rate of NADH formation is directly proportional to the catalytic LDH activity and is determined by measuring the increase in absorbance at 340 nm. This is a kinetic (Rate-A) reaction.

Phosphorus

The method used to measure phosphorus utilizes ammonium molybdate as the color-forming reagent. Measurement of the final product occurs at 340 nm (secondary wavelength 700 nm). Inorganic phosphate forms an ammonium phosphomolybdate complex having the formula $(\text{NH}_4)_3[\text{PO}_4(\text{MoO}_3)_2]$ with ammonium molybdate in the presence of sulfuric acid. The concentration of phosphomolybdate formed is directly proportional to the inorganic phosphate concentration.

Potassium, Chloride, & Sodium

An Ion-Selective Electrode (ISE) makes use of the unique properties of certain membrane materials to develop an electrical potential (electromotive force, EMF) for the measurements of ions in solution. The electrode has a selective membrane in contact with both the test solution and an internal filling solution. The internal filling solution contains the test ion at a fixed concentration. Because of the particular nature of the membrane, the test ions will closely associate with the membrane on each side. The membrane EMF is determined by the difference in concentration of the test ion in the test solution and the internal filling solution. The complete measurement system for a particular ion includes the ISE, a reference electrode, and electronic circuits to measure and process the EMF to give the test ion concentration. The sodium and potassium electrodes are based on neutral carriers and the chloride electrode is based on an ion exchanger.

Chloride

The method used to measure Chloride is an indirect (specimen is diluted (1:31) by the instrument prior to analysis) ion-selective electrode (ISE) method for determination of the serum electrolyte concentrations.

Potassium

Potassium ion concentration is measured by electrolyte activity in solution. This method utilizes an indirect (specimen is diluted by the instrument prior to analysis) ion-selective electrode (ISE) method for determination of the serum electrolyte concentrations.

Sodium

The method used to measure sodium utilizes an indirect (specimen is diluted by the instrument prior to analysis) ion-selective electrode (ISE) method for determination of the serum

electrolyte concentrations.

Total Bilirubin

The method to measure total bilirubin is coupled with 3,5-dichlorophenyl diazonium in the presence of a solubilizing agent in a strongly acidic medium. The intensity of the red azo dye formed is directly proportional to the total bilirubin and can be determined photometrically (546 nm).

Total Calcium

The method used to measure total calcium reacts with 5-nitro-5'-methyl-BAPTA (NM-BAPTA) under alkaline conditions to form a complex. This complex then reacts with EDTA to form a colored product whose intensity is directly proportional to the concentration of calcium in the specimen. It is measured photometrically at 340 nm.

Total Cholesterol

The method used to measure cholesterol is an enzymatic method where esterified cholesterol is converted to cholesterol by cholesterol esterase. The resulting cholesterol is then acted upon by cholesterol oxidase to produce cholest-4-en-3-one and hydrogen peroxide. The hydrogen peroxide then reacts with 4-aminophenazone in the presence of peroxidase to produce a colored product that is measured at 505 nm (secondary wavelength = 700 nm). The final step is known as the Trinder reaction. This method is a single reagent, endpoint reaction that is specific for cholesterol.

Total Protein

The total protein method utilizes the biuret reaction with measurement of the final product at 546 nm. Divalent copper reacts in alkaline solution with protein peptide bonds to form the characteristic purple-colored biuret complex. Sodium potassium tartrate prevents the precipitation of copper hydroxide and potassium iodide prevents auto-reduction of copper. The color intensity is directly proportional to the protein concentration.

Triglycerides

The method used to measure triglyceride is based on the work by Wahlefeld using a lipoprotein lipase from microorganisms for the rapid and complete hydrolysis of triglycerides to glycerol followed by oxidation to dihydroxyacetone phosphate and hydrogen peroxide. The hydrogen peroxide produced then reacts with 4-aminophenazone and 4-chlorophenol under the catalytic action of peroxidase to form a red dyestuff (Trinder endpoint reaction). The color intensity of the red dyestuff formed is directly proportional to the triglyceride concentration and can be measured photometrically.

Uric Acid

In this method uric acid is oxidized by uricase. Then the peroxide produced from this reaction is acted upon by peroxidase in the presence of 4 aminophenazone to produce a measurable colored product. It is a two-point, endpoint reaction, with measurement occurring at 546 nm (secondary wavelength 700 nm).

There were no changes to the lab method in the NHANES August 2021-August 2023 cycle for These series of measurements. However, the lab equipment used for the measurements was updated from [the Cobas 6000 Analyzer to Cobas 8000](#). Please refer to the Analytic Notes section for additional information.

Refer to the Laboratory Method Files section for a detailed description of the laboratory

methods used.

Laboratory Method Files

[Albumin](#) (September 2025)

[ALP](#) (September 2025)

[ALT](#) (September 2025)

[AST](#) (September 2025)

[Bicarbonate](#) (September 2025)

[BUN](#) (September 2025)

[Chloride](#) (September 2025)

[CPK](#) (September 2025)

[Creatinine](#) (September 2025)

[GGT](#) (September 2025)

[Glucose](#) (September 2025)

[Iron](#) (September 2025)

[LDH](#) (September 2025)

[Magnesium](#) (September 2025)

[Phosphorus](#) (September 2025)

[Potassium](#) (September 2025)

[Sodium](#) (September 2025)

[Total Bilirubin](#) (September 2025)

[Total Calcium](#) (September 2025)

[Total Cholesterol](#) (September 2025)

[Total Protein](#) (September 2025)

[Triglycerides](#) (September 2025)

[Uric acid](#) (September 2025)

Laboratory Quality Assurance and Monitoring

Serum specimens are processed, stored, and shipped to the University of Minnesota – Advanced Research Diagnostics Laboratory (ARDL), Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manuals](#) (LPMs). Vials were stored under appropriate refrigerated (2-8°C) conditions until they are shipped to ARDL for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendment mandates. Detailed QA/QC instructions are discussed in the [NHANES LPMs](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split specimens collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

There were 13 additional variables created in this data file to convert the analyzed values into International System of Units (SI). These variables were created using the following formulas:

LBXSAL conversion to LBDSALSI

Albumin in g/dL (LBXSAL) was converted to g/L (LBDSALSI) by multiplying by **10**

LBXSBU conversion to LBDSBUSI

Blood urea nitrogen (BUN) in mg/dL (LBXSBU) was converted to mmol/L (LBDSBUSI) by multiplying by **0.357**

LBXSCA conversion to LBDSCASI

Calcium in mg/dL (LBXSCA) was converted to mmol/L (LBDSCASI) by multiplying by **0.250**

LBXSCH conversion to LBDSCHSI

Cholesterol in mg/dL (LBXSCH) was converted to mmol/L (LBDSCHSI) by multiplying by **0.0259**

LBXSCR conversion to LBDSCRSI

Creatinine in mg/dL (LBXSCR) was converted to μ mol/L (LBDSCRSI) by multiplying by **88.4**

LBXSGL conversion to LBDSGLSI

Glucose in mg/dL (LBXSGL) was converted to mmol/L (LBDSGLSI) by multiplying by **0.0555**

LBXSIR conversion to LBDSIRSI

Iron in μ g/dL (LBXSIR) was converted to μ mol/L (LBDSIRSI) by multiplying by **0.1791**

LBXSPH conversion to LBDSPHSI

Phosphorus in mg/dL (LBXSPH) was converted to mmol/L (LBDSPHSI) by multiplying by **0.323**

LBXSTB conversion to LBDSTBSI

Total bilirubin in mg/dL (LBXSTB) was converted to μ mol/L (LBDSTBSI) by multiplying by **17.1**

LBXSTP conversion to LBDSTPSI

Total protein in g/dL (LBXSTP) was converted to g/L (LBDSTPSI) by multiplying by **10**

LBXSTR conversion to LBDSTRSI

Triglycerides in mg/dL (LBXSTR) were converted to mmol/L (LBDSTRSI) by multiplying by **0.0113**

LBXSUA conversion to LBDSUASI

Uric acid in mg/dL (LBXSUA) was converted to μ mol/L (LBDSUASI) by multiplying by **59.48**

LBXSGB conversion to LBDSGBSI

Globulin in g/dL (LBXSGB) was converted to g/L (LBDSGBSI) by multiplying by **10**

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed.

Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Glucose (LBXSGL)

This glucose value was obtained from the standard battery of biochemical assessments. Use of the laboratory test result from the reference method (LBXGLU), rather than the standard battery of biochemical assessments value (LBXSGL), is generally recommended. These serum glucose values (LBXSGL) reported in this data file should not be used to determine undiagnosed diabetes or prediabetes. Instead, plasma glucose values (LBXGLU) should be used, which are based on the reference analytic method in the **GLU_L** data file. Special weights included in the **GLU_L** data file should be used when analyzing these data.

Total Cholesterol (LBXSCH)

This total cholesterol value was obtained from the standard battery of biochemical assessments. Use of the laboratory test result from the reference method (LBXTC), rather than the standard battery of biochemical assessments value (LBXSCH), is generally recommended. For most serum cholesterol analyses, the appropriate variable to use will be (LBXTC) in the **TCHOL_L** data file. The (LBXSCH) value from the standard biochemistry profile should not be used routinely.

Triglycerides (LBXSTR)

This triglyceride value was obtained from the standard battery of biochemical assessments. Use of the laboratory test result from the reference method (LBXTR), rather than the standard battery of biochemical assessments value (LBXSTR), is generally recommended. For most triglyceride analyses, the appropriate variable to use is (LBXTR) in the **TRIGLY_L** data file. The value from the standard biochemistry profile (LBXSTR) should not be used routinely.

Detection Limits

The detection limits were constant for all of the analytes in the data set. Variable prefixed LBX (ex., LBXSATSI) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection, an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 ($LLOD/\sqrt{2}$). A variable name ending in "LC" (ex. LBDSATLC) is included for these analytes to indicate whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection.

The lower limit of detection (LLOD) for the Standard Biochemistry Profile:

Variable Name	ANALYTE DESCRIPTION	LLOD
LBXSATSI	Alanine Aminotransferase (ALT) (IU/L)	3
LBXSAL	Albumin, refrigerated serum (g/dL)	0.2
LBXSAPSI	Alkaline Phosphatase (ALP) (IU/L)	5
LBXSASSI	Aspartate Aminotransferase (AST) (IU/L)	5
LBXSC3SI	Bicarbonate (mmol/L)	2
LBXSBU	Blood Urea Nitrogen (mg/dL)	1.4
LBXSCLSI	Chloride (mmol/L)	N/A
LBXSCK	Creatine Phosphokinase (CPK) (U/L)	7
LBXSCR	Creatinine, refrigerated serum (mg/dL)	0.06
LBXSGTSI	Gamma-Glutamyl Transferase (GGT) (IU/L)	3
LBXSGL	Glucose, refrigerated serum (mg/dL)	2
LBXSIR	Iron, refrigerated serum (µg/dL)	5
LBXSLDSI	Lactate Dehydrogenase (LDH) (IU/L)	10
LBXMAGN	Magnesium (mg/dL)	0.2
LBXSPH	Phosphorus (mg/dL)	0.3
LBXSKSI	Potassium (mmol/L)	N/A
LBXSNASI	Sodium (mmol/L)	N/A
LBXSTB	Total Bilirubin (mg/dL)	0.15
LBXSCA	Total Calcium (mg/dL)	0.8
LBXSCH	Total Cholesterol, refrigerated serum (mg/dL)	3.86
LBXSTP	Total Protein (g/dL)	0.2
LBXSTR	Triglycerides, refrigerated serum (mg/dL)	9
LBXSUA	Uric Acid (mg/dL)	0.2

Standard Biochemistry Profile regression equations to compare 2017-March 2020 and August 2021-August 2023 data:

A method validation (bridging) studies were performed to compare results from an instrument change in the August 2021-August 2023 cycle: the Cobas 6000 Chemistry Analyzer was upgraded to the Cobas 8000 Chemistry Analyzer. Randomly selected serum samples (n=152) from previous NHANES were measured using both instruments and the results were used to conduct the analysis.

The table below provides the results from the bridging study for each analyte. It includes mean differences (% Diff) and correlation coefficients (r) between the two measurements. When an adjustment is recommended, the regression model used for the adjustment as well as the forward and backward adjustment equations are also included in the table. When the differences between the two measurements were proportional to concentration, Weighted Deming regressions were chosen to adjust the results. When differences were constant across the interval of concentration, a non-weighted Deming regression was chosen. No adjustment is recommended if both the slope and the intercept of the regression equation are not significant (i.e., 95% confidence interval [CI] for slope included 1 and for intercept included 0). All analyses were performed using Analyse-it, v4.30.4.

It is recommended that these equations be used when examining trends among standard biochemistry profile data across 1999-2018 and August 2021-August 2023 cycles, or when combining August 2021-August 2023 data with previous cycles. For more detailed information on the standard biochemistry profile data files in the previous cycles, please refer to the documentations accompanying these datasets.

VARIABLE NAME (unit)	STATISTICAL ADJUSTMENT METHOD	FORWARD EQUATIONS [95% CI]	BACKWARD EQUATIONS [95% CI]	% DIFF	r
LBXSAL (g/dL)	Adjustment Not Recommended	N/A	N/A	1.48	0.969
LBXSAPSI¹ (IU/L)	Weighted Deming	Y_{Cobas 8000} = -0.6745 [-1.513, 0.1641] + 1.106 [1.092, 1.120] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.6098 [-0.1414, 1.361] + 0.9041 [0.8927, 0.9156] * X_{Cobas 8000}	9.68	0.997
LBXSASSI (IU/L)	Adjustment Not Recommended	N/A	N/A	0.59	0.995
LBXSATSI (IU/L)	Deming Regression	Y_{Cobas 8000} = 1.477 [95% CI: 1.203 to 1.751] + 0.9663 [95% CI: 0.9507 to 0.9820] * X_{Cobas 6000}	Y_{Cobas 6000} = -1.529 [95% CI: -1.834 to -1.224] + 1.035 [95% CI: 1.018 to 1.052] * X_{Cobas 8000}	6.96	0.998
LBXSBU³ (mg/dL)	Weighted Deming	Y_{Cobas 8000} = -0.09197 [95% CI: -0.2188 to 0.03483] + 0.9798 [95% CI: 0.9693 to 0.9903] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.09386 [95% CI: -0.03454 to 0.2223] + 1.021 [95% CI: 1.010 to 1.031] * X_{Cobas 8000}	-2.72	0.997
LBXSC3SI (mmol/L)	Weighted Deming	Y_{Cobas 8000} = -1.341 (95% CI: -2.421 to -0.2606] + 0.9788 [95% CI: 0.9340 to 1.024] * X_{Cobas 6000}	Y_{Cobas 6000} = 1.37 (95% CI: 0.3202 to 2.419] + 1.022 [95% CI: 0.9746 to 1.069] * X_{Cobas 8000}	-7.65	0.968
LBXSCLSI³ (mmol/L)	Deming Regression	Y_{Cobas 8000} = -0.5307 [95% CI: -2.293 to 1.232] + 1.017 [95% CI: 1.003 to 1.031] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.522 [95% CI: -1.206 to 2.250] + 0.9836 [95% CI: 0.9699 to 0.9973] * X_{Cobas 8000}	1.05	0.999
LBXSCH (mg/dL)	Adjustment Not Recommended	N/A	N/A	0.61	0.996
LBXSCK² (mg/dL)	Adjustment Not Recommended	N/A	N/A	0.84	0.846
LBXSCLSI³ (mmol/L)	Deming Regression	Y_{Cobas 8000} = -9.942 [95% CI: -19.37 to -0.5140] + 1.088 [95% CI: 0.9965 to 1.180] * X_{Cobas 6000}	Y_{Cobas 6000} = 9.137 [95% CI: 1.232 to 17.04] + 0.919 [95% CI: 0.8415 to 0.9965] * X_{Cobas 8000}	-0.80	0.891
LBXSCR (mg/dL)	Adjustment Not Recommended	N/A	N/A	1.25	0.984
LBXSGB (g/dL)	CALCULATED	N/A	N/A	N/A	N/A
LBXSGL (mg/dL)	Deming Regression	Y_{Cobas 8000} = 0.8089 [95% CI: -0.7978 to 2.416] + 1.034 [95% CI: 1.017 to 1.051] * X_{Cobas 6000}	Y_{Cobas 6000} = -0.7821 [95% CI: -2.350 to 0.7852] + 0.9669 [95% CI: 0.9509 to 0.9830] * X_{Cobas 8000}	4.31	0.999
LBXSGTSI (IU/L)	Deming Regression	Y_{Cobas 8000} = -1.031 [95% CI: -1.879 to -0.1829] + 1.059 [95% CI: 1.022 to 1.096] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.9735 [95% CI: 0.2018 to 1.745] + 0.9444 [95% CI: 0.9110 to 0.9778] * X_{Cobas 8000}	0.74	1.000

LBXSIR (ug/dL)	Deming Regression	Y_{Cobas 8000} = 1.693 [95% CI: 0.3131 to 3.074] + 1.008 [95% CI: 0.9898 to 1.027] * X_{Cobas 6000}	Y_{Cobas 6000} = -1.679 [95% CI: -3.082 to -0.2773] + 0.9918 [95% CI: 0.9736 to 1.010] * X_{Cobas 8000}	3.57	0.997
LBXSKI (mmol/L)	Deming Regression	Y_{Cobas 8000} = -0.2283 [95% CI: -0.5240 to 0.06729] + 1.083 [95% CI: 1.013 to 1.154] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.2108 [95% CI: -0.04559 to 0.4671] + 0.923 [95% CI: 0.8636 to 0.9825] * X_{Cobas} 8000	2.69	0.922
LBXSLDSI (IU/L)	Deming Regression	Y_{Cobas 8000} = -2.513 [95% CI: -4.947 to -0.07941] + 1.021 [95% CI: 1.006 to 1.037] * X_{Cobas 6000}	Y_{Cobas 6000} = 2.46 [95% CI: 0.1125 to 4.808] + 0.979 [95% CI: 0.9643 to 0.9937] * X_{Cobas} 8000	0.50	0.996
LBXNASI (mmol/L)	Deming Regression	Y_{Cobas 8000} = -60.07 [95% CI: -98.03 to -22.10] + 1.432 [95% CI: 1.162 to 1.702] * X_{Cobas 6000}	Y_{Cobas 6000} = 41.95 [95% CI: 23.31 to 60.59] + 0.6983 [95% CI: 0.5667 to 0.8300] * X_{Cobas 8000}	0.60	0.645
LBXSOSSI (mmol/Kg)	CALCULATED	N/A	N/A	N/A	N/A
LBXSPH (mg/dL)	Adjustment Not Recommended	N/A	N/A	-2.23	0.974
LBXSTB (mg/dL)	Deming Regression	Y_{Cobas 8000} = -0.01772 [95% CI: -0.02548 to -0.009959] + 0.9881 [95% CI: 0.9702 to 1.006] * X_{Cobas 6000}	Y_{Cobas 6000} = 0.01793 [95% CI: 0.01038 to 0.02548] + 1.012 [95% CI: 0.9938 to 1.030] * X_{Cobas} 8000	-6.60	0.997
LBXSTP (g/dL)	Adjustment Not Recommended	N/A	N/A	3.13	0.960
LBXSTR (mg/dL)	Weighted Deming	Y_{Cobas 8000} = 0.8567 [95% CI: 0.07287 to 1.641] + 1.012 [95% CI: 1.003 to 1.021] * X_{Cobas 6000}	Y_{Cobas 6000} = -0.8463 [95% CI: -1.628 to -0.06476] + 0.9879 [95% CI: 0.9791 to 0.9967] * X_{Cobas 8000}	2.22	0.999
LBXSUA (mg/dL)	Adjustment Not Recommended	N/A	N/A	-0.04	0.997

- Four of the data points were deemed an extreme outlier, therefore only data from 148 serum samples were used in the analysis for ALP (LBXSAPSI).
- Two of the data points were deemed an extreme outlier, therefore only data from 150 serum samples were used in the analysis for Creatine Phosphokinase (LBXSCK).
- One of the data points was deemed an extreme outlier, therefore only data from 151 serum samples were used in the analysis for BUN (LBXSBU), Chloride (LBXSCLSI).

References

- Wahlefeld, August W. Triglycerides Determination after Enzymatic Hydrolysis. Methods of Enzymatic Analysis (2nd English Edition). 1974. 4:1831-1835.

- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.
- Whitfield JB, Pounder RE, Neale G, et al. Serum γ -glutamyl transpeptidase activity in liver disease. Gut 1972;13:702-708.

Codebook and Frequencies

SEQN - Respondent Sequence Number

Variable Name:	SEQN
SAS Label:	Respondent Sequence Number
English Text:	Respondent Sequence Number
Target:	Both males and females 12 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	6750	6750	
0	No blood sample provided	449	7199	
.	Missing	0	7199	

LBXSATSI - Alanine Aminotransferase (ALT) (IU/L)

Variable Name: LBXSATSI
SAS Label: Alanine Aminotransferase (ALT) (IU/L)
English Text: Alanine Aminotransferase (ALT) (IU/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3 to 350	Range of Values	6321	6321	
.	Missing	878	7199	

LBXSAL - Albumin, refrigerated serum (g/dL)**Variable Name:** LBXSAL**SAS Label:** Albumin, refrigerated serum (g/dL)**English Text:** Albumin, refrigerated serum (g/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.2 to 5.5	Range of Values	6366	6366	
.	Missing	833	7199	

LBDSALSI - Albumin, refrigerated serum (g/L)

Variable Name: LBDSALSI
SAS Label: Albumin, refrigerated serum (g/L)
English Text: Albumin, refrigerated serum(g/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
22 to 55	Range of Values	6366	6366	
.	Missing	833	7199	

LBXSAPSI - Alkaline Phosphatase (ALP) (IU/L)

Variable Name: LBXSAPSI
SAS Label: Alkaline Phosphatase (ALP) (IU/L)
English Text: Alkaline Phosphatase (ALP) (IU/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
20 to 731	Range of Values	6327	6327	
.	Missing	872	7199	

LBXSASSI - Aspartate Aminotransferase (AST) (IU/L)

Variable Name: LBXSASSI
SAS Label: Aspartate Aminotransferase (AST) (IU/L)
English Text: Aspartate Aminotransferase (AST) (IU/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6 to 518	Range of Values	6308	6308	
.	Missing	891	7199	

LBXSC3SI - Bicarbonate (mmol/L)

Variable Name: LBXSC3SI
SAS Label: Bicarbonate (mmol/L)
English Text: Bicarbonate (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15 to 37	Range of Values	6325	6325	
.	Missing	874	7199	

LBXSBU - Blood Urea Nitrogen (mg/dL)**Variable Name:** LBXSBU**SAS Label:** Blood Urea Nitrogen (mg/dL)**English Text:** Blood Urea Nitrogen (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4 to 74	Range of Values	6326	6326	
.	Missing	873	7199	

LBDSBUSI - Blood Urea Nitrogen (mmol/L)

Variable Name: LBDSBUSI
SAS Label: Blood Urea Nitrogen (mmol/L)
English Text: Blood Urea Nitrogen (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.4 to 26.4	Range of Values	6326	6326	
.	Missing	873	7199	

LBXSCLSI - Chloride (mmol/L)

Variable Name: LBXSCLSI
SAS Label: Chloride (mmol/L)
English Text: Chloride (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
82 to 133	Range of Values	6329	6329	
.	Missing	870	7199	

LBXSCK - Creatine Phosphokinase (CPK) (U/L)**Variable Name:** LBXSCK**SAS Label:** Creatine Phosphokinase (CPK) (U/L)**English Text:** Creatine Phosphokinase (CPK) (U/L)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10 to 16837	Range of Values	6321	6321	
.	Missing	878	7199	

LBXSCR - Creatinine, refrigerated serum (mg/dL)**Variable Name:** LBXSCR**SAS Label:** Creatinine, refrigerated serum (mg/dL)**English Text:** Creatinine, refrigerated serum (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.33 to 15.17	Range of Values	6326	6326	
.	Missing	873	7199	

LBDSCRSI - Creatinine, refrigerated serum (umol/L)

Variable Name: LBDSCRSI
SAS Label: Creatinine, refrigerated serum (umol/L)
English Text: Creatinine, refrigerated serum (umol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
29.17 to 1341.03	Range of Values	6326	6326	
.	Missing	873	7199	

LBXSGB - Globulin (g/dL)

Variable Name: LBXSGB**SAS Label:** Globulin (g/dL)**English Text:** Globulin (g/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.7 to 5.7	Range of Values	6364	6364	
.	Missing	835	7199	

LBDSGBSI - Globulin (g/L)

Variable Name: LBDSGBSI
SAS Label: Globulin (g/L)
English Text: Globulin (g/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
17 to 57	Range of Values	6364	6364	
.	Missing	835	7199	

LBXSGl - Glucose, refrigerated serum (mg/dL)**Variable Name:** LBXSGl**SAS Label:** Glucose, refrigerated serum (mg/dL)**English Text:** Glucose, refrigerated serum (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
40 to 898	Range of Values	6357	6357	
.	Missing	842	7199	

LBDSGLSI - Glucose, refrigerated serum (mmol/L)

Variable Name: LBDSGLSI
SAS Label: Glucose, refrigerated serum (mmol/L)
English Text: Glucose, refrigerated serum (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2 to 50	Range of Values	6357	6357	
.	Missing	842	7199	

LBXSGTSI - Gamma Glutamyl Transferase (GGT) (IU/L)

Variable Name: LBXSGTSI
SAS Label: Gamma Glutamyl Transferase (GGT) (IU/L)
English Text: Gamma Glutamyl Transferase (GGT) (IU/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2 to 1541	Range of Values	6327	6327	
.	Missing	872	7199	

LBDSGTLC - GGT Comment Code

Variable Name: LBDSGTLC

SAS Label: GGT Comment Code

English Text: Gamma Glutamyl Transferase (GGT) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	6326	6326	
1	Below lower detection limit	1	6327	
.	Missing	872	7199	

LBXSIR - Iron, refrigerated serum (µg/dL)**Variable Name:** LBXSIR**SAS Label:** Iron, refrigerated serum (µg/dL)**English Text:** Iron, refrigerated serum (µg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
8 to 379	Range of Values	6364	6364	
.	Missing	835	7199	

LBDSIRSI - Iron, refrigerated serum (umol/L)

Variable Name: LBDSIRSI
SAS Label: Iron, refrigerated serum (umol/L)
English Text: Iron, refrigerated serum (umol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 68	Range of Values	6364	6364	
.	Missing	835	7199	

LBXSLDSI - Lactate Dehydrogenase (LDH) (U/L)

Variable Name: LBXSLDSI
SAS Label: Lactate Dehydrogenase (LDH) (U/L)
English Text: Lactate Dehydrogenase (LDH) (U/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
42 to 837	Range of Values	4740	4740	
.	Missing	2459	7199	

LBXMAGN - Magnesium (mg/dL)

Variable Name: LBXMAGN
SAS Label: Magnesium (mg/dL)
English Text: Magnesium (mg/dL)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.7 to 2.8	Range of Values	6324	6324	
.	Missing	875	7199	

LBXSOSI - Osmolality (mmol/Kg)

Variable Name: LBXSOSI
SAS Label: Osmolality (mmol/Kg)
English Text: Osmolality (mmol/Kg)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
240 to 334	Range of Values	6322	6322	
.	Missing	877	7199	

LBXSPH - Phosphorus (mg/dL)**Variable Name:** LBXSPH**SAS Label:** Phosphorus (mg/dL)**English Text:** Phosphorus (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.6 to 7	Range of Values	6322	6322	
.	Missing	877	7199	

LBDSPHSI - Phosphorus (mmol/L)

Variable Name: LBDSPHSI
SAS Label: Phosphorus (mmol/L)
English Text: Phosphorus (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.52 to 2.26	Range of Values	6322	6322	
.	Missing	877	7199	

LBXSKSI - Potassium (mmol/L)**Variable Name:** LBXSKSI**SAS Label:** Potassium (mmol/L)**English Text:** Potassium (mmol/L)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.5 to 6.4	Range of Values	6281	6281	
.	Missing	918	7199	

LBXSNASI - Sodium (mmol/L)

Variable Name: LBXSNASI
SAS Label: Sodium (mmol/L)
English Text: Sodium (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
120 to 167	Range of Values	6366	6366	
.	Missing	833	7199	

LBXSTB - Total Bilirubin (mg/dL)**Variable Name:** LBXSTB**SAS Label:** Total Bilirubin (mg/dL)**English Text:** Total Bilirubin (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.11 to 14.2	Range of Values	6325	6325	
.	Missing	874	7199	

LBDSTBSI - Total Bilirubin (umol/L)

Variable Name: LBDSTBSI
SAS Label: Total Bilirubin (umol/L)
English Text: Total Bilirubin (umol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.42 to 242.82	Range of Values	6193	6193	
.	Missing	1006	7199	

LBDSTBLC - Total Bilirubin Comment Code

Variable Name: LBDSTBLC
SAS Label: Total Bilirubin Comment Code
English Text: Total Bilirubin Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	6193	6193	
1	Below lower detection limit	132	6325	
.	Missing	874	7199	

LBXSCA - Total Calcium (mg/dL)

Variable Name: LBXSCA

SAS Label: Total Calcium (mg/dL)

English Text: Total Calcium (mg/dL)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.5 to 11.7	Range of Values	6362	6362	
.	Missing	837	7199	

LBDSCASI - Total Calcium (mmol/L)

Variable Name: LBDSCASI
SAS Label: Total Calcium (mmol/L)
English Text: Total Calcium (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.9 to 2.9	Range of Values	6362	6362	
.	Missing	837	7199	

LBXSCH - Cholesterol, refrigerated serum (mg/dL)**Variable Name:** LBXSCH**SAS Label:** Cholesterol, refrigerated serum (mg/dL)**English Text:** Total Cholesterol, refrigerated serum (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
63 to 454	Range of Values	6328	6328	
.	Missing	871	7199	

LBDSCHSI - Cholesterol, refrigerated serum (mmol/L)

Variable Name: LBDSCHSI
SAS Label: Cholesterol, refrigerated serum (mmol/L)
English Text: Total Cholesterol, refrigerated serum (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.63 to 11.76	Range of Values	6328	6328	
.	Missing	871	7199	

LBXSTP - Total Protein (g/dL)**Variable Name:** LBXSTP**SAS Label:** Total Protein (g/dL)**English Text:** Total Protein (g/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 9.1	Range of Values	6364	6364	
.	Missing	835	7199	

LBDSTPSI - Total Protein (g/L)

Variable Name: LBDSTPSI
SAS Label: Total Protein (g/L)
English Text: Total Protein (g/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
50 to 91	Range of Values	6364	6364	
.	Missing	835	7199	

LBXSTR - Triglycerides, refrig serum (mg/dL)**Variable Name:** LBXSTR**SAS Label:** Triglycerides, refrig serum (mg/dL)**English Text:** Triglycerides, refrigerated serum (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
24 to 1724	Range of Values	6359	6359	
.	Missing	840	7199	

LBDSTRSI - Triglycerides, refrig serum (mmol/L)

Variable Name: LBDSTRSI
SAS Label: Triglycerides, refrig serum (mmol/L)
English Text: Triglycerides, refrigerated serum (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.27 to 19.48	Range of Values	6359	6359	
.	Missing	840	7199	

LBXSUA - Uric acid (mg/dL)

Variable Name: LBXSUA**SAS Label:** Uric acid (mg/dL)**English Text:** Uric acid (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.1 to 13.2	Range of Values	6329	6329	
.	Missing	870	7199	

LBDSUASI - Uric acid (umol/L)

Variable Name: LBDSUASI
SAS Label: Uric acid (umol/L)
English Text: Uric acid (umol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
65.4 to 785.1	Range of Values	6329	6329	
.	Missing	870	7199	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Complete Blood Count with 5-Part Differential in Whole Blood (CBC_L)

Data File: CBC_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

The complete blood count (CBC) with 5-part differential counts red blood cells (RBCs), white blood cells (WBCs), and platelets, measures hemoglobin; estimates the red cells' volume; and sorts the WBCs into subtypes. A CBC is a routine blood test used to evaluate your overall health and detect a wide range of disorders, including anemia, infection, and leukemia.

These data will be used to estimate deficiencies and toxicities of specific nutrients in the population and subgroups, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to whole blood levels of nutrients. Data will be used for research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion.

Eligible Sample

Examined participants aged 1 year and over were eligible.

Description of Laboratory Methodology

The methods used to derive CBC parameters are based on the Beckman Coulter methodology of counting and sizing, in combination with an automatic diluting and mixing device for sample processing, and a single beam photometer for hemoglobinometry. The WBC differential uses Volume Conductivity Scatter (VCS) technology. The VCS technology uses three measurements to provide cell-by-cell information, the measurements are individual cell volume, high-frequency conductivity, and laser-light scatter. (Beckman Coulter, 2020)

The Beckman Coulter DxH 800 instrument in the NHANES mobile examination center (MEC) produces a CBC on blood specimens and provides a distribution of blood cells for all participants.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[Complete Blood Count](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Whole blood specimens were analyzed in the NHANES Mobile Examination Center.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual](#) (LPM).

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

NHANES uses several methods to monitor the quality of the analyses performed by the MEC analytical laboratory. These methods include performing blind split samples collected during “dry run” sessions in the MEC. NCHS developed a QC protocol for the MEC laboratory, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during the analysis of the specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS on an on-going basis. The reports are reviewed for trends or shifts in the data.

In the MEC, the CBC results are measured in duplicate and averaged. The averaged results are reported to the participant and released in this dataset.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Five derived variables were created in this data file. The variables were created using the following formulas:

$LBDLYMNO = LBXWBCSI * LBXLYPCT / 100$ (round to 1 decimal)
 $LBDMONO = LBXWBCSI * LBXMOPCT / 100$ (round to 1 decimal)
 $LBDNENO = LBXWBCSI * LBXNEPCT / 100$ (round to 1 decimal)
 $LBDEONO = LBXWBCSI * LBXEOPCT / 100$ (round to 1 decimal)
 $LBDBANO = LBXWBCSI * LBXBAPCT / 100$ (round to 1 decimal)

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their

phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, the length of fast and the time of venipuncture.

The laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable ending in "LC" (ex., LBDHGBLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBXHGB provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDHGBLC = 1), an imputed value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower and upper limits of detection for CBC:

Variable Name	Analyte Description	LLOD	ULOD	Range
LBXWBCSI	White blood cell count	0.000	337.700	$\times 10^3$ cells/uL
LBXLYPCT	Lymphocyte percent	0.00	100.00	%
LBXMOPCT	Monocyte percent	0.00	100.00	%
LBXNEPCT	Segmented neutrophils percent	0.00	100.00	%
LBXEOPCT	Eosinophils percent	0.00	100.00	%
LBXBAPCT	Basophils percent	0.00	100.00	%
LBXRBCSI	Red blood cell count	0.00	8.11	$\times 10^6$ cells/uL
LBXHGB	Hemoglobin	0.00	24.00	g/dL
LBXMCVSI	Mean cell volume	50.00	150.00	fL
LBXRDW	Red cell distribution width	10.00	40.00	%
LBXPLTSI	Platelet count	0.0	4857.0	$\times 10^3$ cells/uL
LBXMPSI	Mean platelet volume	5.00	25.00	fL

References

- Beckman Coulter. UniCel DxH Series with System Manager Software: Instructions for Use. April 2020. Downloaded from B26647AG.book on February 1, 2024. [beckmancoulter.com](https://www.beckmancoulter.com)

- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBXWBCSI - White blood cell count (1000 cells/uL)**Variable Name:** LBXWBCSI**SAS Label:** White blood cell count (1000 cells/uL)**English Text:** White blood cell count (1000 cells/uL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.1 to 20.8	Range of Values	7593	7593	
400	400 and over	0	7593	
.	Missing	1134	8727	

LBXLYPCT - Lymphocyte percent (%)

Variable Name: LBXLYPCT
SAS Label: Lymphocyte percent (%)
English Text: Lymphocyte percent (%)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.1 to 78.9	Range of Values	7582	7582	
.	Missing	1145	8727	

LBXMOPCT - Monocyte percent (%)**Variable Name:** LBXMOPCT**SAS Label:** Monocyte percent (%)**English Text:** Monocyte percent (%)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.3 to 25.9	Range of Values	7582	7582	
.	Missing	1145	8727	

LBXNEPCT - Segmented neutrophils percent (%)

Variable Name: LBXNEPCT
SAS Label: Segmented neutrophils percent (%)
English Text: Segmented neutrophils percent (%)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12.4 to 92.1	Range of Values	7582	7582	
.	Missing	1145	8727	

LBXEOPCT - Eosinophils percent (%)

Variable Name: LBXEOPCT
SAS Label: Eosinophils percent (%)
English Text: Eosinophils percent (%)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 46.7	Range of Values	7582	7582	
.	Missing	1145	8727	

LBXBAPCT - Basophils percent (%)**Variable Name:** LBXBAPCT**SAS Label:** Basophils percent (%)**English Text:** Basophils percent (%)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.2	Range of Values	7582	7582	
.	Missing	1145	8727	

LBDLYMNO - Lymphocyte number (1000 cells/uL)

Variable Name: LBDLYMNO
SAS Label: Lymphocyte number (1000 cells/uL)
English Text: Lymphocyte number (1000 cells/uL)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 9.8	Range of Values	7582	7582	
.	Missing	1145	8727	

LBDMONO - Monocyte number (1000 cells/uL)

Variable Name: LBDMONO
SAS Label: Monocyte number (1000 cells/uL)
English Text: Monocyte number (1000 cells/uL)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 1.8	Range of Values	7582	7582	
.	Missing	1145	8727	

LBDNENO - Segmented neutrophils num (1000 cell/uL)**Variable Name:** LBDNENO**SAS Label:** Segmented neutrophils num (1000 cell/uL)**English Text:** Segmented neutrophils number (1000 cell/uL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.5 to 16.1	Range of Values	7582	7582	
.	Missing	1145	8727	

LBDEONO - Eosinophils number (1000 cells/uL)

Variable Name: LBDEONO**SAS Label:** Eosinophils number (1000 cells/uL)**English Text:** Eosinophils number (1000 cells/uL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 6	Range of Values	7582	7582	
.	Missing	1145	8727	

LBDBANO - Basophils number (1000 cells/uL)**Variable Name:** LBDBANO**SAS Label:** Basophils number (1000 cells/uL)**English Text:** Basophils number (1000 cells/uL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 0.3	Range of Values	7582	7582	
.	Missing	1145	8727	

LBXRBCSI - Red blood cell count (million cells/uL)

Variable Name: LBXRBCSI
SAS Label: Red blood cell count (million cells/uL)
English Text: Red blood cell count (million cells/uL)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.44 to 7.08	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXHGB - Hemoglobin (g/dL)**Variable Name:** LBXHGB**SAS Label:** Hemoglobin (g/dL)**English Text:** Hemoglobin (g/dL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6.3 to 18.6	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXHCT - Hematocrit (%)**Variable Name:** LBXHCT**SAS Label:** Hematocrit (%)**English Text:** Hematocrit (%)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
19.9 to 54.9	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXMCVSI - Mean cell volume (fL)

Variable Name: LBXMCVSI
SAS Label: Mean cell volume (fL)
English Text: Mean cell volume (fL)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
52.5 to 116.7	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXMC - Mean Cell Hgb Conc. (g/dL)**Variable Name:** LBXMC**SAS Label:** Mean Cell Hgb Conc. (g/dL)**English Text:** Mean cell hemoglobin concentration (g/dL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
27.8 to 40.5	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXMCHSI - Mean cell hemoglobin (pg)

Variable Name: LBXMCHSI
SAS Label: Mean cell hemoglobin (pg)
English Text: Mean cell hemoglobin (pg)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15.6 to 41.1	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXRDW - Red cell distribution width (%)**Variable Name:** LBXRDW**SAS Label:** Red cell distribution width (%)**English Text:** Red cell distribution width (%)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11.2 to 37.5	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXPLTSI - Platelet count (1000 cells/uL)**Variable Name:** LBXPLTSI**SAS Label:** Platelet count (1000 cells/uL)**English Text:** Platelet count (1000 cells/uL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
46 to 787	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXMPSI - Mean platelet volume (fL)**Variable Name:** LBXMPSI**SAS Label:** Mean platelet volume (fL)**English Text:** Mean platelet volume (fL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5.5 to 12.9	Range of Values	7593	7593	
.	Missing	1134	8727	

LBXNRBC - Nucleated red blood cells (/100 WBC)**Variable Name:** LBXNRBC**SAS Label:** Nucleated red blood cells (/100 WBC)**English Text:** Nucleated red blood cells (/100 WBC)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4	Range of Values	7582	7582	
.	Missing	1145	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Cotinine and Hydroxycotinine - Serum (COT_L)

Data File: COT_L.xpt

First Published: August 2026

Last Revised: NA

Component Description

The specific aims of the component are: 1) to measure the prevalence and extent of tobacco use; 2) to estimate the extent of exposure to environmental tobacco smoke (ETS), and determine trends in exposure to ETS; and 3) to describe the relationship between tobacco use (as well as exposure to ETS) and chronic health conditions, including respiratory and cardiovascular diseases.

Cotinine and trans-3'-hydroxycotinine (hydroxycotinine) are the primary metabolites of nicotine. The concentrations of cotinine and hydroxycotinine in body fluids can be used as markers for active smoking and as indices for secondhand smoke (SHS) exposure. Because their concentrations are greater and their elimination half-lives significantly longer, these metabolites are generally preferred over nicotine itself as biomarkers. Cotinine, the primary proximal metabolite of nicotine, is generally regarded as the marker of choice. The estimated elimination half-life (or the amount of time it takes your body to remove half) of cotinine is about 15-20 hours; by contrast, the half-life of nicotine is only 0.5-3 hours. The half-life of hydroxycotinine is approximately 5-6 hours, but when hydroxycotinine is generated from cotinine, its elimination half-life becomes similar to that of cotinine.

Cotinine and hydroxycotinine can be measured in serum, urine, and saliva—the half-life of cotinine in all three fluids is essentially the same. Cotinine concentrations tend to be three to eight times higher in urine than in serum; however, plasma or serum is the fluid of choice for studies requiring a quantitative assessment of exposure. For that reason, serum was chosen as the matrix for the National Health and Nutrition Examination Survey (NHANES) cotinine analyses. In serum, hydroxycotinine concentrations tend to be two to four times lower than cotinine concentrations.

The tobacco component for NHANES also included questionnaire items on current and past use of cigarettes, pipes, cigars, and smokeless tobacco. Questions were asked regarding exposure to ETS at home, at work, and in utero among children. In addition, use of nicotine replacement products (e.g., gum and patch) was collected using questionnaires.

Eligible Sample

Examined participants aged 3 years and older were eligible

Description of Laboratory Methodology

Serum cotinine and hydroxycotinine are measured by an isotope-dilution high-performance liquid chromatography/atmospheric pressure chemical ionization tandem mass spectrometric (ID HPLC-APCI MS/MS) method. Briefly, the serum sample is spiked with methyl-D3-cotinine and methyl-D3-hydroxycotinine as internal standards. The sample is basified and then applied to a supported liquid extraction (SLE) plate. The analytes are extracted with an isopropanol/methylene chloride mixture, the organic extract is concentrated, and the residue is injected onto a C18 HPLC column. The eluent from these injections is monitored by APCI-

MS/MS. The m/z 80 product ion from the m/z 177 quasi-molecular ion is measured for cotinine and the m/z 80 product ion from the m/z 193 quasi-molecular ion is measured for hydroxycotinine. Additional ions for the internal standards and for confirmation are also monitored for the respective compounds. Analyte concentrations are derived from the area ratios of native-to-labeled compounds in the sample by comparisons to a standard curve.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Cotinine and Hydroxycotinine](#) (August 2026)

Laboratory Quality Assurance and Monitoring

Serum specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, and Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a quality control protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' quality control and quality assurance performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample weight variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information such as fasting status, the length of fast, and the time of venipuncture.

The laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDCOTLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. For analytes with analytic results below the lower limit of detection (ex. LBDCOTLC = 1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 ($LLOD/\sqrt{2}$). The other variable prefixed LBX (ex., LBXCOT) provides the analytic result for the analyte.

The lower limit of detection (LLOD in ng/mL) for Cotinine and Hydroxycotinine in serum:

VARIABLE	SAS LABEL	LLOD
LBXCOT	Cotinine, serum	0.010
LBXHCOT	Hydroxycotinine, serum	0.004

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. Statist. Med. (2008) 27(20):4094-40106.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent Sequence Number

Variable Name:	SEQN
SAS Label:	Respondent Sequence Number
English Text:	Respondent sequence number
Target:	Both males and females 3 YEARS - 150 YEARS

LBXCOT - Cotinine, Serum (ng/mL)**Variable Name:** LBXCOT**SAS Label:** Cotinine, Serum (ng/mL)**English Text:** Cotinine, Serum (ng/mL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.007 to 1090	Range of Values	7091	7091	
.	Missing	1402	8493	

LBDCOTLC - Cotinine, Serum Comment Code

Variable Name: LBDCOTLC
SAS Label: Cotinine, Serum Comment Code
English Text: Cotinine, Serum Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	4367	4367	
1	Below lower detection limit	2724	7091	
.	Missing	1402	8493	

LBXHCOT - Hydroxycotinine, Serum (ng/mL)

Variable Name: LBXHCOT
SAS Label: Hydroxycotinine, Serum (ng/mL)
English Text: Hydroxycotinine, Serum (ng/mL)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.003 to 804	Range of Values	7092	7092	
.	Missing	1401	8493	

LBDHCOLC - Hydroxycotinine, Serum Comment Code

Variable Name: LBDHCOLC
SAS Label: Hydroxycotinine, Serum Comment Code
English Text: Hydroxycotinine, Serum Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	4251	4251	
1	Below lower detection limit	2841	7092	
.	Missing	1401	8493	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Fatty Acids - Washed RBCs (FAR_L)

Data File: FAR_L.xpt

First Published: December 2025

Last Revised: NA

Component Description

The analysis of red blood cell (RBC) fatty acids (FA) is an indicator of long term (4 months) FA status. Evaluation of RBC fatty acid content has become increasingly favored as a measure of n-3 polyunsaturated fatty acids (PUFA) intake. PUFA alter membrane physical characteristics and the activity of membrane-bound proteins. In membranes, they interact with ion channels and can be converted into bioactive eicosanoid (Harris, et al., 2008).

The American Heart Association recommendation for those with no history of coronary heart disease (CHD) is to eat two fatty fish servings per week and for those with a history of CHD to consume 1 g/day of two omega-3 fatty acids, eicosapentaenoic (EPA) and docosahexaenoic (DHA) acid (Kris-Etherton, et al., 2002). The Omega-3 Index is a measure of EPA (C20:5 n-3) and DHA (C22:6 n-3) in the blood, and expressed as a percent of 21 fatty acids, in specifically the red blood cell membranes (Harris, et al., 2004). The Omega-3 Index has been suggested for use to identify those at risk for death from coronary heart disease (Harris, et. al., 2004).

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

Esterified FA are hydrolyzed from washed RBC membranes using sequential treatment with mineral acid and base in the presence of heat. Using a modification of Lagerstedt et al., total FA are hexane-extracted from the matrix (100 µL 50:50 saline diluted RBC) along with an internal standard solution containing thirteen stable isotopically-labeled FA to account for recovery (Lagerstedt et. al., 2001). The extract is derivatized with pentafluorobenzyl bromide (PFBBR) in the presence of triethylamine to form pentafluorobenzyl esters. The reaction mixture is injected onto a capillary gas chromatograph column to resolve individual FA of interest from other matrix constituents. FA are detected using electron capture negative-ion mass spectrometry within 34 minutes. Six saturated FA (SFA), four monounsaturated FA (MUFA), and eleven polyunsaturated FA (PUFA) (21 FA in total) are measured using selected ion monitoring. Quantitation is accomplished by comparing the peak area ratio of the analyte to internal standard in the unknown sample with the peak area ratio of a known amount of analyte to internal standard in a calibrator solution.

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

Laboratory Method Files

[Fatty Acids – RBC](#) (December 2025)

Laboratory Quality Assurance and Monitoring

Washed red blood cell specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

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Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

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The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the Demographics file and other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. The lower limit of detection (LLOD, in mg/L) for the RBC fatty acids are shown below.

Variable Name	Analyte Description	LLOD
LBXPAN	alpha-Linolenic acid (C18:3n3) (%)	0.03
LBXP1A	Arachidic acid (C20:0) (%)	0.30
LBXPRA	Arachidonic acid (C20:4n6) (%)	0.08
LBXPDA	Docosanoic acid (C22:0) (%)	0.10
LBXPHA	Docosahexaenoic acid (C22:6n3) (%)	0.20
LBXPD3	Docosapentaenoic acid 3 (C22:5n3) (%)	0.07
LBXPD6	Docosapentaenoic acid 6 (C22:5n6) (%)	0.08
LBXPTA	Docosatetraenoic acid (C22:4n6) (%)	0.05
LBXPED	11,14-Eicosadienoic acid (C20:2n6) (%)	0.05
LBXP1E	11-Eicosenoic acid (C20:1n9) (%)	0.06
LBXPPE	Eicosapentaenoic acid (C20:5n3) (%)	0.40
LBXPLG	gamma-Linolenic acid (C18:3n6) (%)	0.07
LBXPGH	homo-gamma-Linolenic acid (C20:3n6) (%)	0.06
LBXP1G	Tetracosanoic acid (C24:0) (%)	0.10
LBXPNL	Linoleic acid (C18:2n6) (%)	0.30
LBXPMR	Myristic acid (C14:0) (%)	1.00
LBXPNR	15-Tetracosenoic acid (C24:1n9) (%)	0.05
LBXPOL	Oleic acid (C18:1n9) (%)	0.30
LBXPPL	Palmitoleic acid (C16:1n7) (%)	0.05
LBXPPM	Palmitic acid (C16:0) (%)	3.00
LBXPST	Stearic acid (C18:0) (%)	9.00

This component does not report data as mass concentrations (mg/L). Instead, data are reported as weight percentages of total fatty acids (see section on Calculating Percent RBC Fatty Acid). Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDPANLC) indicates whether the mass concentration result used in the weight percentage calculation was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXPAN) provides the analytic weight percentage result for that analyte. For analytes with mass concentration results below the lower limit of detection (ex., LBDPANLC=1), the weight percentage values were set to missing because the weight percentage data cannot be compared to the LLOD. 15 mass concentration results in the dataset were below the LLOD and corresponding weight percentage values were set to missing and their comment variables set to 1.

Calculating Percent RBC Fatty Acid

Fatty acid concentrations measured in mg/L are summed for the total FA concentration (mg/L). Individual fatty acid weight percentages are calculated by dividing the individual fatty acid concentrations (mg/L) by the total FA (mg/L). The fatty acid data is expressed as weight percentages of total fatty acids.

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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 241728.85724	Range of Values	8068	8068	
.	Missing	0	8068	

LBXPAN - alpha-Linolenic acid (C18:3n-3) (%)**Variable Name:** LBXPAN**SAS Label:** alpha-Linolenic acid (C18:3n-3) (%)**English Text:** alpha-Linolenic acid (C18:3n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.038 to 0.611	Range of Values	7176	7176	
.	Missing	892	8068	

LBDPANLC - alpha-Linolenic acid (C18:3n-3) cmt code

Variable Name: LBDPANLC
SAS Label: alpha-Linolenic acid (C18:3n-3) cmt code
English Text: alpha-Linolenic acid (C18:3n-3) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7176	7176	
1	below detectable limit	0	7176	
.	Missing	892	8068	

LBXP1A - Arachidic acid (C20:0) (%)**Variable Name:** LBXP1A**SAS Label:** Arachidic acid (C20:0) (%)**English Text:** Arachidic acid (C20:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.157 to 0.817	Range of Values	7133	7133	
.	Missing	935	8068	

LBDP1ALC - Arachidic acid (C20:0) comment code

Variable Name: LBDP1ALC
SAS Label: Arachidic acid (C20:0) comment code
English Text: Arachidic acid (C20:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7133	7133	
1	below detectable limit	0	7133	
.	Missing	935	8068	

LBXPRA - Arachidonic acid (C20:4n-6) (%)**Variable Name:** LBXPRA**SAS Label:** Arachidonic acid (C20:4n-6) (%)**English Text:** Arachidonic acid (C20:4n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.17 to 21.2	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPRALC - Arachidonic acid (C20:4n-6) comment code

Variable Name: LBDPRALC
SAS Label: Arachidonic acid (C20:4n-6) comment code
English Text: Arachidonic acid (C20:4n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPDA - Docosanoic acid (C22:0) (%)**Variable Name:** LBXPDA**SAS Label:** Docosanoic acid (C22:0) (%)**English Text:** Docosanoic acid (C22:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.629 to 3.26	Range of Values	7165	7165	
.	Missing	903	8068	

LBDPDALC - Docosanoic acid (C22:0) comment code

Variable Name: LBDPDALC
SAS Label: Docosanoic acid (C22:0) comment code
English Text: Docosanoic acid (C22:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	

LBXPHA - Docosahexaenoic acid (C22:6n-3) (%)**Variable Name:** LBXPHA**SAS Label:** Docosahexaenoic acid (C22:6n-3) (%)**English Text:** Docosahexaenoic acid (C22:6n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.337 to 10.4	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPHALC - Docosahexaenoic acid (C22:6n-3) cmt code

Variable Name: LBDPHALC

SAS Label: Docosahexaenoic acid (C22:6n-3) cmt code

English Text: Docosahexaenoic acid (C22:6n-3) comment code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXPD3 - Docosapentaenoic acid 3 (C22:5n-3) (%)**Variable Name:** LBXPD3**SAS Label:** Docosapentaenoic acid 3 (C22:5n-3) (%)**English Text:** Docosapentaenoic acid 3 (C22:5n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.381 to 7.5	Range of Values	7210	7210	
.	Missing	858	8068	

LBDPD3LC - Docosapentaenoic acid 3 (C22:5n-3) cmt**Variable Name:** LBDPD3LC**SAS Label:** Docosapentaenoic acid 3 (C22:5n-3) cmt**English Text:** Docosapentaenoic acid 3 (C22:5n-3) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7210	7210	
1	below detectable limit	0	7210	
.	Missing	858	8068	

LBXPD6 - Docosapentaenoic acid (C22:5n-6) (%)**Variable Name:** LBXPD6**SAS Label:** Docosapentaenoic acid (C22:5n-6) (%)**English Text:** Docosapentaenoic acid (C22:5n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.116 to 1.66	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPD6LC - Docosapentaenoic acid (C22:5n-6) cmt cd**Variable Name:** LBDPD6LC**SAS Label:** Docosapentaenoic acid (C22:5n-6) cmt cd**English Text:** Docosapentaenoic acid (C22:5n-6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPTA - Docosatetraenoic acid (C22:4n-6) (%)**Variable Name:** LBXPTA**SAS Label:** Docosatetraenoic acid (C22:4n-6) (%)**English Text:** Docosatetraenoic acid (C22:4n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.563 to 8.89	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPTALC - Docosatetraenoic acid (C22:4n-6) cmt cd

Variable Name: LBDPTALC
SAS Label: Docosatetraenoic acid (C22:4n-6) cmt cd
English Text: Docosatetraenoic acid (C22:4n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPED - 11,14-Eicosadienoic acid (C20:2n6) (%)

Variable Name: LBXPED**SAS Label:** 11,14-Eicosadienoic acid (C20:2n6) (%)**English Text:** 11,14-Eicosadienoic acid (C20:2n6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.123 to 0.628	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPEDLC - 11,14-Eicosadienoic acid(C20:2n6) cmt cd**Variable Name:** LBDPEDLC**SAS Label:** 11,14-Eicosadienoic acid(C20:2n6) cmt cd**English Text:** 11,14-Eicosadienoic acid (C20:2n6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXP1E - 11-Eicosenoic acid (C20:1n-9) (%)**Variable Name:** LBXP1E**SAS Label:** 11-Eicosenoic acid (C20:1n-9) (%)**English Text:** 11-Eicosenoic acid (C20:1n-9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.109 to 0.764	Range of Values	7035	7035	
.	Missing	1033	8068	

LBDP1ELC - 11-Eicosenoic acid (C20:1n-9) cmt code

Variable Name: LBDP1ELC
SAS Label: 11-Eicosenoic acid (C20:1n-9) cmt code
English Text: 11-Eicosenoic acid (C20:1n-9) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7035	7035	
1	below detectable limit	0	7035	
.	Missing	1033	8068	

LBXPPE - Eicosapentaenoic acid (C20:5n-3) (%)**Variable Name:** LBXPPE**SAS Label:** Eicosapentaenoic acid (C20:5n-3) (%)**English Text:** Eicosapentaenoic acid (C20:5n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.063 to 6.92	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPPELC - Eicosapentaenoic acid (C20:5n-3) cmt cd

Variable Name: LBDPPELC
SAS Label: Eicosapentaenoic acid (C20:5n-3) cmt cd
English Text: Eicosapentaenoic acid (C20:5n-3) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	2	7215	
.	Missing	853	8068	

LBXPLG - gamma-Linolenic acid (C18:3n-6) (%)**Variable Name:** LBXPLG**SAS Label:** gamma-Linolenic acid (C18:3n-6) (%)**English Text:** gamma-Linolenic acid (C18:3n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.016 to 0.168	Range of Values	7176	7176	
.	Missing	892	8068	

LBDPLGLC - gamma-Linolenic acid (C18:3n-6) cmt code

Variable Name: LBDPLGLC
SAS Label: gamma-Linolenic acid (C18:3n-6) cmt code
English Text: gamma-Linolenic acid (C18:3n6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7176	7176	
1	below detectable limit	0	7176	
.	Missing	892	8068	

LBXPGH - homo-gamma-Linolenic acid(C20:3n-6) (%)**Variable Name:** LBXPGH**SAS Label:** homo-gamma-Linolenic acid(C20:3n-6) (%)**English Text:** homo-gamma-Linolenic acid(C20:3n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.535 to 4.6	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPGHLC - homo-gamma-Linolenic acid (C20:3n-6) cmt**Variable Name:** LBDPGHLC**SAS Label:** homo-gamma-Linolenic acid (C20:3n-6) cmt**English Text:** homo-gamma-Linolenic acid (C20:3n-6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXP1G - Tetracosanoic acid (C24:0) (%)**Variable Name:** LBXP1G**SAS Label:** Tetracosanoic acid (C24:0) (%)**English Text:** Tetracosanoic acid (C24:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.05 to 9.48	Range of Values	7165	7165	
.	Missing	903	8068	

LBDP1GLC - Tetracosanoic acid (C24:0) comment code

Variable Name: LBDP1GLC
SAS Label: Tetracosanoic acid (C24:0) comment code
English Text: Tetracosanoic acid (C24:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	

LBXPNL - Linoleic acid (C18:2n-6) (%)**Variable Name:** LBXPNL**SAS Label:** Linoleic acid (C18:2n-6) (%)**English Text:** Linoleic acid (C18:2n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.04 to 15.6	Range of Values	7173	7173	
.	Missing	895	8068	

LBDPNLLC - Linoleic acid (C18:2n-6) comment code

Variable Name: LBDPNLLC
SAS Label: Linoleic acid (C18:2n-6) comment code
English Text: Linoleic acid (C18:2n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7173	7173	
1	below detectable limit	0	7173	
.	Missing	895	8068	

LBXPMR - Myristic acid (C14:0) (%)**Variable Name:** LBXPMR**SAS Label:** Myristic acid (C14:0) (%)**English Text:** Myristic acid (C14:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.135 to 1.13	Range of Values	7153	7153	
.	Missing	915	8068	

LBDPMRLC - Myristic acid (C14:0) comment code

Variable Name: LBDPMRLC
SAS Label: Myristic acid (C14:0) comment code
English Text: Myristic acid (C14:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7153	7153	
1	below detectable limit	13	7166	
.	Missing	902	8068	

LBXPNR - 15-Tetracosenoic acid (C24:1n9) (%)**Variable Name:** LBXPNR**SAS Label:** 15-Tetracosenoic acid (C24:1n9) (%)**English Text:** 15-Tetracosenoic acid (C24:1n9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.84 to 10.6	Range of Values	7166	7166	
.	Missing	902	8068	

LBDPNRLC - 15-Tetracosenoic acid(C24:1n9) comt code

Variable Name: LBDPNRLC
SAS Label: 15-Tetracosenoic acid(C24:1n9) comt code
English Text: 15-Tetracosenoic acid (C24:1n9) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7166	7166	
1	below detectable limit	0	7166	
.	Missing	902	8068	

LBXPOL - Oleic acid (C18:1n-9) (%)**Variable Name:** LBXPOL**SAS Label:** Oleic acid (C18:1n-9) (%)**English Text:** Oleic acid (C18:1n-9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.93 to 19.1	Range of Values	7125	7125	
.	Missing	943	8068	

LBDPOLL - Oleic acid (C18:1n-9) comment code

Variable Name: LBDPOLL

SAS Label: Oleic acid (C18:1n-9) comment code

English Text: Oleic acid (C18:1n-9) comment code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7125	7125	
1	below detectable limit	0	7125	
.	Missing	943	8068	

LBXPPL - Palmitoleic acid (16:1n-7) (%)**Variable Name:** LBXPPL**SAS Label:** Palmitoleic acid (16:1n-7) (%)**English Text:** Palmitoleic acid (16:1n-7) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.025 to 2.17	Range of Values	7084	7084	
.	Missing	984	8068	

LBDPPLLC - Palmitoleic acid (16:1n-7) comment code

Variable Name: LBDPPLLC
SAS Label: Palmitoleic acid (16:1n-7) comment code
English Text: Palmitoleic acid (16:1n-7) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7084	7084	
1	below detectable limit	0	7084	
.	Missing	984	8068	

LBXPPM - Palmitic acid (C16:0) (%)**Variable Name:** LBXPPM**SAS Label:** Palmitic acid (C16:0) (%)**English Text:** Palmitic acid (C16:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15.4 to 31.1	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPPMLC - Palmitic acid (C16:0) comment code

Variable Name: LBDPPMLC
SAS Label: Palmitic acid (C16:0) comment code
English Text: Palmitic acid (C16:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXPST - Stearic acid (C18:0) (%)**Variable Name:** LBXPST**SAS Label:** Stearic acid (C18:0) (%)**English Text:** Stearic acid (C18:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10.1 to 21.8	Range of Values	7165	7165	
.	Missing	903	8068	

LBDPSTLC - Stearic acid (C18:0) comment code

Variable Name: LBDPSTLC
SAS Label: Stearic acid (C18:0) comment code
English Text: Stearic acid (C18:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Fasting Questionnaire (FASTQX_L)

Data File: FASTQX_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

The fasting questionnaire is administered to determine the fasting status of the survey participant. Questions include but are not limited to: length of "food" fast, whether the participant had gum or mints, coffee or tea, or alcohol or dietary supplements prior to their laboratory examination.

Eligible Sample

Examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

The MEC phlebotomist administered the questionnaire to determine the fasting status of the survey participant.

There were no changes to the questionnaire or lab site for this component in the NHANES August 2021 - August 2023 cycle.

Laboratory Quality Assurance and Monitoring

Detailed instructions on specimen collection and processing are discussed in the NHANES Laboratory Procedures Manual (LPM). The NHANES quality control and quality assurance protocols (QA/QC) meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the NHANES LPM.

Data Processing and Editing

Fasting questionnaire data were recorded directly into a computerized database; all forms used in the MEC were automated.

Analytic Notes

Refer to the August 2021 – August 2023 Laboratory Data Overview for general information on NHANES laboratory data.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES August 2021 – August 2023 Demographics File contains demographic data, health indicators, and other related information collected

during household interviews as well as the sample weight variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This Fasting Questionnaire File includes auxiliary information such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Use PHAFSTHR and PHAFSTMN to determine the duration of fasting in hours and minutes, respectively. When calculating a fasting time in hours and minutes, pay careful attention to the other times in hours and minutes. For example, the survey participants were asked: "When was the last time you ate or drank anything other than plain water? Do not include diet soda or black coffee or tea with artificial sweeteners like Sweet'N Low, NutraSweet, Equal, or Splenda." Survey participants were then asked questions PHQ020 to PHQ060: "Have you had any of the following since the last time you ate or drank anything other than plain water? Coffee or tea with cream or sugar (include milk or non-dairy creamers); Alcohol, such as beer, wine, or liquor; Gum, breath mints, lozenges, cough drops, or cold remedies; Antacids, laxatives, or anti-diarrheal; Dietary supplements such as vitamins and minerals."

If you are using fasting status in any of your analyses, please consider the answers to PHACOFHR/PHACOFMN, PHAALCHR/PHAALCMN, PHAGUMHR/PHAGUMMN/ PHAANTHR/PHAANTMN, and PHASUPHR/PHASUPMN with the times in hours and minutes in addition to the PHAFSTHR and PHAFSTMN times.

Use PHDSESN to determine the session in which the survey participant was examined.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 150 YEARS

PHQ020 - Coffee or tea with cream or sugar?**Variable Name:** PHQ020**SAS Label:** Coffee or tea with cream or sugar?**English Text:** Coffee or tea with cream or sugar? [Include milk or non-dairy creamers.]**English Instructions:** Have you had any of the following since " today at" or "yesterday at" fasting time in PHQ010.**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	29	29	
2	No	8607	8636	PHQ030
.	Missing	91	8727	

PHACOFHR - Coffee/tea fast time (hours)

Variable Name: PHACOFHR

SAS Label: Coffee/tea fast time (hours)

English Text: The time (in hours) since the examinee last drank coffee or tea with cream or sugar.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 26	Range of Values	29	29	
.	Missing	8698	8727	

PHACOFMN - Coffee/tea fast time (minutes)

Variable Name: PHACOFMN

SAS Label: Coffee/tea fast time (minutes)

English Text: The time (in minutes) since the examinee last drank coffee or tea with cream or sugar.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 59	Range of Values	29	29	
.	Missing	8698	8727	

PHQ030 - Alcohol, such as beer, wine, or liquor?

Variable Name: PHQ030

SAS Label: Alcohol, such as beer, wine, or liquor?

English Text: Alcohol, such as beer, wine, or liquor?

English Instructions: Have you had any of the following since "today at" or "yesterday at" the fasting time in PHQ010.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	25	25	
2	No	8611	8636	PHQ040
.	Missing	91	8727	

PHAALCHR - Alcohol fast time (hours)

Variable Name: PHAALCHR

SAS Label: Alcohol fast time (hours)

English Text: The time (in hours) the examinee last drank alcohol suchs as beer, wine, or liquor.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 43	Range of Values	25	25	
.	Missing	8702	8727	

PHAALCMN - Alcohol fast time (minutes)

Variable Name: PHAALCMN

SAS Label: Alcohol fast time (minutes)

English Text: The time (in minutes) the examinee last drank alcohol such as beer, wine, or liquor.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4 to 57	Range of Values	25	25	
.	Missing	8702	8727	

PHQ040 - Gum, mints, lozenges or cough drops

Variable Name: PHQ040

SAS Label: Gum, mints, lozenges or cough drops

English Text: Gum, breath mints, lozenges or cough drops, or other cough or cold remedies?

English Instructions: Have you had any of the following since "today at" or "yesterday at" the fasting time in PHQ010?

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	311	311	
2	No	8324	8635	PHQ050
.	Missing	92	8727	

PHAGUMHR - Gum, mints cough drops fast time (hours)

Variable Name: PHAGUMHR

SAS Label: Gum, mints cough drops fast time (hours)

English Text: The time (in hours) the examinee last had gum, breath mints, lozenges or cough drops, or other cough or cold remedies.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 18	Range of Values	311	311	
.	Missing	8416	8727	

PHAGUMMN - Gum, mints, cough fast time (minutes)**Variable Name:** PHAGUMMN**SAS Label:** Gum, mints, cough fast time (minutes)**English Text:** The time (in minutes) the examinee last had gum, breath mints, lozenges or cough drops, or other cough or cold remedies.**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 58	Range of Values	311	311	
.	Missing	8416	8727	

PHQ050 - Antacids, laxatives, or anti-diarrheals?

Variable Name: PHQ050

SAS Label: Antacids, laxatives, or anti-diarrheals?

English Text: Antacids, laxatives, or anti-diarrheals?

English Instructions: Have you had any of the following since "today at" or "yesterday at" the fasting time in PHQ010.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	105	105	
2	No	8531	8636	PHQ060
.	Missing	91	8727	

PHAANTHR - Antacids, laxatives fast time (hours)

Variable Name: PHAANTHR

SAS Label: Antacids, laxatives fast time (hours)

English Text: The time (in hours) the examinee last had antacids, laxatives or anti-diarrheals.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 17	Range of Values	105	105	
.	Missing	8622	8727	

PHAANTMN - Antacids, laxatives fast time (minutes)

Variable Name: PHAANTMN

SAS Label: Antacids, laxatives fast time (minutes)

English Text: The time (in minutes) the examinee last had antacids, laxatives or anti-diarrheals.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 59	Range of Values	105	105	
.	Missing	8622	8727	

PHQ060 - Dietary supplements?

Variable Name: PHQ060

SAS Label: Dietary supplements?

English Text: Dietary supplements such as vitamins and minerals? [Include multivitamins and single nutrient supplements.]

English Instructions: Have you had any of the following since "today at" or "yesterday at" the fasting time in PHQ010.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	219	219	
2	No	8417	8636	PHAFSTHR
.	Missing	91	8727	

PHASUPHR - Dietary supplements fast time (hours)

Variable Name: PHASUPHR

SAS Label: Dietary supplements fast time (hours)

English Text: The time (in hours) the examinee last had dietary supplements such as vitamins and minerals.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 20	Range of Values	219	219	
.	Missing	8508	8727	

PHASUPMN - Dietary supplements fast time (minutes)

Variable Name: PHASUPMN

SAS Label: Dietary supplements fast time (minutes)

English Text: The time (in minutes) the examinee last had dietary supplements such as vitamins and minerals.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 59	Range of Values	219	219	
.	Missing	8508	8727	

PHAFSTHR - Total length of 'food fast' (hours)

Variable Name: PHAFSTHR

SAS Label: Total length of 'food fast' (hours)

English Text: The time (in hours) between when the examinee last ate or drank anything other than water and the time of the venipuncture.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 29	Range of Values	8609	8609	
30	30 hours or more	23	8632	
.	Missing	95	8727	

PHAFSTMN - Total length of 'food fast' (minutes)

Variable Name: PHAFSTMN

SAS Label: Total length of 'food fast' (minutes)

English Text: The time (in minutes) between when the examinee last ate or drank anything other than water and the time of the venipuncture.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 59	Range of Values	8632	8632	
.	Missing	95	8727	

PHDSESNZ - Session in which SP was examined

Variable Name: PHDSESNZ
SAS Label: Session in which SP was examined
English Text: Session in which SP was examined
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	morning	4756	4756	
1	afternoon/evening	3971	8727	
.	Missing	0	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Ferritin (FERTIN_L)

Data File: FERTIN_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Ferritin is a protein that contains iron and is found in red blood cells. Ferritin levels are used primarily in evaluating the body's iron metabolism and levels of iron storage or reserves, which can be too low or too high. Low storage of iron can lead to iron deficiency anemia. High levels of iron storage, also called iron overload, occurs when excess iron is accumulated in the body, primarily the liver.

The objectives of this component are: 1) to provide data for monitoring secular trends in measures of nutritional status in the U.S. population; 2) to evaluate the effects of people's habits and behaviors, such as physical activity and the use of alcohol, tobacco, and dietary supplements on nutritional status; and 3) to evaluate the effect of changes in nutrition and public health policies, including welfare reform legislation, food fortification policy, and child nutrition programs, on the nutritional status of the U.S. population.

These data will be used to estimate deficiencies and toxicities of specific nutrients in the population and subgroups, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to serum levels of nutrients. Data will be used for research to further define nutrient requirements, as well as optimal levels for disease prevention and health promotion.

Eligible Sample

Examined participants 1-5 years old and 12-49 years old females were eligible.

Description of Laboratory Methodology

The method for measurement of Ferritin on the Roche Cobas® e601 is a sandwich principle with a total duration time of 18 minutes. The 1st incubation uses 10 µL of sample, a ferritin-specific antibody and a labeled ferritin-specific antibody to form a sandwich complex. The 2nd incubation occurs after the addition of microparticles that cause the complex to bind to the solid phase. The reaction mixture is aspirated into the measuring cell where the microparticles are magnetically captured onto the surface of the electrode. Unbound substances are then removed. Application of a voltage to the electrode then induces chemiluminescent emission which is measured by a photomultiplier. Results are determined via a calibration curve.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

Ferritin (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual](#) (LPM). Vials were stored under appropriate frozen (-30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable LBDFERSI was created using the formula:

LBDFERSI: The ferritin value in ng/mL (LBXFER) was converted to $\mu\text{g/L}$ (LBDFERSI) by multiplying LBXFER by 1.00 (rounded to 3 significant figures).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBDFERLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXFER) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDFERLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD in ng/mL) for LBXFER:

Variable Name	SAS Label	LLOD
LBXFER	Ferritin	0.5 ng/mL

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 5 YEARS
Target:	Females only 12 YEARS - 49 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6213.3002375 to 253478.77765	Range of Values	2051	2051	
0	No blood sample provided	513	2564	
.	Missing	0	2564	

LBXFER - Ferritin(ng/mL)

Variable Name: LBXFER
SAS Label: Ferritin(ng/mL)
English Text: Ferritin(ng/mL)
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.05 to 984	Range of Values	1950	1950	
.	Missing	614	2564	

LBDFERSI - Ferritin($\mu\text{g/L}$)

Variable Name: LBDFERSI
SAS Label: Ferritin($\mu\text{g/L}$)
English Text: Ferritin($\mu\text{g/L}$)
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.05 to 984	Range of Values	1950	1950	
.	Missing	614	2564	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Folate Forms - Total & Individual - Washed RBCs (FFMR_L)

Data File: FFMR_L.xpt

First Published: January 2026

Last Revised: NA

Component Description

Folate belongs to the group of water-soluble B vitamins that occur naturally in food. Prolonged folate deficiency leads to megaloblastic anemia. Low folate status has been causally linked to an increased risk in women of reproductive age to have an offspring with neural tube defects. Low folate status also increases plasma homocysteine levels, a potential risk factor for chronic diseases, such as cardiovascular disease or cognitive function. Potential roles of folate and other B vitamins in modulating the risk for diseases (e.g., heart disease, cancer, and cognitive impairment) are under investigation. While serum folate is an indicator of recent intake, red blood cell (RBC) folate is an indicator of long-term status.

These data will be used to estimate deficiencies and other health effects of specific nutrients in the population and subgroup, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to the levels of nutrients. Data will be used in research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion. Data will also be used to evaluate the effect of changes in nutrition and public health policies, including welfare reform legislation, food fortification policy, and child nutrition programs on the nutritional status of the U.S. population.

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

RBC folate status can be determined directly by measuring folate forms in washed RBCs (Stamm et al., 2018) or indirectly by measuring folate forms in a whole blood (WB) lysate (Fazili et al., 2004; Fazili et al., 2005). The direct measurement requires the addition of exogenous γ -glutamyl hydrolase (exo-GGH) to deconjugate folate polyglutamates to monoglutamates (Stamm et al., 2018). It also requires the measurement of hemoglobin (Hb) in the RBC lysate to correct for residual moisture (~20%) in the washed RBCs. By using the mean corpuscular hemoglobin content (MCHC), RBC folate can be calculated. The indirect measurement can either utilize the endogenous plasma-based γ -glutamyl hydrolase (no addition of enzyme; 4 h incubation at 37°C) or preferably, for faster deconjugation, use the addition of exo-GGH (30 min incubation at ambient temperature [+15°C to +30°C]). To calculate RBC folate, it requires the measurement of serum folate to correct for the serum folate contribution to WB folate and hematocrit (Hct) to normalize to the proportion of packed red cells.

For this NHANES component, RBC folate status was determined directly by measuring folate forms in washed RBCs.

Five folate forms, 5-methylTHF, pteroylglutamic acid (Folic Acid, PGA), tetrahydrofolate (THF), 5-formyl-tetrahydrofolic acid (5-formylTHF), 5,10-methenyltetrahydrofolic acid (5,10-methenylTHF), and one oxidation product of 5-methylTHF called MeFox (pyrazino-s-triazine

derivative of 4- α -hydroxy-5-methylTHF) are measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (HPLC-MS/MS). The current method is a modification of previously published methods (Fazili et al., 2004; Fazili, et al., 2005) to add the measurement of an oxidation product (Fazili et al., 2013;143:108-113) and to scale down the amount of specimen needed and increase the sample throughput (Fazili et al., 2013;405:4549-60). The method distinguishes 5-formylTHF from 10-formylTHF on the basis of mass transitions; however, during HPLC-MS/MS in the acidic mobile phase, 10-formylTHF converts within minutes to 5,10-methenylTHF and trace amounts of 5-formylTHF, THF, and 10-formyl-folic acid. Thus, this method is not capable of quantifying 10-formylTHF; formylated folates are quantified as 5-formylTHF and 5,10-methenylTHF, which are stable under these conditions (Pfeiffer et al., 2004).

The assay is performed by combining specimen (150 μ L of WB lysate or RBC lysate) with an internal standard mixture containing exo-GGH enzyme (5 μ g per mL of WB lysate or RBC lysate) and incubating at ambient temperature (+15°C to +30°C) for 30 min to deconjugate folate polyglutamates to monoglutamates prior to folate extraction. Ammonium formate buffer (1%) is added to the samples and extraction and clean-up is performed by automated 96-probe solid phase extraction (SPE) using 96-well phenyl SPE plates which takes ~1 h for a 96-well plate. Folate forms are separated within 4 min using isocratic mobile phase conditions and measured by HPLC-MS/MS (7 min to next injection). Quantitation is based on peak area ratios interpolated against a five-point aqueous linear calibration curve using 1/x² weighting.

RBC folate form concentrations (nmol/L) measured in an RBC lysate are calculated as RBC folate form = (RBC lysate folate form / Hb) * MCHC.

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

Laboratory Method Files

[Folate Forms - Total & Individual – Washed RBCs](#) (January 2026)

Laboratory Quality Assurance and Monitoring

Washed red blood cell specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured QA evaluation during unscheduled visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences’ QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

RBC Folate Forms for NHANES 2021-2023

In NHANES 2021-2023, washed red blood cells (RBC) were diluted approximately 1+1 with saline and then 1+10 with ascorbic acid solution to generate RBC lysates. Several folate forms were measured in RBC lysates by isotope-dilution high-performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS). Instrument-measured lysate concentrations were converted to concentrations expressed in nmol/L RBC by dividing by the Hb content measured in the same aliquot and multiplying by the participant-specific MCHC to account for the ‘dilution’ effect from washed RBC to lysate (Table 1, LBXRF1SI to LBXRF6SI). For individual RBC folate forms with lysate concentrations below the limit of detection (LOD), an imputed value equal to the LOD divided by the square root of 2 was used prior to calculating the RBC concentration (nmol/L RBC). RBC folate form concentrations were not calculated when the Hb or MCHC value was missing. Total RBC folate (LBDRF7SI) was calculated as the sum of individual folate forms LBXRF1SI through LBXRF5SI. LBXRF6SI was

not included in the total folate calculation, because this folate form is not biologically active and due to evidence that it may already be present in vivo (Pfeiffer, et al., 2015). RBC total folate concentration was not calculated when one or more individual RBC folate form results were missing.

Table 1. Folate forms measured by LC-MS/MS

Analyte	Abbreviation	Variable Name
5-Methyltetrahydrofolate	5-methylTHF	LBXRF1SI
Pteroylglutamic acid (Folic Acid)	PGA	LBXRF2SI
5-Formyltetrahydrofolate	5-formylTHF	LBXRF3SI
Tetrahydrofolate	THF	LBXRF4SI
5,10-Methenyltetrahydrofolate	5,10-methenylTHF	LBXRF5SI
Pyrazino-s-triazine derivative of 4-a-hydroxy-5-methyltetrahydrofolate	MeFox	LBXRF6SI
Total washed RBC folate (sum of biologically active folate forms)	tFOL	LBDRF7SI

Demographic and Other Related Variables

The analysis of NHANES laboratory data may require additional demographic variables. The [NHANES August 2021- August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews.

This laboratory data file can be linked to the Demographics file and other NHANES data files in the August 2021-August 2023 cycle using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The lower limit of detection (LLOD) for the five folate forms and the oxidation product of 5-methylTHF applies to instrument measurements performed on RBC lysates, not to the reported concentrations of folate forms in RBC (nmol/L RBC) that are presented in the data. Reported RBC folate concentrations are calculated from the instrument-measured lysate values using each participant's Hb and MCHC. As a result, even when the underlying instrument measurement is below the detection limit, the reported values expressed in nmol/L RBC will be larger than the instrument LLOD values. Because values below the LLOD were addressed during the calculation of RBC concentrations from lysate instrument measurements, data analysts do not need to apply the LLOD values in subsequent analyses. However, an indicator variable is provided. The variable name ending in "LC" (ex., LBDRF1LC) indicates whether the instrument-measured lysate value was below the LLOD (0 = at or above LLOD; 1 = below LLOD). Measurements indicated as below the LLOD retain the increased uncertainty associated with instrument results below the detection limit, even when the reported RBC concentration is greater than the LLOD. The corresponding variable prefixed LBX (ex., LBXRF1SI) reports the calculated concentration in nmol/L RBC.

Total folate was calculated as the sum of individual folate forms (RF1–RF5); therefore, a lower limit of detection does not apply to total folate.

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBDRF7SI - Total Folate, RBC (nmol/L)

Variable Name: LBDRF7SI
SAS Label: Total Folate, RBC (nmol/L)
English Text: Total folate, washed Red Blood Cells (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
125 to 6360	Range of Values	7190	7190	
.	Missing	878	8068	

LBXRF1SI - 5-Methyl-tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF1SI
SAS Label: 5-Methyl-tetrahydrofolate, RBC (nmol/L)
English Text: 5-Methyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
28 to 5960	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF1LC - 5-Methyl-tetrahydrofolate, RBC cmt

Variable Name: LBDRF1LC
SAS Label: 5-Methyl-tetrahydrofolate, RBC cmt
English Text: 5-Methyl-tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7190	7190	
1	Below lower detection limit	0	7190	
.	Missing	878	8068	

LBXRF2SI - Folic acid, RBC (nmol/L)

Variable Name: LBXRF2SI
SAS Label: Folic acid, RBC (nmol/L)
English Text: Folic acid, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.664 to 72.5	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF2LC - Folic acid, RBC cmt

Variable Name: LBDRF2LC**SAS Label:** Folic acid, RBC cmt**English Text:** Folic acid, RBC comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	3230	3230	
1	Below lower detection limit	3960	7190	
.	Missing	878	8068	

LBXRF3SI - 5-Formyl-tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF3SI
SAS Label: 5-Formyl-tetrahydrofolate, RBC (nmol/L)
English Text: 5-Formyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.92 to 45.5	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF3LC - 5-Formyl-tetrahydrofolate, RBC cmt

Variable Name: LBDRF3LC
SAS Label: 5-Formyl-tetrahydrofolate, RBC cmt
English Text: 5-Formyl-tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2985	2985	
1	Below lower detection limit	4205	7190	
.	Missing	878	8068	

LBXRF4SI - Tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF4SI
SAS Label: Tetrahydrofolate, RBC (nmol/L)
English Text: Tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.84 to 1360	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF4LC - Tetrahydrofolate, RBC cmt

Variable Name: LBDRF4LC
SAS Label: Tetrahydrofolate, RBC cmt
English Text: Tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7101	7101	
1	Below lower detection limit	89	7190	
.	Missing	878	8068	

LBXRF5SI - 5,10-Methenyl-tetrahydrofolate (nmol/L)

Variable Name: LBXRF5SI
SAS Label: 5,10-Methenyl-tetrahydrofolate (nmol/L)
English Text: 5,10-Methenyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.56 to 1020	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF5LC - 5,10-Methenyl-tetrahydrofolate, RBC cmt**Variable Name:** LBDRF5LC**SAS Label:** 5,10-Methenyl-tetrahydrofolate, RBC cmt**English Text:** 5,10-Methenyl-tetrahydrofolate, RBC comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	5291	5291	
1	Below lower detection limit	1899	7190	
.	Missing	878	8068	

LBXRF6SI - MeFox oxidation product, RBC (nmol/L)

Variable Name: LBXRF6SI
SAS Label: MeFox oxidation product, RBC (nmol/L)
English Text: MeFox oxidation product, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.57 to 1630	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF6LC - MeFox oxidation product, RBC cmt

Variable Name: LBDRF6LC
SAS Label: MeFox oxidation product, RBC cmt
English Text: MeFox oxidation product, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7190	7190	
1	Below lower detection limit	0	7190	
.	Missing	878	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Folate - RBC (FOLATE_L)

Data File: FOLATE_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Folate belongs to the group of water-soluble B vitamins that occur naturally in food. It is required in cellular one carbon metabolism and hematopoiesis. Prolonged folate deficiency leads to megaloblastic anemia (Bailey, 2015). Low folate status has been shown to increase the risk of women of childbearing age to have an offspring with neural tube defects. Low folate status also increases plasma homocysteine levels, a potential risk factor for cardiovascular disease, in the general population. Potential roles of folate and other B vitamins in modulating the risk for diseases (e.g., heart disease, cancer, and cognitive impairment) are currently being studied.

The measurement of total folate (TFOL) provides information on the folate status of the individual. Serum folate is an indicator of short-term status, while red blood cell (RBC) folate is an indicator of long-term status. These data will be used to estimate deficiencies and toxicities of specific nutrients in the population and subgroup, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to serum levels of nutrients. Data will be used in research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion.

Eligible Sample

All examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

Population folate status in the NHANES August 2021–August 2023 survey cycle was assessed by a combination of two analytical methods: whole-blood folate was measured by microbiologic assay, while serum folate forms were measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS), this is the same method used in NHANES 2011–March 2020. RBC folate was then calculated using the data from both assays.

Whole Blood Folate

Microbiological assays have been used for many years to estimate the concentration of folate in blood and other tissues. In the 1990s, more robust and reliable procedures were developed that use microtiter plates for higher throughput and a cryopreserved antibiotic resistant microorganism to avoid having to work under aseptic conditions. The described procedure is an adaptation of this method and was introduced by the National Center for Environmental Health (NCEH) to NHANES in 2007. Diluted whole blood is added to an assay medium containing *Lactobacillus rhamnosus* (formerly known as *Lactobacillus casei*) (NCIB 10463) and all of the nutrients necessary for the growth of *L. rhamnosus* except folate. The inoculated medium is incubated for 42 hours at 37°C. Since the growth of *L. rhamnosus* is proportional to the amount of total folate present in whole blood sample; the folate level can be assessed by measuring the turbidity of the inoculated medium at 590 nm in a PowerWave X340 Microplate

reader (Bio-Tek Instrument). The assay was calibrated with 5-methyl-tetrahydrofolate from Merck Cie (Eprova).

Serum Total Folate

Serum total folate was calculated as the sum of individual folate forms. Five folate forms, 5-methyl-tetrahydrofolate, folic acid, 5-formyl-tetrahydrofolate, tetrahydrofolate, and 5,10-methenyl-tetrahydrofolate are measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) (Fazili, et. al., 2013). The assay is performed by combining specimen (150 µL serum) with an ammonium formate buffer and an internal standard mixture. Sample extraction and clean-up is performed by automated 96-probe solid phase extraction (SPE) using 96-well phenyl SPE plates and takes ~1 h for a 96-well plate. Folate forms are separated within 6 min using isocratic mobile phase conditions and measured by LC-MS/MS. Quantitation is based on peak area ratios interpolated against a five-point aqueous linear calibration curve using 1/x² weighting.

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab for this component during the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Folate - RBC](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Whole blood and blood serum are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the NHANES [Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured QA evaluation during unscheduled visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable (LBDRFO) was created using the following formula:

LBDRFO: The RBC folate value in nmol/L RBC (LBDRFOSI) was converted to ng/mL RBC (LBDRFO) by dividing LBDRFOSI by 2.265 (rounded to 3 significant figures).

Analytic Notes

Red blood cell (RBC) folate in NHANES August 2021 – August 2023 was calculated from the whole blood folate concentration as measured by microbiologic assay by adjusting for RBC volume and correcting for serum total folate concentration, which was calculated as the sum of individual folate forms. The amounts of individual serum folate forms were measured by liquid chromatography-tandem mass spectrometry (LC-MS/MS). For folate forms with results lower than limit of detection (LOD), an imputed value of LOD divided by the square root of 2 was used. Serum folate forms used to calculate serum total folate concentration were: 5-methyl-tetrahydrofolate, folic acid, 5-formyl-tetrahydrofolate, tetrahydrofolate, and 5,10-methenyl-tetrahydrofolate. For more detailed information regarding folate forms data in NHANES August 2021–August 2023, please refer to the documentation accompanying the Folate Forms – Serum ([FOLFMS_L](#)) file.

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, the time of venipuncture, and the conditions precluding venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

An exact lower limit of detection (LLOD) for RBC folate cannot be calculated because the value is a composite of whole blood folate, serum folate, and hematocrit. Therefore, there is no LLOD for the calculated value of RBC folate. Furthermore, the LOD of this method for whole blood folate depends on the dilution factor (i.e., $\text{LOD} = 44 \text{ nmol/L}$ whole blood if whole blood hemolysate is only diluted 1/40; assuming a hematocrit of 40%, this would correspond to a RBC folate concentration of 110 nmol/L RBC).

References

- Bailey LB, Stover PJ, McNulty H, Fenech MF, Gregory III JF, Mills JL, Pfeiffer CM, Fazili Z, Zhang M, et al. Biomarkers of nutrition for development – folate review. *J Nutr.* 2015;145:1636S–80S.
- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094–40106.
- Fazili Z, Whitehead RD Jr, Paladugula N, Pfeiffer CM. A high-throughput LC-MS/MS method suitable for population biomonitoring measures five serum folate vitamers and one oxidation product. *Anal Bioanal Chem.* 2013;405:4549–60.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493–501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBDRFO - RBC folate (ng/mL)**Variable Name:** LBDRFO**SAS Label:** RBC folate (ng/mL)**English Text:** RBC folate (ng/mL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
63.6 to 3240	Range of Values	7506	7506	
.	Missing	1221	8727	

LBDRFOSI - RBC folate (nmol/L)

Variable Name: LBDRFOSI
SAS Label: RBC folate (nmol/L)
English Text: RBC folate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
144 to 7330	Range of Values	7506	7506	
.	Missing	1221	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Serum Folate Forms - Total & Individual - Serum (FOLFMS_L)

Data File: FOLFMS_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Folate belongs to the group of water-soluble B vitamins that occur naturally in food. It is required in cellular one carbon metabolism and hematopoiesis (Bailey, 2015). Prolonged folate deficiency leads to megaloblastic anemia. Low folate status has been shown to increase the risk of women of childbearing age to have an offspring with neural tube defects. Low folate status also increases plasma homocysteine levels, a potential risk factor for cardiovascular disease, in the general population. Potential roles of folate and other B vitamins in modulating the risk for diseases (e.g., heart disease, cancer, and cognitive impairment) are currently being studied.

The measurement of total folate (TFOL) provides information on the folate status of the individual. Serum folate is an indicator of short-term status, while red blood cell (RBC) folate is an indicator of long-term status. These data will be used to estimate deficiencies and toxicities of specific nutrients in the population and subgroup, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to serum levels of nutrients. Data will be used in research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion.

Eligible Sample

Examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

Six folate forms, 5-methyltetrahydrofolate, folic acid, tetrahydrofolate, 5-formyltetrahydrofolate, 5,10-methenyltetrahydrofolate, and an oxidation product of 5-methyltetrahydrofolate called MeFox (pyrazino-s-triazine derivative of 4- α -hydroxy-5-methyltetrahydrofolate) are measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) (Fazili, et. al., 2013). The assay is performed by combining specimen (150 μ L serum) with an ammonium formate buffer and an internal standard mixture. Sample extraction and clean-up is performed by automated 96-probe solid phase extraction (SPE) using 96-well phenyl SPE plates and takes $\sim$ 1 h for a 96-well plate. Folate forms are separated within 6 min using isocratic mobile phase conditions and measured by LC-MS/MS. Quantitation is based on peak area ratios interpolated against a five-point aqueous linear calibration curve using $1/x^2$ weighting.

This method distinguishes 5-formylTHF from 10-formylTHF on the basis of mass transitions, however, during LC-MS/MS in the acidic mobile phase, 10-formylTHF converts within minutes to 5,10-methenylTHF and trace amounts of 5-formylTHF, THF, and 10-formyl-folic acid. Thus, this method is not capable of quantifying 10-formylTHF; formylated folates are quantified as 5-formylTHF and 5,10-methenylTHF, which are stable under these conditions (Pfeiffer, et. al., 2004). Because concentrations of the three minor folate forms THF, 5-formylTHF, and 5,10-methenylTHF are typically below the limit of detection (LOD) and can be a result of folate

interconversions at slightly acidic pH during sample preparation, it is recommended to express the sum of these three forms as non-methyl folate (Pfeiffer, et. al., 2015).

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component during the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Folate Forms](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured QA evaluation during unscheduled visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences’ QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable (LBDFOT) was created using the following formula:

LBDFOT: The serum folate value in nmol/L (LBDFOTSI) was converted to ng/mL (LBDFOT) by dividing LBDFOTSI by 2.265 (rounded to 3 significant figures).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Serum Folate Forms for NHANES August 2021-August 2023

In NHANES August 2021-August 2023, a comprehensive list of serum folate forms were measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) (Table 1). Serum total folate (LBDFOTSI) was calculated by adding LBXSF1SI-LBXS5SI. LBXS6SI was not included in the total folate calculation, due to evidence that it may already be present in vivo (Pfeiffer, et. al., 2015). An imputed value of LOD divided by the square root of 2 was used for individual folate forms with results that were less than the limit of detection (LOD).

Please refer to the Analytic Notes for the 2011-2012 Folate Forms – Serum ([FOLFMS_G](#)) file for additional details on the comparability in serum total folate and folate forms measured between NHANES 2011–2020 and the previous survey cycles.

Table 1. Folate forms measured by LC-MS/MS

Analyte	Abbreviation	Variable Name
5-Methyltetrahydrofolate	5-methylTHF	LBXSF1SI
Pteroylglutamic acid (Folic Acid)	Folic acid	LBXSF2SI
5-Formyltetrahydrofolate	5-formylTHF	LBXSF3SI
Tetrahydrofolate	THF	LBXSF4SI
5,10-Methenyltetrahydrofolate	5,10-methenylTHF	LBXSF5SI
Pyrazino-s-triazine derivative of 4-a-hydroxy-5-methyltetrahydrofolate	MeFox	LBXSF6SI
Serum total folate (sum of folate forms)	tFOL	LBDFOTSI

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The August 2021-August 2023 Demographics File contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The Fasting Questionnaire File includes auxiliary information, such as fasting status, the time of venipuncture, and the conditions precluding venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDSF1LC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXSF1SI) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDSF1LC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD, in nmol/L) for the 6 folate forms are shown below. Because total folate is calculated from the sum of folate forms SF1-SF5, a lower limit of detection does not apply.

Variable Name	Analyte Description	LLOD
LBXSF1SI	5-Methyltetrahydrofolate	0.13
LBXSF2SI	Folic acid	0.14
LBXSF3SI	5-Formyltetrahydrofolate	0.20
LBXSF4SI	Tetrahydrofolate	0.25
LBXSF5SI	5,10-Methenyltetrahydrofolate	0.20
LBXSF6SI	MeFox	0.10
LBDFOISI	Total folate, serum	n/a

References

- Bailey LB, Stover PJ, McNulty H, Fenech MF, Gregory III JF, Mills JL, Pfeiffer CM, Fazili Z, Zhang M, et al. Biomarkers of nutrition for development – folate review. J Nutr. 2015;145:1636S–80S.
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- Fazili Z, Whitehead RD Jr, Paladugula N, Pfeiffer CM. A high-throughput LC-MS/MS method suitable for population biomonitoring measures five serum folate vitamers and one oxidation product. Anal Bioanal Chem. 2013;405:4549–60.
- Pfeiffer C, Sternberg M, Fazili M, Lacher D, Zhang M, Johnson C, Hammer H, Baily R, Rader J, Yamini S, Berry RJ, Yetley E. British Journal of Nutrition (2015) 113:1965:1977.
- Pfeiffer CM, Fazili Z, McCoy LF, Gunter EW. Determination of folate vitamers in human serum by stable-isotope dilution tandem mass spectrometry and comparison to radioassay and microbiologic assay. Clin Chem. 2004;50(2):423-432.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 1 YEARS - 150 YEARS

LBDFOTSI - Serum total folate (nmol/L)

Variable Name: LBDFOTSI
SAS Label: Serum total folate (nmol/L)
English Text: Serum total folate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.89 to 445	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDFOT - Serum total folate (ng/mL)**Variable Name:** LBDFOT**SAS Label:** Serum total folate (ng/mL)**English Text:** Serum total folate (ng/mL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.72 to 196	Range of Values	7348	7348	
.	Missing	1379	8727	

LBXSF1SI - 5-Methyl-tetrahydrofolate (nmol/L)

Variable Name: LBXSF1SI
SAS Label: 5-Methyl-tetrahydrofolate (nmol/L)
English Text: 5-Methyl-tetrahydrofolate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.7 to 338	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF1LC - 5-Methyl-tetrahydrofolate cmt

Variable Name: LBDSF1LC
SAS Label: 5-Methyl-tetrahydrofolate cmt
English Text: 5-Methyl-tetrahydrofolic comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7348	7348	
1	Below lower detection limit	0	7348	
.	Missing	1379	8727	

LBXSF2SI - Folic acid (nmol/L)

Variable Name: LBXSF2SI
SAS Label: Folic acid (nmol/L)
English Text: Folic acid (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.056 to 317	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF2LC - Folic acid cmt

Variable Name: LBDSF2LC
SAS Label: Folic acid cmt
English Text: Folic acid comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7272	7272	
1	Below lower detection limit	76	7348	
.	Missing	1379	8727	

LBXSF3SI - 5-Formyl-tetrahydrofolate (nmol/L)

Variable Name: LBXSF3SI
SAS Label: 5-Formyl-tetrahydrofolate (nmol/L)
English Text: 5-Formyl-tetrahydrofolate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.141 to 1.48	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF3LC - 5-Formyl-tetrahydrofolate cmt

Variable Name: LBDSF3LC
SAS Label: 5-Formyl-tetrahydrofolate cmt
English Text: 5-Formyl-tetrahydrofolate comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	17	17	
1	Below lower detection limit	7331	7348	
.	Missing	1379	8727	

LBXSF4SI - Tetrahydrofolate (nmol/L)

Variable Name: LBXSF4SI
SAS Label: Tetrahydrofolate (nmol/L)
English Text: Tetrahydrofolate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.177 to 31	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF4LC - Tetrahydrofolate cmt

Variable Name: LBDSF4LC
SAS Label: Tetrahydrofolate cmt
English Text: Tetrahydrofolate comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7033	7033	
1	Below lower detection limit	315	7348	
.	Missing	1379	8727	

LBXSF5SI - 5,10-Methenyl-tetrahydrofolate (nmol/L)

Variable Name: LBXSF5SI
SAS Label: 5,10-Methenyl-tetrahydrofolate (nmol/L)
English Text: 5,10-Methenyl-tetrahydrofolate (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.141 to 7.14	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF5LC - 5,10-Methenyl-tetrahydrofolate cmt

Variable Name: LBDSF5LC
SAS Label: 5,10-Methenyl-tetrahydrofolate cmt
English Text: 5,10-Methenyl-tetrahydrofolate comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	270	270	
1	Below lower detection limit	7078	7348	
.	Missing	1379	8727	

LBXSF6SI - Mefox oxidation product (nmol/L)

Variable Name: LBXSF6SI
SAS Label: Mefox oxidation product (nmol/L)
English Text: Mefox oxidation product (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.062 to 38.9	Range of Values	7348	7348	
.	Missing	1379	8727	

LBDSF6LC - Mefox oxidation product cmt

Variable Name: LBDSF6LC
SAS Label: Mefox oxidation product cmt
English Text: Mefox oxidation product comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7348	7348	
1	Below lower detection limit	0	7348	
.	Missing	1379	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Glycohemoglobin (GHB_L)

Data File: GHB_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

According to the National Diabetes Statistics Report, in 2021, diabetes was the eighth leading cause of death in the United States (Centers for Disease Control and Prevention, 2024). More than 38 million Americans are living with diabetes, where almost 30 million were diagnosed and nearly 9 million were undiagnosed (American Diabetes Association, 2023). Also, more than 97 million are living with prediabetes, which is a serious health condition that increases a person's risk of type-2 diabetes and other chronic diseases (American Diabetes Association, 2023). The prevalence of diabetes and overweight, one of the major risk factors for diabetes, continues to increase. Recognized and accredited programs are available for people to prevent or manage diabetes, including the National Diabetes Prevention Program (Centers for Disease Control and Prevention, 2024) and diabetes self-management education and support services (Centers for Disease Control and Prevention, 2024).

Diabetes testing provides population data to: 1) determine a national estimate of diabetes prevalence (diagnosed and undiagnosed); 2) identify the risk factors; 3) permit a national cohort to be established for follow-up studies of this condition; and 4) provide critical information to clinicians and public health officials for the development of preventive care and community-based interventions.

Eligible Sample

Examined participants aged 12 years and older were eligible.

Description of Laboratory Methodology

There was a change to the lab equipment for the glycohemoglobin test in August 2021-August 2023. During the cycle, the laboratory instrument changed from the Tosoh G8 to the Bio-Rad D-100.

Tosoh G8

In this assay, the stable (SA1c) and labile (LA1c) A1c forms can be individually resolved on the chromatogram without manual pretreatment, allowing accurate measurement of the stable form of HbA1c. The analyzer dilutes the whole blood specimen with a hemolysis solution, and then injects a small volume of the treated specimen onto the HPLC analytical column. Separation is achieved by utilizing differences in ionic interactions between the cation exchange group on the column resin surface and the hemoglobin components. The hemoglobin fractions (A1c, A1b, F, LA1c, SA1c, A0 and H-Var) are subsequently removed from the column material by step-wise elution using elution buffers each with a different salt concentration. The separated hemoglobin components pass through the photometer flow cell where the analyzer measures changes in absorbance at 415 nm. The analyzer integrates and reduces the raw data, and then calculates the relative percentages of each hemoglobin fraction. Analysis requires three minutes. If a specimen showed a deterioration peak, hemoglobin variant, or a LA1c results $\geq 5\%$ and/or LA1c results $>$ half SA1c during the regular test, it would be

retested by a second method, ultra 2 HPLC. In August 2021-August 2023, only 1.9% of the blood specimens required to be retested by the Ultra 2 HPLC method. A lab instrumentation change was made for this secondary method during the data collection period.

Bio-Rad D-100

The D-100 test utilizes principles of ion-exchange high-performance liquid chromatography (HPLC). The samples are automatically diluted on the D-100 and injected into the analytical cartridge. The D-100 delivers a programmed buffer gradient of increasing ionic strength to the cartridge, where the hemoglobins are separated based on their ionic interactions with the cartridge material. The separated hemoglobins then pass through the flow cell, where changes in the absorbance at 415 nm are measured. The D-100 software collects raw data from each analysis and calculates HbA1c values based on a bi-level calibration curve. The HbA1c area is calculated using an exponentially modified Gaussian (EMG) algorithm. A sample report and a chromatogram are generated for each sample.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, or lab site for this component in the NHANES August 2021 - August 2023 cycle.

Laboratory Method Files

[Glycohemoglobin_G8](#) (September 2024)

[Glycohemoglobin_D-100](#) (September 2024)

[Glycohemoglobin_Premier](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Whole blood specimens were processed, stored, and shipped to the University of Missouri-Columbia, MO for analysis.

Detailed instructions on specimen collection and processing are discussed in the NHANES [Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate refrigerated (2-8°C) conditions until they were shipped to University of Missouri for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et, al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

Since this data is reported in percent, the use of lower limits of detection (LLODs) isn't applicable.

No Correction Needed for Glycohemoglobin Results for NHANES August 2021–August 2023

A method validation (bridging) study was performed to compare results from a laboratory instrument change that occurred during the August 2021–August 2023 survey cycle. The Tosoh G8 was upgraded to the Bio-Rad D-100 during the cycle. Randomly selected whole blood samples (n=165 NHANES, n=36 non-NHANES, total n=201) were measured using both instruments and the results were used to conduct the analysis. No substantial differences were found between results from the two instruments. On average, GHB values measured from the Tosoh G8 were only 2.3% higher than values from the Bio-Rad D-100, which is below the total allowable error of 3% in Westgard's Desirable Biological Variation Database (Westgard, 2014). Data from the bridging study also indicated the correlation coefficient (r) between the measurements was 0.999. Therefore, the NHANES August 2021–August 2023 GHB data did not have to be adjusted.

References

- Centers for Disease Control and Prevention (2024). Diabetes Self-Management Education and Support. About Diabetes Self-Management Education and Support | Diabetes | CDC. Accessed 09 September 2024.
- Centers for Disease Control and Prevention (2024). National Diabetes Prevention Program. About the National Diabetes Prevention Program | National Diabetes Prevention Program | CDC. Accessed 09 September 2024.
- Centers for Disease Control and Prevention. National Diabetes Statistics Report (2024).(2023). National Diabetes Statistics Report | Diabetes | CDC. Accessed 09 September 2024.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.
- Westgard QC (2014). Desirable Biological Variation Database specifications. Accessed from Desirable Biological Variation Database specifications – Westgard on August 5, 2024.

Codebook and Frequencies

SEQN - Sequence number

Variable Name:	SEQN
SAS Label:	Sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	6750	6750	
0	No blood sample provided	449	7199	
.	Missing	0	7199	

LBXGH - Glycohemoglobin (%)**Variable Name:** LBXGH**SAS Label:** Glycohemoglobin (%)**English Text:** Glycohemoglobin (%)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.2 to 17.1	Range of Values	6715	6715	
0	No Lab Result	0	6715	
.	Missing	484	7199	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Plasma Fasting Glucose (GLU_L)

Data File: GLU_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

According to the National Diabetes Statistics Report, in 2021, diabetes was the eighth leading cause of death in the United States (Centers for Disease Control and Prevention 2024). More than 38 million Americans are living with diabetes, where almost 30 million were diagnosed and nearly 9 million were undiagnosed (American Diabetes Association, 2023). Also, more than 97 million are living with prediabetes, which is a serious health condition that increases a person's risk of type-2 diabetes and other chronic diseases (American Diabetes Association, 2023). The prevalence of diabetes and overweight, one of the major risk factors for diabetes, continues to increase. Recognized and accredited programs are available for people to prevent or manage diabetes, including the National Diabetes Prevention Program (Centers for Disease Control and Prevention, 2024) and diabetes self-management education and support services (Centers for Disease Control and Prevention, 2024).

Diabetes testing provides population data to: 1) determine a national estimate of diabetes prevalence (diagnosed and undiagnosed); 2) identify the risk factors; 3) permit a national cohort to be established for follow-up studies of this condition; and 4) provide critical information to clinicians and public health officials for the development of preventive care and community-based interventions.

Eligible Sample

All examined participants 12 years and older were eligible.

Description of Laboratory Methodology

Glucose

In this enzymatic method glucose is converted to glucose-6-phosphate (G-6-P) by hexokinase in the presence of ATP, a phosphate donor. Glucose-6-phosphate dehydrogenase then converts the G-6-P to gluconate-6-P in the presence of NADP+. As the NADP+ is reduced to NADPH during this reaction, the resulting increase in absorbance at 340 nm (secondary wavelength = 700 nm) is measured. This is an endpoint reaction that is specific for glucose.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[Plasma Fasting Glucose](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Plasma specimens are processed, stored, and shipped to the University of Missouri-Columbia, Columbia, MO for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedure Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C) conditions until they are shipped to University of Missouri-Columbia for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable was created using the following formulas:

LBXGLU and LBDGLUSI:

The plasma fasting glucose value in mg/dL (LBXGLU) was converted to mmol/L (LBDGLUSI) by multiplying by 0.05551 (rounded to 3 decimals).

Analytic Notes

Refer to the [August 2021–August 2023 Laboratory Data Overview](#) for general information on NHANES laboratory data.

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. For example, in August 2021–August 2023, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Analysts should evaluate the extent of

missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Subsample Weights

Plasma glucose was measured in a fasting subsample of participants 12 years and older. Special sample weights are required to analyze these data properly. Specific sample weights for this subsample are included in this data file and should be used when analyzing these data.

Participants included in the fasting subsample but did not provide a blood specimen (n=324) have an assigned sample weight value of "0" in their records. In addition, participants who provided blood specimen but did not meet the 8 to less than 24 hour fasting criteria (n=311) have the sample weight value assigned as "0" (WTSAF2YR=0) as well. Plasma fasting glucose measurements for these 311 participants are included in the dataset. However, to include these data in the analysis, a reweighting would be required.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBDGLULC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXGLU) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDGLULC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD, in mg/dL) for plasma fasting glucose:

Variable Name	Analyte Description	LLOD
LBXGLU	Plasma fasting glucose	2.0

Comparability with previous cycles

There were equipment changes during the 2015-2016, 2007-2008, and 2005-2006 survey cycles. For analysis involving 2015-2016 data and data collected prior to 2007-2008 cycle, please refer to the documentation accompanying those data for additional information on any needed adjustments to the data prior to analysis. If August 2021–August 2023 data are compared or combined with the 2015-2016 data, no adjustments are needed for equipment changes. As mentioned in the 2015-2016 documentation, most of the 2015-2016 samples

were measured using the Cobas C311, which is the same instrument used in 2017-2023. Some of the samples were collected using the Cobas C501. Results for these specimens have already been adjusted using the forward regression equation for comparability. Please see the 2015-2016 plasma fasting glucose documentation for more information about using backwards or forwards regression models to account for equipment changes between cycles (GLU_I).

References

- Centers for Disease Control and Prevention (2024). Diabetes Self-Management Education and Support. About Diabetes Self-Management Education and Support | Diabetes | CDC. Accessed 09 September 2024.
- Centers for Disease Control and Prevention (2024). National Diabetes Prevention Program. About the National Diabetes Prevention Program | National Diabetes Prevention Program | CDC. Accessed 09 September 2024.
- Centers for Disease Control and Prevention. National Diabetes Statistics Report (2024).(2023). National Diabetes Statistics Report | Diabetes | CDC. Accessed 09 September 2024.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 12 YEARS - 150 YEARS

WTSAF2YR - Fasting Subsample 2 Year MEC Weight

Variable Name: WTSAF2YR
SAS Label: Fasting Subsample 2 Year MEC Weight
English Text: Fasting Subsample 2 Year MEC Weight
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11459.183689 to 561922.42956	Range of Values	3361	3361	
0	No Lab Result or Not Fasting for 8 to <24 hours	635	3996	
.	Missing	0	3996	

LBXGLU - Fasting Glucose (mg/dL)**Variable Name:** LBXGLU**SAS Label:** Fasting Glucose (mg/dL)**English Text:** Fasting Glucose (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
59 to 561	Range of Values	3672	3672	
.	Missing	324	3996	

LBDGLUSI - Fasting Glucose (mmol/L)

Variable Name: LBDGLUSI
SAS Label: Fasting Glucose (mmol/L)
English Text: Fasting Glucose (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.28 to 31.1	Range of Values	3672	3672	
.	Missing	324	3996	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Cholesterol - High-Density Lipoprotein (HDL_L)

Data File: HDL_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Heart disease is the leading cause of death in the United States (Murphy, et. al., 2018). Blood lipid levels are fundamental measures included in NHANES that can be used for cardiovascular risk assessment. The goals of the NHANES blood lipids measurements include: 1) monitoring the prevalence and trends in major cardiovascular conditions and overall risk factors in the U.S.; 2) evaluating prevention and treatment programs targeting cardiovascular disease in the U.S.; and 3) monitoring the status of hyperlipidemia.

In 2018, new Blood Cholesterol Guidelines were released by the American College of Cardiology and American Heart Association Task Force on Clinical Practice Guidelines, which aim to reduce the risk of atherosclerotic cardiovascular disease through cholesterol management (Grundy, et. al., 2018). The blood lipids measurements in NHANES include total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), and triglycerides. The present file provides data on HDL-C. Data on total cholesterol are provided in Cholesterol - Total TCHOL_L, and LDL-C and triglyceride data are provided in TRIGLY_L.

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

High-Density Lipoprotein (HDL)

This automated HDLC4 assay uses detergents, cholesterol esterase (CHER), cholesterol oxidase (CHOD), and peroxidase to form a colored pigment that is measured optically. The HDLC4 assay meets the 1998 National Institutes of Health (NIH) / National Cholesterol Education Program (NCEP) goals for precision and accuracy.

In this method, non-HDL lipoproteins such as LDL, VLDL, and chylomicrons are combined with polyanions and a detergent, forming a water-soluble complex. In this complex, the enzymatic reaction of CHER and CHOD towards non-HDL lipoproteins is blocked. Finally, only HDL-particles can react with CHER and CHOD. The concentration of HDL-cholesterol is determined enzymatically by CHER and CHOD. Cholesterol esters are broken down quantitatively into free cholesterol and fatty acids by CHER.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Serum HDL-C levels were calculated from directly measured values of total cholesterol, triglycerides, and LDL-C. Please see below the Data Processing and Editing section for more

details. For laboratory methods used for total cholesterol, LDL-C, and triglycerides, please refer to the accompanying documentation: TCHOL_L and TRIGLY_L

There were no changes to the lab method or lab site for this component in the NHANES August 2021 - 2023 cycle. However, the lab equipment used changed from the Cobas 6000 Analyzer to the Cobas 8000 during the cycle.

Laboratory Method Files

[HDL- Cholesterol Laboratory Procedure Manual](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the University of Minnesota, Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to University of Minnesota for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One derived variable was created in this data file. The variable was created using the following formula:

LBDHDDSI

HDL-cholesterol in mg/dL (LBXHDD) was converted to mmol/L (LBDHDDSI) by multiplying by 0.02586.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analyst should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limit was constant for the analyte in the data set.

The lower limit of detection (LLOD, in mg/dL) for HDL-cholesterol:

Variable Name	Analyte Description	LLOD
LBXHDD	HDL Cholesterol	3

No Correction Need for HDL-Cholesterol Results for NHANES August 2021 - August 2023

A method validation (bridging) study was performed to compare results from an instrument change in the August 2021-August 2023 cycle. The Cobas 6000 Chemistry Analyzer was

upgraded to the Cobas 8000 Chemistry Analyzer during the cycle. Randomly selected serum samples (n=178) from previous NHANES were measured using both instruments and the results were used to conduct the analysis. On average, HDL-Cholesterol values measured from the Cobas 6000 were 0.91% higher than values from the Cobas 8000 (p = 0.0001). Data from the bridging study indicated the correlation coefficient (r) between the measurements was 0.997. Regression analyses were performed using Analyse-it, v4.30.4. The weighted Deming regression equation did not show a significant slope or intercept (95% confidence interval [CI] for slope included 1 and for intercept included 0 (mg/L):

Cobas 8000 = 0.05591 + 0.9894 * (Cobas 6000); 95% CI of slope (0.9780 to 1.001) and intercept (-0.5311 to 0.6429).

Therefore, the NHANES August 2021-August 2023 HDL-Cholesterol values did not have to be adjusted.

References

- Grundy SM, Stone NJ, Bailey AL, Beam C, Birtcher KK, Blumenthal RS, Braun LT, de Ferranti S, Faiella-Tommasino J, Forman DE, Goldberg R, Heidenreich PA, Hlatky MA, Jones DW, Lloyd-Jones D, Lopez-Pajares N, Ndumele CE, Orringer CE, Peralta CA, Saseen JJ, Smith SC Jr, Sperling L, Virani SS, Yeboah J., 2018. AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2019;139:e1082–e1143.
- Murphy SL, Xu JQ, Kochanek KD, Arias, E. Mortality in the United States, 2017. NCHS Data Brief, no 328. Hyattsville, MD: National Center for Health Statistics. 2018. <https://www.cdc.gov/nchs/data/databriefs/db328-h.pdf> Accessed May, 2019.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent Sequence Number

Variable Name:	SEQN
SAS Label:	Respondent Sequence Number
English Text:	Respondent Sequence Number
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBDHDD - Direct HDL-Cholesterol (mg/dL)

Variable Name: LBDHDD

SAS Label: Direct HDL-Cholesterol (mg/dL)

English Text: Direct HDL-Cholesterol (mg/dL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
22 to 159	Range of Values	6890	6890	
.	Missing	1178	8068	

LBDHDDSI - Direct HDL-Cholesterol (mmol/L)

Variable Name: LBDHDDSI
SAS Label: Direct HDL-Cholesterol (mmol/L)
English Text: Direct HDL-Cholesterol (mmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.57 to 4.11	Range of Values	6890	6890	
.	Missing	1178	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis A (HEPA_L)

Data File: HEPA_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES provides the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. In addition, NHANES provides the means to better define the epidemiology of other hepatitis viruses. NHANES testing for markers of infection with hepatitis viruses is used to determine secular trends in infection rates across most age and racial/ethnic groups and provides a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS, 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among persons with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in *Healthy People* and the Health and Human Services (HHS) Viral Hepatitis National Strategic Plan, which in turn support U.S. and global viral hepatitis elimination goals. The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as persons experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors (HHS Healthy People, 2022 and National Academies of Science, Engineering and Medicine, 2017).

Viral hepatitis data from NHANES are available beginning with NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted during 1988-1994 for hepatitis C, hepatitis D, and hepatitis E.

Hepatitis A virus (HAV) infection is a cause of morbidity and socio-economic loss in many parts of the world. Transmission is typically via the fecal-oral route associated with contaminated water or food. In areas where sanitation is poor, infections often occur early in life. In childhood, HAV infection is generally mild or asymptomatic and results in lifelong immunity.

With improved sanitation and hygiene, infections are delayed and consequently the number of adolescents and adults susceptible to the virus increases. In adolescents and adults, HAV infection is more serious leading to hepatitis and an increased mortality rate.

Anti-HAV IgM is detectable during the acute stage of illness, while anti-HAV IgG may be present for many years after recovery or following vaccination. The presence of anti-HAV (IgG or IgM) in human serum or plasma is indicative of past or present infection with hepatitis A virus (HAV) or vaccination against HAV. The test for total anti-HAV is primarily used to determine exposure to HAV either naturally or due to vaccination. No test can differentiate between exposure naturally or due to vaccination.

The total anti-HAV test is used for the NHANES viral hepatitis component.

Eligible Sample

Examined participants aged 2 years and older were eligible.

Description of Laboratory Methodology

Hepatitis A antibody (anti-HAV) is measured using the VITROS Anti-HAV Total assay. The test is performed using the VITROS Immunodiagnostic Products Anti-HAV Total Reagent Pack and the VITROS Immunodiagnostic Products Anti-HAV Total Calibrator on the VITROS 3600 Immunodiagnostic System.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Hepatitis A Antibody Laboratory Procedure Manual](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedure Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures

and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The assay used in this study cannot differentiate between natural infection and vaccination; therefore, seropositivity for anti-HAV reflects either natural or vaccine-induced immunity.

This data is qualitative. The use of lower limits of detection (LLODs) isn't applicable.

References

- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>.
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>.
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 2 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7583	7583	
0	No blood sample provided	1028	8611	
.	Missing	0	8611	

LBXHA - Hepatitis A antibody

Variable Name: LBXHA

SAS Label: Hepatitis A antibody

English Text: Hepatitis A antibody

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	3791	3791	
2	Negative	3511	7302	
3	Indeterminate	64	7366	
.	Missing	1245	8611	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis B: Core antibody, Surface antigen, and Hepatitis D: antibody (HEPBD_L)

Data File: HEPBD_L.xpt

First Published: February 2026

Last Revised: NA

Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES will provide the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. In addition, NHANES provides the means to better define the epidemiology of other hepatitis viruses. NHANES testing for markers of infection with hepatitis viruses has been used to determine secular trends in infection rates across most age and racial/ethnic groups and has provided a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among people with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in Healthy People and the HHS Viral Hepatitis National Strategic Plan, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Science, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with the NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D and hepatitis E.

An estimated 300 million people worldwide are persistent carriers of hepatitis B virus (HBV). Infection with HBV results in a wide spectrum of acute and chronic liver diseases that may lead to cirrhosis and hepatocellular carcinoma. Co-infection with hepatitis D virus (HDV) in persons

with acute or chronic hepatitis B virus (HBV) infection can lead to fulminant [hepatitis](#). This file addresses HBV and HDV only.

Transmission of HBV occurs by percutaneous exposure to blood products and contaminated instruments, sexual contact and perinatally from HBV-infected mothers to their unborn child.

HBV infection produces an array of unique antigens and antibody responses that, in general, follow distinct serological patterns.

Hepatitis B surface antigen (HBsAg), derived from the viral envelope, is the first antigen to appear following infection and can be detected serologically as an aid in the laboratory diagnosis of acute HBV infection.

Hepatitis B core antibody (anti-HBc) is detectable shortly after the appearance of hepatitis B surface antigen (HBsAg). As the appearance of anti-HBsAg may be delayed after HBsAg clearance, anti-HBc is sometimes the only serological marker for HBV infection and potentially infectious blood. Anti-HBc is found in acute and chronic hepatitis B patients and also indicates past resolved infection.

The Delta antigen/antibody system (HDAg/Anti-HDV) is related to HBV infection but immunologically distinct from its known reactivities; it is the expression of the Delta virus (Hepatitis D Virus, HDV), a cause of severe liver disease in HBsAg carriers. HDV is a 35-37nm particle containing low molecular weight RNA and HDAg, with an outer coat of HBsAg obtained from HBV. HDV is a defective virus and its replication requires helper functions provided by HBV. HDAg has been detected in liver and in serum and induces a specific antibody response (anti-HDV antibodies) in both the IgG and IgM classes.

Tests for anti-HBc, HBsAg, and anti-HDV are conducted as part of the NHANES viral hepatitis component. In August 2025, a new variable (SSLBDHD) was added to reflect retesting of HDV using a different assay after concerns were raised about potential false positive results during original testing (see Description of Laboratory Methodology section for additional information).

Eligible Sample

Examined participants aged 6 years or older were eligible.

Description of Laboratory Methodology

Hepatitis B core antibody (anti-HBc)

The VITROS Anti-HBc assay is performed using the VITROS Anti-HBc Reagent Pack and VITROS Immunodiagnostic Products Anti-HBc Calibrator on the VITROS ECi/ECiQ or VITROS 3600 Immunodiagnostic System.

A competitive immunoassay technique is used. This involves the reaction of anti-HBc in the sample with hepatitis B core antigen (HBcAg) coated wells. Unbound sample is removed by washing. Horseradish peroxidase (HRP)-labeled antibody conjugate (mouse monoclonal anti-HBc) is then allowed to react with the remaining exposed HBcAg on the well surface. Unbound conjugate is removed by washing.

The bound HRP conjugate is measured by a luminescent reaction. A reagent containing luminogenic substrates (a luminol derivative and a peracid salt) and an electron transfer agent is added to the wells. The HRP in the bound conjugate catalyzes the oxidation of the luminol derivative, producing light. The electron transfer agent (a substituted acetanilide) increases the level of light produced and prolongs its emission. The light signals are read by the system. The amount of HRP conjugate bound is indicative of the concentration of anti HBc present.

Hepatitis B surface antigen (HBsAg)

The VITROS HBsAg test is performed using the VITROS HBsAg Reagent Pack and VITROS Immunodiagnostic Products HBsAg Calibrator on the VITROS ECi/ECiQ Immunodiagnostic Systems and the VITROS 3600 Immunodiagnostic System. An immunometric immunoassay technique is used, which involves the simultaneous reaction of HBsAg in the sample with mouse monoclonal anti-HBs antibody coated onto the wells and a horseradish peroxidase (HRP)-labeled mouse monoclonal anti-HBs antibody in the conjugate. Unbound conjugate is removed by washing.

The bound HRP conjugate is measured by a luminescent reaction. A reagent containing luminogenic substrates (a luminol derivative and a peracid salt) and an electron transfer agent is added to the wells. The HRP in the bound conjugate catalyzes the oxidation of the luminol derivative, producing light. The electron transfer agent (a substituted acetanilide) increases the level of light produced and prolongs its emission. The light signals are read by the system. The amount of HRP conjugate bound is indicative of the level of HBsAg present in the sample.

Hepatitis D antibody (anti-HDV)

The International Immunodiagnostics HDV Ab (IID HDV Ab) assay is a competitive enzyme immunoassay (EIA) for the determination of antibodies to Hepatitis D Virus in human serum with a "one-step" methodology. Anti-HDV antibodies, if present in the sample, compete with a virus-specific polyclonal IgG, labeled with Horseradish peroxidase (HRP), for a fixed amount of recombinant HDV protein coated on the microplate. The concentration of the bound enzyme on the solid phase is inversely proportional to the amount of anti-HDV antibodies in the sample and its activity is detected by adding the chromogen/substrate in the second incubation.

From 2007-2008 through 2017-2018, NHANES anti-HDV testing was performed with a DiaSorin enzyme immunoassay designated as Research Use Only in the United States because it was not approved for clinical use by the U.S. Food and Drug Administration. Concerns about high rates of false positivity from this assay were raised after NHANES results showed a higher-than-expected number of positive results for hepatitis D antibody for certain cycles of data collection (Soriano, et. al., 2019). The DiaSorin assay was removed from the market in 2019.

There were no changes to the Hepatitis B core antibody or the Hepatitis B surface antigen lab methods, lab equipment, or lab site for this component in the August 2021- August 2023 cycle.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Hepatitis B Core Antibody](#) (February 2026)

[Hepatitis B Surface Antigen](#) (February 2026)

[Hepatitis D Antibody](#) (February 2026)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (-30°C)

conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected

during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

The age range and constraints for Hepatitis B and D testing are as follows:

Hepatitis B

The hepatitis B core antibody test is performed on all examined participants aged 6 years and older while the hepatitis B surface antibody test is performed on all examined participants aged 2 years old and older. The Hepatitis B surface antigen is tested only when the Hepatitis B core antibody test is positive. Participant results are coded positive for surface antigen if the surface antigen test is positive; they are coded negative for surface antigen if the test for surface antigen is negative or if the test for hepatitis B core antibody is negative.

Hepatitis D

The Hepatitis Delta Virus (HDV) is a RNA defective virus; and an infection with HDV only occurs in the presence of acute or chronic HBV infection. In NHANES, the test for antibody to HDV is performed on all examined participants 6 years and older who test positive or indeterminate for anti-HBc and HBsAg. The denominator for anti-HDV includes all anti-HBc negative samples, the anti-HBc positive samples that were subsequently found to be HBsAg negative, and the samples that were anti-HBc positive and HBsAg positive that were subsequently found to be anti-HDV negative.

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) is not applicable.

References

- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 241728.85724	Range of Values	8068	8068	
.	Missing	0	8068	

LBXHBC - Hepatitis B core antibody

Variable Name: LBXHBC

SAS Label: Hepatitis B core antibody

English Text: Hepatitis B core antibody

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	327	327	
2	Negative	6792	7119	
3	Indeterminate	1	7120	
.	Missing	948	8068	

LBDHBG - Hepatitis B surface antigen

Variable Name: LBDHBG

SAS Label: Hepatitis B surface antigen

English Text: Hepatitis B surface antigen

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	25	25	
2	Negative	7093	7118	
3	Indeterminate	0	7118	
.	Missing	950	8068	

LBDHD - Hepatitis D antibody (anti-HDV)

Variable Name: LBDHD

SAS Label: Hepatitis D antibody (anti-HDV)

English Text: Hepatitis D antibody (anti-HDV)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	0	0	
2	Negative	7118	7118	
3	Indeterminate	0	7118	
.	Missing	950	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis B Surface Antibody (HEPB_S_L)

Data File: HEPB_S_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES provides the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. NHANES testing for markers of infection with hepatitis viruses is used to determine secular trends in infection rates across most age and racial/ethnic groups, and provides a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among persons with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in *Healthy People* and the Health and Human Services (HHS) *Viral Hepatitis National Strategic Plan*, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Sciences, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D, and hepatitis E.

An estimated 300 million people worldwide are persistent carriers of hepatitis B virus (HBV). Infection with HBV results in a wide spectrum of acute and chronic liver diseases that may lead to cirrhosis and hepatocellular carcinoma. Transmission of HBV occurs by percutaneous exposure to blood products and contaminated instruments, sexual contact and perinatally from HBV-infected mothers to their unborn child. In 2022, U.S. vaccination guidelines were updated

to recommend hepatitis B vaccination for all infants, unvaccinated children aged <19 years, adults aged 19 through 59 years, and adults aged 60 years and older with risk factors for hepatitis B.

HBV infection produces an array of unique antigens and antibody responses that, in general, follow distinct serological patterns.

Testing for hepatitis B surface antibody contributes information important to distinguish immunity from vaccination among persons who lack antibodies to hepatitis B core antigen. Assay results may be used as an aid in the determination of susceptibility to hepatitis B virus (HBV) infection for individuals prior to or following HBV vaccination, or where vaccination status is unknown.

The hepatitis B surface antibody test is used in NHANES to assess immunity due to vaccination.

Eligible Sample

Examined participants aged 2 years or older were eligible.

Description of Laboratory Methodology

Hepatitis B surface antibody (anti-HBs) is measured using the VITROS Anti-HBs Quantitative assay. The test is performed using the VITROS Anti-HBs Quantitative Reagent Pack and VITROS Immunodiagnostic Products Anti-HBs Calibrators on the VITROS 3600 Immunodiagnostic System.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Hepatitis B Surface Antibody Laboratory Procedure Manual](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedure Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both

the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) isn't applicable.

References

- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 2 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7583	7583	
0	No blood sample provided	1028	8611	
.	Missing	0	8611	

LBXHBS - Hepatitis B Surface Antibody

Variable Name: LBXHBS

SAS Label: Hepatitis B Surface Antibody

English Text: Hepatitis B Surface Antibody

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	2042	2042	
2	Negative	5324	7366	
3	Indeterminate	0	7366	
.	Missing	1245	8611	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis C: RNA (HCV-RNA), Confirmed Antibody (INNO-LIA), & Genotype (HEPC_L)

Data File: HEPC_L.xpt

First Published: February 2026

Last Revised: NA

Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES provides the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. NHANES testing for markers of infection with hepatitis viruses is used to determine secular trends in infection rates across most age and racial/ethnic groups, and provides a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Recommendations have also been developed for the prevention and control of hepatitis C virus (HCV) infection. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among people with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in *Healthy People* and the Health and Human Services (HHS) *Viral Hepatitis National Strategic Plan*, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Sciences, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D, and hepatitis E.

In 2013, CDC revised its guidelines for Hepatitis C (HCV) testing because of 1) changes in the availability of certain commercial HCV antibody tests; 2) evidence that many people who are

identified as reactive by an HCV antibody test might not subsequently be evaluated to determine if they have current HCV infection; and 3) there have been significant advances in the development of antiviral agents with improved efficacy against HCV.¹

The flow chart for the new Hepatitis C algorithm can be found in the laboratory method file or by following the MMWR link found in the references.

Eligible Sample

Examined participants aged 6 years or older were eligible.

Description of Laboratory Methodology

Hepatitis C RNA (HCV-RNA)

The COBAS AmpliPrep COBAS TaqMan Test, is a nucleic acid amplification test for the quantitation of Hepatitis C Virus RNA in human serum or plasma on the COBAS TaqMan 48 or the COBAS 5800 Analyzer.

Hepatitis C Confirmed Antibody (INNO-LIA)

In 2012 and earlier years, the hepatitis C virus testing algorithm began with an antibody screening test, and antibody screening reactive samples were then tested with a recombinant/synthetic immunoblot assay as the antibody confirmation test. Samples with positive antibody confirmation results were then tested with a nucleic acid amplification test for hepatitis C virus RNA. HCV-RNA positive samples were then tested with a line probe assay to determine the virus genotype from six genotypes, and subtype for genotype 1, the most prevalent genotype in the U.S.

Beginning in 2013, samples reactive to the antibody screening test were first tested for RNA. Then, all RNA positive samples were tested for genotype, and only RNA negative samples were tested with an antibody confirmation test. This change in the testing algorithm was made to align the NHANES HCV testing algorithm with a 2013 update to CDC's guidance on testing for HCV infection by clinicians and laboratorians. Additionally, the antibody confirmation test changed to a 3rd generation line immunoassay in 2013 because the test used in 2012 and earlier was no longer commercially available and the stored supply of these test kits had been exhausted.

For the antibody confirmation test, the manual 16-hour sample incubation test procedure is used. 10ul of specimens and controls are diluted in 1mL diluent in test troughs to ensure test strips will slide easily and remain submerged during incubation. Specimens and controls are incubated with agitation by rocker at 34 rpm overnight (16±2 hours) at room temperature (18–25°). After a 3X/5min each wash step on microtiter plate shaker, conjugate is added with a 30-minute incubation at room temperature, followed by aspiration, 3X/5min each wash step and 30-minute incubation in substrate solution also on microtiter plate shaker. Finally, strips are agitated with stop solution for 10–30 minutes at room temperature. The microtiter plate shaker is set for 160 rpm and used for all the washes and solution incubations. Strips are then dried completely in a dry incubator at 37°C for 30 minutes then mounted on the line immunoassay score data reporting sheet with inverted tape for reading within one hour using the reading card. The strips have lines for a background control, three reference lines for antibody (level ±, human IgG, weak positive; level 1+ human IgG, moderate positive; and level 3+, anti-human Ig, strong positive), and six lines for HCV antigens (C1, C2, E2, NS3, NS4, and NS5). Samples with a positive antibody confirmation test result are reported as positive. Samples with a negative antibody confirmation test result are reported as negative. Samples with an indeterminate antibody confirmation test result are released as missing. Only specimens insufficient in quantity are deemed indeterminate.

Hepatitis C genotype

The GenMark Dx eSensor HCVg Direct Test is designed to genotype a panel of eight (8) prevalent HCV types/subtypes (1a, 1b, 2a/c, 3, 4, 5 and 6) in human serum or EDTA plasma samples. Subtype information is available in the majority of cases.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to laboratory methods, lab equipment, or lab site for this component in the August 2021-August 2023 cycle.

Laboratory Method Files

[Hepatitis C Confirmed Antibody Laboratory Procedure Manual](#) (February 2026)

[Hepatitis C Genotype Laboratory Procedure Manual](#) (February 2026)

[Hepatitis C virus RNA Laboratory Procedure Manual](#) (February 2026)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) is not applicable.

References

- Centers for Disease Control and Prevention. Recommendations for prevention and control of hepatitis C virus (HCV) infection and HCV-related chronic disease. MMWR 1998;47(No. RR-19):1-39. Available from: <https://www.cdc.gov/mmwr/PDF/RR/RR4719.pdf>.
- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>
- Testing for HCV Infection: An Update of Guidance for Clinicians and Laboratorian. Morbidity and Mortality Weekly Report. Centers for Disease Control and Prevention. May 7, 2013. <<https://www.cdc.gov/mmwr/pdf/wk/mm62e0507a2.pdf>>.
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBXHCR - Hepatitis C RNA

Variable Name: LBXHCR

SAS Label: Hepatitis C RNA

English Text: Hepatitis C RNA

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	27	27	
2	Negative	113	140	
3	Negative Screening HCV Antibody	6975	7115	
.	Missing	953	8068	

LBDHCI - Hepatitis C Antibody (confirmed)

Variable Name: LBDHCI

SAS Label: Hepatitis C Antibody (confirmed)

English Text: Hepatitis C Antibody (confirmed)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	87	87	
2	Negative	17	104	
3	Negative Screening HCV Antibody	6975	7079	
4	Positive HCV RNA	27	7106	
.	Missing	962	8068	

LBXHCG - Hepatitis C Genotype

Variable Name: LBXHCG

SAS Label: Hepatitis C Genotype

English Text: Hepatitis C Genotype

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Genotype 1a	10	10	
2	Genotype 1b	3	13	
3	Gen 1 other than a/b/not determined	1	14	
4	Genotype 2	0	14	
5	Genotype 3	4	18	
6	Genotype 4	0	18	
7	Genotype 5	0	18	
8	Genotype 6	0	18	
9	Genotype undetermined	0	18	
.	Missing	8050	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis E: IgG & IgM Antibodies (HEPE_L)

Data File: HEPE_L.xpt

First Published: February 2026

Last Revised: NA

Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES provides the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. NHANES testing for markers of infection with hepatitis viruses is used to determine secular trends in infection rates across most age and racial/ethnic groups, and provides a national picture of the epidemiologic determinants of these infections. In addition, NHANES provides the means to better define the epidemiology of other hepatitis viruses.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among people with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in *Healthy People* and the Health and Human Services (HHS) *Viral Hepatitis National Strategic Plan*, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Sciences, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D, and hepatitis E.

Infection with hepatitis E virus (HEV) has been responsible for large water-borne epidemics of acute disease in developing countries and for acute sporadic disease in industrialized

developed countries. In immunocompromised individuals, infection with HEV may also cause chronic infection, which may progress to fibrosis and cirrhosis.

Eligible Sample

Examined participants aged 6 years or older were eligible.

Description of Laboratory Methodology

Hepatitis E IgG Antibody (IgG Anti-HEV)

Abia HEV IgG enzyme immunoassay, which was formerly DS-EIA-ANTI-HEV-G, is an enzyme immunoassay kit intended for the detection of IgG antibodies to hepatitis E virus in human serum or plasma. During an initial incubation, HEV antibody in the sample binds with HEV antigen coated onto wells of a polystyrene stripped plate. Unbound sample is removed by washing. During a second incubation, horseradish peroxidase-labeled antibody conjugate (monoclonal mouse antibodies against human IgG) binds to any human IgG from the sample that was captured on the well. Unbound conjugate is removed by washing and a substrate solution containing tetramethylbenzidine is added to produce color. The reaction is stopped by adding a sulphuric acid solution and the optical density (OD) of each well is read. The presence or absence of IgG antibodies to hepatitis E virus is determined by the ratio of the OD of each sample to the calculated cut-off value. A sample is considered negative if the sample OD value is less than the cut-off and is considered positive if the sample OD value is greater than or equal to the cut-off.

Hepatitis E IgM Antibody (IgM Anti-HEV)

Abia HEV IgM enzyme immunoassay, which was formerly DS-EIA-ANTI-HEV-M is an enzyme immunoassay kit intended for the detection of IgM antibodies to hepatitis E virus in human serum or plasma. During an initial incubation, HEV antibody in the sample binds with HEV antigen coated onto wells of a polystyrene stripped plate. Unbound sample is removed by washing. During a second incubation, horseradish peroxidase-labeled antibody conjugate (monoclonal mouse antibodies against human IgM) binds to any human IgM from the sample that was captured on the well. Unbound conjugate is removed by washing and a substrate solution containing tetramethylbenzidine is added to produce color. The reaction is stopped by adding a sulphuric acid solution and the OD of each well is read. The presence or absence of IgM antibodies to hepatitis E virus is determined by the ratio of the OD of each sample to the calculated cut-off value. A sample is considered negative if the sample OD value is less than the cut-off and is considered positive if the sample OD value is greater than or equal to the cut-off.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to laboratory methods, lab equipment, or lab site for this component in the August 2021-August 2023 cycle.

Laboratory Method Files

[Hepatitis E IgG Antibody](#) (February 2026)

[Hepatitis E IgM Antibody](#) (February 2026)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and

Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) is not applicable.

References

- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBDHEG - Hepatitis E IgG (anti-HEV)

Variable Name: LBDHEG

SAS Label: Hepatitis E IgG (anti-HEV)

English Text: Hepatitis E IgG (anti-HEV)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	836	836	
2	Negative	6284	7120	
3	Indeterminate	1	7121	
.	Missing	947	8068	

LBDHEM - Hepatitis E IgM (anti-HEV)

Variable Name: LBDHEM

SAS Label: Hepatitis E IgM (anti-HEV)

English Text: Hepatitis E IgM (anti-HEV)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	128	128	
2	Negative	708	836	
3	Indeterminate	0	836	
.	Missing	7232	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

High-Sensitivity C-Reactive Protein (HSCRCP_L)

Data File: HSCRCP_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

C-reactive protein (CRP) is an acute phase protein synthesized in the liver. It is involved in the activation of complement, enhancement of phagocytosis, and detoxification of substances released from damaged tissue. It is one of the most sensitive, though nonspecific, indicators of inflammation. CRP levels may rise within six hours of an inflammatory stimulus. Measurement of CRP concentrations by this highly sensitive method is performed primarily to ascertain the level of cardiovascular disease risk in individuals who have no existing inflammatory conditions. Increases in CRP concentration are non-specific and should be used in conjunction with traditional clinical laboratory evaluation of acute coronary syndromes.

Eligible Sample

Examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

This is a two-reagent, immunoturbidimetric system. The specimen is first combined with a Tris buffer, then incubated. The second reagent (latex particles coated with mouse anti-human CRP antibodies) is then added. In the presence of circulating CRP the latex particles aggregate, forming immune complexes. These complexes cause an increase in light scattering that is proportional to the CRP concentration. The light absorbance resulting from this light scatter is read against a stored CRP standard curve. The concentration of CRP is determined from this line. Turbidity is measured at a primary wavelength of 546 nm (secondary wavelength 800 nm).

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method or lab site for this component in the NHANES August 2021-August 2023 cycle. During the cycle, the laboratory equipment changed from the Cobas 6000 Analyzer to the Cobas 8000.

Laboratory Method Files

[HS-CRP Laboratory Procedure Manual](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens are processed, stored, and shipped to the University of Minnesota – Advanced Research Diagnostics Laboratory (ARDL), Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to the University of Minnesota for testing.

The NHANES quality control and quality assurance protocols (QA/QC) meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analyst should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be

used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limit was constant for the analyte in the data set. Two variables are provided for this analyte. The variable name ending in "LC" (ex., LBDHRPLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. For the analyte with analytic results below the lower limit of detection (LBDHRPLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt [2]). The variable prefixed LBX (ex., LBXHSCR_P) provides the analytic result for that analyte.

The lower limit of detection (LLOD, in mg/L) for High-Sensitivity C-Reactive Protein:

Variable Name	Analyte Description	LLOD
LBXHSCR_P	High Sensitivity C-Reactive Protein (mg/L)	0.15

High Sensitivity C-Reactive Protein (HSCR_P) regression equations to compare August 2021-August 2023 and 2017-March 2020 data:

A method validation (bridging) study was performed to compare results from results from a laboratory instrument change that occurred during the August 2021-August 2023 survey cycle. The Cobas 6000 Chemistry Analyzer was upgraded to the Cobas 8000 Chemistry Analyzer during the cycle. Randomly selected serum samples (n=150) from previously NHANES were measured using both instruments and the results were used to conduct the analysis. On average, HSCR_P values measured with the Cobas 6000 (2017-March 2020 instrument) were 14.28% higher than values measured with the Cobas 8000 (August 2021-August 2023 instrument) (p = 0.0001). Data from the bridging study indicated the correlation coefficient (r) between the measurements was 0.996. Regression analyses were performed using Analyse-it, v4.30.4. Given that the data showed proportional differences in variability, a weighted Deming regression was chosen to describe the relation between the HSCR_P results (mg/dL) from the two instruments as below:

Forward: Cobas 8000 = -0.08653 + 0.9672 * (Cobas 6000); 95% CI of slope (0.9388 to 0.9956) and intercept (-0.1157 to -0.05737).

Backward: Cobas 6000 = 0.08946 + 1.034 * (Cobas 8000); 95% CI of slope (1.003 to 1.065) and intercept (0.06146 to 0.1175).

These regression equations are provided for analytic use and should be applied according to the analytic aims and interests. The backwards equation can be used to ensure comparability

between the August 2021-August 2023 cycle and previous years that were measured using the Cobas 6000 Analyzer values (2017-March 2020). The forward equation may be used if the interest is ensuring comparability with data collected using the Cobas 8000 Analyzer (August 2021-August 2023 and later) when comparing or combining the August 2021-August 2023 values with other years of data. Please note that HSCR_P was first measured in the continuous NHANES in the 2015-2016 cycle, and it was not measured with the Cobas 6000 Analyzer then. For analysis involving 2015-2016 data, please refer to the documentation accompanying the 2017-2018 data ([HSCR_P_J](#)) for additional adjustments.

References

- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent Sequence Number

Variable Name:	SEQN
SAS Label:	Respondent Sequence Number
English Text:	Respondent Sequence Number
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBXHSCR_P - HS C-Reactive Protein (mg/L)**Variable Name:** LBXHSCR_P**SAS Label:** HS C-Reactive Protein (mg/L)**English Text:** High-Sensitivity C-Reactive Protein (hs-CRP) (mg/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.11 to 165.63	Range of Values	7282	7282	
.	Missing	1445	8727	

LBDHRPLC - HS C-Reactive Protein Comment Code

Variable Name: LBDHRPLC

SAS Label: HS C-Reactive Protein Comment Code

English Text: High-Sensitivity C-Reactive Protein (hs-CRP) Comment Code

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7096	7096	
1	Below lower detection limit	186	7282	
.	Missing	1445	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Mercury: Inorganic, Ethyl, and Methyl - Blood (IHGEM_L)

Data File: IHGEM_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Inorganic, Ethyl and Methyl Mercury

Uncertainties exist regarding levels of exposure to methyl mercury from fish consumption and potential health effects resulting from this exposure. Past estimates of exposure to methyl mercury have been obtained from results of food consumption surveys and measures of methyl mercury in fish. Measures of a biomarker of exposure are needed for improved exposure assessments.

Blood measures of total and inorganic mercury are important for evaluating exposure to mercury in interior latex paints.

Eligible Sample

All examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

Inorganic, Ethyl and Methyl Mercury

The quantification of InHg, MeHg, and EtHg in whole blood samples is performed using a triple spike isotope dilution (TSID) method employing gas chromatography (GC) to separate the species followed by introduction into an ICP-DRC-MS for detection. TSID is a specialized extension of the Isotope Dilution (ID) technique. TSID measures individual chemical species (inorganic, methyl and ethyl mercury species) in samples using ID principles. The blood sample is spiked with known amounts of each Hg species that have been enriched with isotopic variants of the target element of interest.

The first step of this method involves the addition ("spiking") of enriched isotopes ($^{199}\text{Hg}^{2+}$, $\text{CH}_3^{200}\text{Hg}^+$, and $\text{C}_2\text{H}_5^{201}\text{Hg}^+$ or $\text{C}_2\text{H}_5^{198}\text{Hg}^+$, $^{199}\text{Hg}^2$, and $\text{CH}_3^{201}\text{Hg}^+$) to the blood sample. Each Hg species spike is labeled with an enriched Hg isotope such that its isotopic pattern is unique to the species' chemical identity (i.e. the manner of isotope spiking is "species specific"). Next, the spiked sample is digested in tetramethylammonium hydroxide (TMAH) which disassociates bound mercury species from proteins, polypeptides, and other biomolecules. The digested blood sample with freed mercury species is chemically reacted ("derivatized") with a reagent that adds 3-carbon chains (n-propyl groups) to the mercury atom of each species molecule without compromising species identity. This type of chemical derivatization results in loss of ionic charge and reduced polarity making each mercury species molecule volatile so that it can escape the liquid phase and accumulate in the gaseous phase ("headspace") directly above the sample. Derivatization is performed inside a partially filled vial sealed with a rubber septa cap that can be penetrated by a needle.

Solid Phase Microextraction (SPME) is a sampling technique that uses a thin polymer fiber with a hydrophobic coating. The method described here uses a SPME fiber with a 100 µm coating of polydimethylsiloxane (PDMS). The SPME assembly consists of the fiber inserted in a stainless-steel needle. A key design feature is that the fiber can be mechanically withdrawn into the needle during vial septum penetration and then pushed out to expose the fiber to the headspace. During headspace exposure (the "extraction" step), the gaseous derivatized Hg species adsorbs onto the PDMS coating of the SPME fiber. When other factors are held constant, the adsorbed mass increases as a function of sample concentration. After a predetermined time, the SPME fiber is retracted into the injection needle, and the needle is withdrawn from the sample vial. Subsequently, the needle moves to the injector port of the programmable temperature gradient gas chromatograph, (GC) and on programmatic command, performs a programmed temperature ramp injection sequence. This action transfers the propylated inorganic, methyl and ethyl Hg species to the head of a 30 m capillary GC column which, using He as the carrier gas, ramps the column temperature to 280°C. The order of chromatographic separation of the Hg species is based on increasing molecular weight: methylpropylmercury (derivatized methyl Hg), ethylpropylmercury (derivatized ethyl Hg), followed by dipropylmercury (derivatized inorganic Hg). Hg species exiting the GC column are seen as chromatographic peaks detected using an inductively coupled argon plasma (ICP) as the ion source and a quadrupole mass spectrometer (MS) for mass specific quantification. Species identification is based on chromatographic retention time. Species-specific isotope ratios are calculated from integrated peak areas derived from m/z signals corresponding to ¹⁹⁹Hg, ²⁰⁰Hg, ²⁰¹Hg, and ²⁰²Hg isotopes or ¹⁹⁸Hg, ¹⁹⁹Hg, ²⁰¹Hg, and ²⁰²Hg isotopes. The ICP-MS is equipped with a Dynamic Reaction Cell (DRC™) for minimizing polyatomic interferences. Operating the ICP-MS in DRC mode has an added benefit of enhancing Hg signal strength through an effect known as "collisional focusing" (Baranov, et. al., 1999 and Tanner et. al., 2000).

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to lab methods, lab equipment, or lab site for this component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[Mercury: Inorganic, Ethyl, and Methyl - Blood](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Whole blood specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences’ QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Three calculated variables were created in this data file. The variable were created using the following formulas:

LBDIHG: The inorganic mercury value in µg/L (LBXIHG) was converted to nmol/L (LBDIHGSI) by multiplying LBXIHG by 4.99 (Round to 2 decimal points).

LBDBGE: The ethyl mercury value in µg/L (LBXBGE) was converted to nmol/L (LBDBGE) by multiplying LBXBGE by 4.99 (Round to 2 decimal points).

LBDBGM: The methyl mercury value in µg/L (LBXBGM) was converted to nmol/L (LBDBGM) by multiplying LBXBGM by 4.99 (Round to 2 decimal points).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional

phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBXIHGLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXIHG) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBXIHGLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD, in µg/L) for Inorganic, Ethyl and Methyl Mercury:

Variable Name	Analyte Description	LLOD
LBXIHG	Mercury, inorganic	0.21
LBXBGE	Mercury, ethyl	0.064
LBXBGM	Mercury, methyl	0.26

References

- Baranov VI, Tanner SD. A dynamic reaction cell for inductively coupled plasma mass spectrometry (ICP-DRC-MS). Part 1. The rf-field energy contribution in thermodynamics of ion-molecule reactions. J. Anal. At. Spectrom. 1999;14:1133-1142.
- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. Statist. Med. (2008) 27(20):4094-40106.
- Tanner S, Baranov VI, Vollkopf U. A dynamic reaction cell for inductively coupled plasma mass spectroscopy (ICP-DRC-MS). Part III. Optimization and analytical performance. J. Anal. At. Spectrom. 2000;15:1261-1269.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem. 1981 Mar; 27(3):493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBXIHG - Mercury, inorganic (ug/L)**Variable Name:** LBXIHG**SAS Label:** Mercury, inorganic (ug/L)**English Text:** Inorganic mercury, blood (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.15 to 17.14	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDIHGSI - Mercury, inorganic (nmol/L)

Variable Name: LBDIHGSI
SAS Label: Mercury, inorganic (nmol/L)
English Text: Mercury, inorganic (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.75 to 85.53	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDIHGLC - Mercury, inorganic comment code

Variable Name: LBDIHGLC
SAS Label: Mercury, inorganic comment code
English Text: Mercury, inorganic comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	942	942	
1	Below lower detection limit	6625	7567	
.	Missing	1160	8727	

LBXBGE - Mercury, ethyl (ug/L)**Variable Name:** LBXBGE**SAS Label:** Mercury, ethyl (ug/L)**English Text:** Mercury, ethyl (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.05 to 0.25	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDBGESI - Mercury, ethyl (nmol/L)

Variable Name: LBDBGESI
SAS Label: Mercury, ethyl (nmol/L)
English Text: Mercury, ethyl (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.25 to 1.25	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDBGELC - Mercury, ethyl comment code

Variable Name: LBDBGELC
SAS Label: Mercury, ethyl comment code
English Text: Mercury, ethyl comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	10	10	
1	Below lower detection limit	7557	7567	
.	Missing	1160	8727	

LBXBGM - Mercury, methyl (ug/L)**Variable Name:** LBXBGM**SAS Label:** Mercury, methyl (ug/L)**English Text:** Mercury, methyl (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.18 to 36.1	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDBGMSI - Mercury, methyl (nmol/L)

Variable Name: LBDBGMSI
SAS Label: Mercury, methyl (nmol/L)
English Text: Mercury, methyl (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.9 to 180.14	Range of Values	7567	7567	
.	Missing	1160	8727	

LBDBGMLC - Mercury, methyl comment code

Variable Name: LBDBGMLC
SAS Label: Mercury, methyl comment code
English Text: Mercury, methyl comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	4324	4324	
1	Below lower detection limit	3243	7567	
.	Missing	1160	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Insulin (INS_L)

Data File: INS_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Insulin is the primary hormone responsible for controlling glucose metabolism, and its secretion is determined by plasma glucose concentration. The insulin molecule is synthesized in the pancreas as pro-insulin and is later cleaved to form C-peptide and insulin. The principal function of insulin is to control the uptake and utilization of glucose in the peripheral tissues. Insulin concentrations are severely reduced in insulin-dependent diabetes mellitus (IDDM) and some other conditions, while insulin concentrations are raised in non-insulin-dependent diabetes mellitus (NIDDM), obesity, and some endocrine disorders.

Eligible Sample

All examined participants 12 years and older, in the NHANES August 2021 – August 2023 sample, were eligible.

Description of Laboratory Methodology

Insulin

The AIA-PACK IRI is a two-site immunoenzymometric assay, which is performed on a Tosoh AIA System analyzer. This assay is designed for the quantitative measurement of Insulin (IRI) in human serum. Insulin present in the test sample is bound with monoclonal antibody immobilized on a magnetic solid phase and enzyme-labeled monoclonal antibody in the AIA-PACK. The magnetic beads are washed to remove unbound enzyme-labeled monoclonal antibody and are then incubated with a fluorogenic substrate, 4-methylumbelliferyl phosphate (4MUP). The amount of enzyme-labeled monoclonal antibody that binds to the beads is directly proportional to the insulin concentration in the test sample. A standard curve is constructed, and unknown sample concentrations are calculated using this curve.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Insulin](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the University of Missouri-Columbia, Columbia, MO for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedure Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C)

conditions until they are shipped to University of Missouri-Columbia for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPMs](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable LBDINSI was created using the following formula:

LBDINSI: The insulin value in $\mu\text{U/mL}$ (LBXIN) was converted to pmol/L (LBDINSI) by multiplying by 6.0 (rounded to 2 decimals).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Subsample Weights

Insulin was measured in a fasting subsample of participants 12 years and older. Specific sample weights (WTSF2YR) for this subsample are included in this data file and should be used when analyzing these data.

Participants included in the fasting subsample but did not provide a blood specimen (n=324) have an assigned sample weight value of "0" in their records. In addition, participants who provided blood specimen but did not meet the 8 to less than 24 hour fasting criteria (n=311) have the sample weight value assigned as "0" (WTSAF2YR =0) as well. Twenty-one out of these 311 participants did not provide enough volume to have their insulin tested. Insulin measurements for the other 290 participants are included in the dataset. However, to include these data in the analysis, a reweighting is required. There are another 168 participants in the fasting subsample that did not provide enough blood; therefore, they did not have insulin measured. However, these participants have data in at least one other fasting subsample tests (e.g., plasma fasting glucose). Therefore, they have a sample weight larger than "0," regardless of missing their insulin test results.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021 - August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBDINLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXIN) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDINLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD, in $\mu\text{U/mL}$) for insulin is:

Variable Name	Analyte Description	LLOD
LBXIN	Insulin	0.5

References

- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTSAF2YR - Fasting Subsample 2 Year MEC Weight

Variable Name: WTSAF2YR
SAS Label: Fasting Subsample 2 Year MEC Weight
English Text: Fasting Subsample 2 Year MEC Weight
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11459.183689 to 561922.42956	Range of Values	3361	3361	
0	No Lab Result or Not Fasting for 8 to <24 hours	635	3996	
.	Missing	0	3996	

LBXIN - Insulin (uU/mL)

Variable Name: LBXIN
SAS Label: Insulin (uU/mL)
English Text: Insulin (uU/mL)
Target: Both males and females 12 YEARS - 150 YEARS
Hard Edits: 0 to 9999.99

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.35 to 699.9	Range of Values	3510	3510	
.	Missing	486	3996	

LBDINSI - Insulin (pmol/L)

Variable Name: LBDINSI**SAS Label:** Insulin (pmol/L)**English Text:** Insulin (pmol/L)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.1 to 4199.4	Range of Values	3510	3510	
.	Missing	486	3996	

LBDINLC - Insulin Comment Code

Variable Name: LBDINLC

SAS Label: Insulin Comment Code

English Text: Insulin Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	Detectable result	3509	3509	
1	Below detectable limit	1	3510	
.	Missing	486	3996	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Lead, Cadmium, Total Mercury, Selenium, & Manganese - Blood (PBCD_L)

Data File: PBCD_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Lead

Lead is a known environmental toxin that has been shown to affect deleteriously the nervous, hematopoietic, endocrine, renal, and reproductive systems. In young children, lead exposure is a particular hazard because children more readily absorb lead than adults, and children's developing nervous systems also make them more susceptible to the effects of lead. The primary sources of exposure for children are lead-laden paint chips and dust as a result of deteriorating lead-based paint. The risk for lead exposure is disproportionately higher for children who are poor, non-Hispanic black, living in large metropolitan areas, or living in older housing. Among adults, the most common high exposure sources are occupational. Blood lead levels measured in previous NHANES cycles have been the cornerstone of lead exposure surveillance in the U.S. The data have been used to document the burden and dramatic decline of elevated blood lead levels, to promote the reduction of lead use, and to help to redefine national lead poisoning prevention guidelines, standards, and abatement activities.

Cadmium

Blood cadmium reflects both recent and cumulative exposures. Cadmium is absorbed via inhalation and ingestion. Occupational exposure is the most common cause of elevated cadmium levels. Inhalation of cigarette smoke is a predominant source of exposure in smokers whose blood cadmium levels have been observed to be about twice as high compared to nonsmokers. For nonsmokers who are not exposed to cadmium in the workplace, ingestion through food is the largest source of exposure. With chronic exposure, cadmium accumulates in the liver and kidneys where it is bound to metallothionein, an inducible metal binding protein. The kidney is a critical target and shows the earliest sign of cadmium toxicity. Cadmium can produce lung, pituitary gland and kidney tumors in animals and has been associated with lung cancer in humans in occupational epidemiologic studies. Both International Agency for Research on Cancer (IARC) and National Toxicology Program (NTP) consider cadmium a human carcinogen.

Manganese

The greatest demand for manganese is for the production of iron and steel. In addition, it is a key component of low-cost stainless steel and certain aluminum alloys. At low concentrations, it is used to decolorize glass, while at higher concentrations; it is used to make violet-colored glass. Manganese dioxide, besides being a useful pigment, is a catalyst and a component of certain dry cell batteries. Potassium permanganate is a potent oxidizer and disinfectant. Manganese (in the form of manganese ions) is an essential trace nutrient in all known forms of life. On the other hand, excess manganese is toxic.

Total Mercury

Uncertainties exist regarding levels of exposure to methyl mercury from fish consumption and potential health effects resulting from this exposure. Past estimates of exposure to methyl mercury have been obtained from results of food consumption surveys and measures of methyl mercury in fish. Measures of a biomarker of exposure are needed for improved exposure assessments. Blood mercury levels will be assessed in two subpopulations particularly vulnerable to the health effects from mercury exposure: children 1-5 years old and women of childbearing age. Blood measures of total and inorganic mercury will be important for evaluation of exposure from exposure to mercury in interior latex paints.

Selenium

Selenium salts are toxic in large amounts, but trace amounts are necessary for cellular function in many organisms, including all animals. Selenium is a component of the antioxidant enzymes glutathione peroxidase and thioredoxin reductase (which indirectly reduce certain oxidized molecules in animals and some plants). It is also found in three deiodinase enzymes, which convert one thyroid hormone to another. Selenium requirements in plants differ by species, with some plants requiring relatively large amounts, and others apparently requiring none.

Eligible Sample

Examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

This method directly measures lead (Pb), cadmium (Cd), total mercury (Hg), manganese (Mn), and selenium (Se) content of whole blood specimens using mass spectrometry after a simple dilution sample preparation step.

During the sample dilution step, a small volume of whole blood is extracted from a larger whole blood patient specimen after the entire specimen is mixed (vortexed) to create a uniform distribution of cellular components. This mixing step is important because some metals (e.g., Pb) are known to be associated mostly with the red blood cells in the specimen and a uniform distribution of this cellular material must be produced before a small volume extracted from the larger specimen will accurately reflect the average metal concentration of all fractions of the larger specimen. Coagulation is the process in which blood forms solid clots from its cellular components. If steps are not taken to prevent this process from occurring, i.e., addition of anti-coagulant reagents such as EDTA in the blood collection tube prior to blood collection, blood will immediately begin to form clots once leaving the body and entering the tube. These clots prevent the uniform distribution of cellular material in the blood specimen even after rigorous mixing, making a representative sub-sample of the larger specimen unattainable. It is important that prior to or during sample preparation the analyst identify any sample having clots or micro-clots (small clots). Clotted samples are not analyzed by this method due to the inhomogeneity concerns (i.e., all results for the sample are processed as "not reportable").

Dilution of the blood in the sample preparation step prior to analysis is a simple dilution of 1 part sample + 1 part water + 48 parts diluent. The effects of the chemicals in the diluent are to release metals bound to red blood cells making them available for ionization, reduce ionization suppression by the biological matrix, prevent clogging of the sample introduction system pathways by undissolved biological solids, and allow introduction of internal standards to be utilized in the analysis step. Tetramethylammonium hydroxide (TMAH, 0.4% v/v) and Triton X-100TM (0.05%) in the sample diluent solubilizes blood components. Triton X-100TM also helps prevent biological deposits on internal surfaces of the instrument's sample introduction system and reduce collection of air bubbles in sample transport tubing. Ammonium pyrrolidine dithiocarbamate (APDC) in the sample diluent (0.01%) aids in solubilizing metals released from the biological matrix. Ethyl alcohol in the sample diluent (1%) aids solubility of blood components and aids in aerosol generation by reduction of the surface tension of the solution. The internal standards, rhodium, iridium, and tellurium, are at a constant concentration in all blanks, calibrators, QC, and samples. Monitoring the instrument

signal ratio of a metal to its internal standard allows correction for instrument noise and drift, and sample-to-sample matrix differences.

Liquid samples are introduced into the mass spectrometer through the inductively coupled plasma (ICP) ionization source. The liquid diluted blood sample is forced through a nebulizer, which converts the bulk liquid into small droplets in an argon aerosol. The smaller droplets from the aerosol are selectively passed through the spray chamber by a flowing argon stream into the ICP. By coupling radio-frequency power into flowing argon, plasma is created in which the predominant species are positive argon ions and electrons and has a temperature of 6000-8000 K. The small aerosol droplets pass through a region of the plasma and the thermal energy vaporizes the liquid droplets, atomizes the molecules of the sample and then ionizes the atoms. The ions, along with the argon, enter the mass spectrometer through an interface that separates the ICP (at atmospheric pressure, ~760 torr) from the mass spectrometer (operating at a pressure of 10⁻⁵ torr). The ions first pass through a focusing region, then the dynamic reaction cell (DRC), the quadrupole mass filter, and finally are selectively counted in rapid sequence at the detector allowing individual isotopes of an element to be determined.

Generally, the DRC operates in one of two modes. In 'vented' (or 'standard') mode the cell is not pressurized and ions pass through the cell to the quadrupole mass filter unaffected. In 'DRC' mode, the cell is pressurized with a gas for the purpose of causing collisions and/or reactions between the fill gas and the incoming ions. In general, collisions or reactions with the incoming ions selectively occur to either eliminate an interfering ion, change the ion of interest to a new mass, which is free from interference, or collisions between ions in the beam and the DRC gas can focus the ion beam to the middle of the cell and increase the ion signal. In this method, the instrument is operated in DRC mode when analyzing for manganese, mercury, and selenium. For selenium, the DRC is pressurized with methane gas (CH₄, 99.999%) which reduces the signal from 40Ar²⁺ while allowing the 80Se⁺ ions to pass relatively unaffected through the DRC on toward the analytical quadrupole and detector. Manganese and mercury are both measured when the DRC is pressurized with oxygen gas (O₂, 99.999%). They are analyzed at the same flow rate of oxygen to the DRC cell to avoid lengthening analysis time due to pause delays that would be necessary if different gas flows were used for the two analytes. The oxygen reduces the ion signal from several interfering ions (37Cl¹⁸O⁺, 40Ar¹⁵N⁺, 38Ar¹⁶O¹H⁺, 54Fe¹H⁺) while allowing the Mn⁺ ion stream to pass relatively unaffected through the DRC on toward the analytical quadrupole and detector. In the case of mercury, collisional focusing of the mercury ions occurs, increasing the observed mercury signal at the detector by approximately a factor of two (2x).

Once ions pass through the DRC cell and electrically selected for passage through the analytical quadrupole, electrical signals resulting from the ions striking the discrete dynode detector are processed into digital information that is used to indicate the intensity of the ions. The intensity of ions detected while aspirating an unknown sample is correlated to an elemental concentration through comparison of the analyte: internal standard signal ratio with that obtained when aspirating calibration standards. This method was originally based on the method by Lutz (Lutz et. al., 1991). The DRC portions of the method are based on work published by Tanner (Tanner, et. al., 1999; 2002).

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab equipment or lab site for this component, however there was a change to the lab methods for the August 2021–August 2023 cycle.

Laboratory Method Files

[Blood Lead, Cadmium, Total Mercury, Selenium and Manganese \(September 2024\)](#)

Laboratory Quality Assurance and Monitoring

Whole blood specimens were processed, stored, and shipped to the National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Environmental Health Laboratory Sciences QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Five additional variables were created for this data file. The variables were created using the following formulas:

LBDBCDSI: The analyte cadmium value in $\mu\text{g/L}$ (LBXBCD) was converted to nmol/L (LBDBCDSI) by multiplying LBXBCD by 8.897 (Round 3 decimal points).

LBDBPBSI: The analyte lead value in $\mu\text{g/dL}$ (LBXBPB) was converted to $\mu\text{mol/L}$ (LBDBPBSI) by multiplying LBXBPB by 0.0483 (Round 3 decimal points).

LBDBMNSI: The analyte manganese value in $\mu\text{g/L}$ (LBXBMN) was converted to nmol/L (LBDBMNSI) by multiplying LBXBMN by 18.202 (Round 2 decimal points).

LBDBSESI: The analyte selenium value in $\mu\text{g/L}$ (LBXBSE) was converted to $\mu\text{mol/L}$ (LBDBSESI) by multiplying LBXBSE by 0.0127 (Round 1 decimal points).

LBPTHGSI: The analyte mercury value in $\mu\text{g/L}$ (LBXTHG) was converted to nmol/L (LBPTHGSI) by multiplying LBXTHG by 4.99 (Round 2 decimal points).

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021–August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDBCDLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed URX (ex., LBXBCD) provides the analytic result for the analyte. For analytes with analytic results below the lower limit of detection (ex. LBDBCDLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]). The lower limit of detection (LLOD, in µg/L) for cadmium, manganese, total mercury and selenium, and (LLOD, in ug/dL) for lead:

Variable Name	Analyte Description	LLOD
LBXBCD	Cadmium, blood	0.065
LBXBPB	Lead, blood	0.049
LBXMN	Manganese, blood	0.52
LBXTHG	Mercury, total, blood	0.17
LBXBSE	Selenium, blood	9.90

References

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- Lutz, T.M., P.M.V. Nirel, and B. Schmidt, Whole-blood analysis by ICP-MS. *Applications of Plasma Source Mass Spectrometry*, ed. G. Holland and A.N. Eaton. 1991, Cambridge: Royal Soc Chemistry. 96-100.
- Tanner, S.D., Baranov, Vladimir I, Theory, Design, and Operation of a Dynamic Reaction Cell for ICP-MS. *Atomic Spectroscopy*, 1999. 20(2): p. 45-52.
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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem.* 1981 Mar; 27(3):493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBXBPB - Blood lead (ug/dL)**Variable Name:** LBXBPB**SAS Label:** Blood lead (ug/dL)**English Text:** Blood lead (ug/dL)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.085 to 48.07	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBPBSI - Blood lead (umol/L)

Variable Name: LBDBPBSI
SAS Label: Blood lead (umol/L)
English Text: Blood lead (umol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.004 to 2.322	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBPBLC - Blood lead comment code

Variable Name: LBDBPBLC
SAS Label: Blood lead comment code
English Text: Blood lead comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7586	7586	
1	Below lower detection limit	0	7586	
.	Missing	1141	8727	

LBXBCD - Blood cadmium (ug/L)**Variable Name:** LBXBCD**SAS Label:** Blood cadmium (ug/L)**English Text:** Blood cadmium (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.046 to 8.379	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBCDSI - Blood cadmium (nmol/L)

Variable Name: LBDBCDSI
SAS Label: Blood cadmium (nmol/L)
English Text: Blood cadmium (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.409 to 74.548	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBCDLC - Blood cadmium comment code

Variable Name: LBDBCDLC
SAS Label: Blood cadmium comment code
English Text: Blood cadmium comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7257	7257	
1	Below lower detection limit	329	7586	
.	Missing	1141	8727	

LBXTHG - Blood mercury, total (ug/L)**Variable Name:** LBXTHG**SAS Label:** Blood mercury, total (ug/L)**English Text:** Blood mercury, total (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.12 to 42.49	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDTHGSI - Blood mercury, total (nmol/L)

Variable Name: LBDTHGSI
SAS Label: Blood mercury, total (nmol/L)
English Text: Blood mercury, total (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.6 to 212.03	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDTHGLC - Blood mercury, total comment code

Variable Name: LBDTHGLC
SAS Label: Blood mercury, total comment code
English Text: Blood mercury, total comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	5610	5610	
1	Below lower detection limit	1976	7586	
.	Missing	1141	8727	

LBXBSE - Blood selenium (ug/L)**Variable Name:** LBXBSE**SAS Label:** Blood selenium (ug/L)**English Text:** Blood selenium(ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
86 to 867.7	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBSESI - Blood selenium (umol/L)

Variable Name: LBDBSESI
SAS Label: Blood selenium (umol/L)
English Text: Blood selenium (umol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.09 to 11.02	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBSELC - Blood selenium comment code

Variable Name: LBDBSELC
SAS Label: Blood selenium comment code
English Text: Blood selenium comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7586	7586	
1	Below lower detection limit	0	7586	
.	Missing	1141	8727	

LBXBMN - Blood manganese (ug/L)**Variable Name:** LBXBMN**SAS Label:** Blood manganese (ug/L)**English Text:** Blood manganese (ug/L)**Target:** Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.48 to 63.17	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBMNSI - Blood manganese (nmol/L)

Variable Name: LBDBMNSI
SAS Label: Blood manganese (nmol/L)
English Text: Blood manganese (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
45.14 to 1149.82	Range of Values	7586	7586	
.	Missing	1141	8727	

LBDBMNLC - Blood manganese comment code

Variable Name: LBDBMNLC
SAS Label: Blood manganese comment code
English Text: Blood manganese comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7586	7586	
1	Below lower detection limit	0	7586	
.	Missing	1141	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS_L)

Data File: PFAS_L.xpt

First Published: August 2026

Last Revised: NA

Component Description

Perfluoroalkyl and polyfluoroalkyl substances (PFAS) are used in multiple commercial applications, including surfactants, lubricants, paints, polishes, food packaging, and fire-retarding foams. Certain PFAS are used in the manufacture of polymers used in many industrial and consumer products, including soil, stain, grease, and water-resistant coatings on textiles and carpet; uses in the automotive, mechanical, aerospace, chemical, electrical, medical, and building/construction industries; personal care products; and non-stick coatings on cookware. Some PFAS are ubiquitous contaminants found in humans and animals worldwide.

Synthesis of PFAS employed electrochemical fluorination (ECF) or fluorotelomerization. ECF, used from the 1950s until the early 2000s, yielded branched and linear isomers. By contrast, fluorotelomerization produces almost exclusively linear compounds (Vyas, et. al., 2007). The structural isomer patterns of perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) in humans may be useful for understanding routes and sources of exposure (Benskin, et. al., 2010). Therefore, concentrations of linear PFOA (n-PFOA), the sum of branched isomers of PFOA (Sb-PFOA, branched PFOA isomers), linear PFOS (n-PFOS), and the sum of perfluoromethylheptane sulfonate isomers (Sm-PFOS, monomethyl branched PFOS isomers) were measured in serum.

Eligible Sample

Examined participants aged 12 years and older were eligible.

Description of Laboratory Methodology

Online solid phase extraction coupled to high performance liquid chromatography-turboionspray ionization-tandem mass spectrometry (online SPE-HPLC-TIS-MS/MS) is used for the quantitative detection of PFAS: 2-(N-methyl-perfluorooctane sulfonamido) acetate (Me-PFOA-AcOH), perfluorohexane sulfonate (PFHxS), n-perfluorooctane sulfonate (n-PFOS), the sum of perfluoromethylheptane sulfonate isomers (Sm-PFOS, monomethyl branched isomers of PFOS), n-perfluorooctanoate (n-PFOA), the sum of branched perfluorooctanoate isomers (Sb-PFOA, branched PFOA isomers), perfluorononanoate (PFNA), perfluorodecanoate (PFDeA), perfluoroundecanoate (PFUA), and perfluorododecanoate (PFDoA) 21. Briefly, after dilution with formic acid, one aliquot of 50 µL of serum is injected into a commercial column switching system allowing for concentration of the analytes on solid-phase extraction column. Separation of the analytes from each other and from other serum components is achieved with high-performance liquid chromatography. Detection and quantification are done using negative-ion TurboIonSpray ionization, a variant of electrospray ionization, tandem mass spectrometry. This method allows for rapid detection of these PFAS in human serum with limits of detection in the low parts per billion (ppb or ng/mL) range.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Perfluoroalkyl and Polyfluoroalkyl Substances](#) (August 2026)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et al. 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. For example, in 2017-2018, approximately 80% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults age 18 and

older provided a blood specimen. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for details on the use of sample weights and other analytic issues.

Subsample weights

Serum PFAS were measured in a one-half subsample of participants 12 years and over. Special sample weights are required to analyze these data properly. Specific sample weights for this subsample are included in this data file and should be used when analyzing these data.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021- August 2023 Demographic File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "L" (ex., LBDPFHSL) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. For analytes with analytic results below the lower limit of detection (ex., LBDPFHSL=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by square root of 2 ($LLOD/\sqrt{2}$). The other variable prefixed LBX (ex., LBXPFS) provides the analytic result for that analyte.

The lower limit of detection (LLOD, in ng/mL) for each PFAS:

Variable Name	Analyte Description	LLOD
LBXPFDE	Perfluorodecanoic acid (PFDeA) (ng/mL)	0.10
LBXPFHS	Perfluorohexane sulfonic acid (PFHxS) (ng/mL)	0.10
LBXMPAH	2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH) (ng/mL)	0.10
LBXPFNA	Perfluorononanoic acid (PFNA) (ng/mL)	0.10
LBXPFUA	Perfluoroundecanoic acid (PFUA) (ng/mL)	0.10
LBXNFOA	n-perfluorooctanoic acid (n-PFOA) (ng/mL)	0.10
LBXBFOA	Branch perfluorooctanoic acid isomers (Sb-PFOA) (ng/mL)	0.10
LBXNFOS	n-perfluorooctane sulfonic acid (n-PFOS) (ng/mL)	0.10
LBXMFOS	Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) (ng/mL)	0.10

References

- Benskin JP, De Silva AO, Martin JW. 2010. Isomer Profiling of Perfluorinated Substances as a Tool for Source Tracking: A Review of Early Findings and Future Applications. Rev Environ Contam Toxicol:111-160.
- Caudill SP, Schleicher RL, Pirkle JL. Multi-rule quality control for the age-related eye disease study. Statist Med 2008;27:4094-4106.
- Vyas SM, Kania-Korwel I, Lehmler HJ. 2007. Differences in the isomer composition of perfluorooctanesulfonyl (PFOS) derivatives. J Environ Sci Health A Tox Hazard Subst Environ Eng 42:249-255.
- Westgard JO, Barry PL, Hunt MR, Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem. 1981 Mar;27(3):493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTSPF2YR - PFAS subsample weight

Variable Name: WTSPF2YR
SAS Label: PFAS subsample weight
English Text: PFAS subsample weight
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12575.1443 to 634502.90873	Range of Values	2551	2551	
0	No blood sample provided	1067	3618	
.	Missing	0	3618	

LBXPFDE - Perfluorodecanoic acid (ng/mL)**Variable Name:** LBXPFDE**SAS Label:** Perfluorodecanoic acid (ng/mL)**English Text:** Perfluorodecanoic acid (PFDeA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 16.2	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFDEL - Perfluorodecanoic acid Comment Code

Variable Name: LBDPFDEL
SAS Label: Perfluorodecanoic acid Comment Code
English Text: Perfluorodecanoic acid (PFDeA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	1130	1130	
1	Below lower detection limit	1421	2551	
.	Missing	1067	3618	

LBXPFHS - Perfluorohexane sulfonic acid (ng/mL)**Variable Name:** LBXPFHS**SAS Label:** Perfluorohexane sulfonic acid (ng/mL)**English Text:** Perfluorohexane sulfonic acid (PFHxS) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 26.4	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFHSL - Perfluorohexane sulfonic acid Comt Code

Variable Name: LBDPFHSL

SAS Label: Perfluorohexane sulfonic acid Comt Code

English Text: Perfluorohexane sulfonic acid (PFHxS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2500	2500	
1	Below lower detection limit	51	2551	
.	Missing	1067	3618	

LBXMPAH - 2-(N-methyl-PFOSA)acetic acid (ng/mL)**Variable Name:** LBXMPAH**SAS Label:** 2-(N-methyl-PFOSA)acetic acid (ng/mL)**English Text:** 2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH)
(ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 3.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDMPAHL - 2-(N-methyl-PFOSA) acetic acid Comt Code**Variable Name:** LBDMPAHL**SAS Label:** 2-(N-methyl-PFOSA) acetic acid Comt Code**English Text:** 2-(N-methylperfluorooctanesulfonamido)acetic acid (Me-PFOSA-AcOH)
Comment Code**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	571	571	
1	Below lower detection limit	1980	2551	
.	Missing	1067	3618	

LBXPFNA - Perfluorononanoic acid (ng/mL)**Variable Name:** LBXPFNA**SAS Label:** Perfluorononanoic acid (ng/mL)**English Text:** Perfluorononanoic acid (PFNA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 3.2	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFNAL - Perfluorononanoic acid Comment Code

Variable Name: LBDPFNAL
SAS Label: Perfluorononanoic acid Comment Code
English Text: Perfluorononanoic acid (PFNA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2414	2414	
1	Below lower detection limit	137	2551	
.	Missing	1067	3618	

LBXPFUA - Perfluoroundecanoic acid (ng/mL)**Variable Name:** LBXPFUA**SAS Label:** Perfluoroundecanoic acid (ng/mL)**English Text:** Perfluoroundecanoic acid (PFUA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 4.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDPFUAL - Perfluoroundecanoic acid Comment Code

Variable Name: LBDPFUAL
SAS Label: Perfluoroundecanoic acid Comment Code
English Text: Perfluoroundecanoic acid (PFUA) Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	656	656	
1	Below lower detection limit	1895	2551	
.	Missing	1067	3618	

LBXNFOA - n-perfluorooctanoic acid (ng/mL)**Variable Name:** LBXNFOA**SAS Label:** n-perfluorooctanoic acid (ng/mL)**English Text:** n-perfluorooctanoic acid (n-PFOA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 19.5	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDNFOAL - n-perfluorooctanoic acid Comment Code

Variable Name: LBDNFOAL

SAS Label: n-perfluorooctanoic acid Comment Code

English Text: n-perfluorooctanoic acid (n-PFOA) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2532	2532	
1	Below lower detection limit	19	2551	
.	Missing	1067	3618	

LBXBFOA - Br. perfluorooctanoic acid iso (ng/mL)**Variable Name:** LBXBFOA**SAS Label:** Br. perfluorooctanoic acid iso (ng/mL)**English Text:** Branch perfluorooctanoic acid isomers (Sb-PFOA) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 4.8	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDBFOAL - Br. perfluorooctanoic acid iso Comt Code

Variable Name: LBDBFOAL

SAS Label: Br. perfluorooctanoic acid iso Comt Code

English Text: Branch perfluorooctanoic acid isomers Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	260	260	
1	Below lower detection limit	2291	2551	
.	Missing	1067	3618	

LBXNFOS - n-perfluorooctane sulfonic acid (ng/mL)**Variable Name:** LBXNFOS**SAS Label:** n-perfluorooctane sulfonic acid (ng/mL)**English Text:** n-perfluorooctane sulfonic acid (n-PFOS) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 144	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDNFOSL - n-perfluorooctane sulfonic Comt Code

Variable Name: LBDNFOSL

SAS Label: n-perfluorooctane sulfonic Comt Code

English Text: n-perfluorooctane sulfonic acid (n-PFOS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2536	2536	
1	Below lower detection limit	15	2551	
.	Missing	1067	3618	

LBXMFOS - Sm-PFOS (ng/mL)

Variable Name: LBXMFOS
SAS Label: Sm-PFOS (ng/mL)
English Text: Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) (ng/mL)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 23.1	Range of Values	2551	2551	
.	Missing	1067	3618	

LBDMFOSL - Sm-PFOS Comment Code

Variable Name: LBDMFOSL

SAS Label: Sm-PFOS Comment Code

English Text: Perfluoromethylheptane sulfonic acid isomers (Sm-PFOS) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2488	2488	
1	Below lower detection limit	63	2551	
.	Missing	1067	3618	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Cholesterol - Total (TCHOL_L)

Data File: TCHOL_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Heart disease is the leading cause of death in the United States (Murphy, et. al., 2018). Blood lipid levels are fundamental measures included in NHANES that can be used for cardiovascular risk assessment. The goals the NHANES blood lipids measurements include: 1) monitoring the prevalence and trends in major cardiovascular conditions and overall risk factors in the U.S.; 2) evaluating prevention and treatment programs targeting cardiovascular disease in the U.S.; and 3) monitoring the status of hyperlipidemia.

In 2018, new Blood Cholesterol Guidelines were released, by the American College of Cardiology and American Heart Association Task Force on Clinical Practice Guidelines, which aim to reduce the risk of atherosclerotic cardiovascular disease through cholesterol management (Grundy, et. al., 2018). The blood lipids measurements in NHANES include total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), and triglycerides. The present file provides data on total cholesterol. Data on LDL-C and triglycerides are provided in the Cholesterol - Low - Density Lipoprotein and Triglycerides (TRIGLY_L) files, and HDL-C data are provided in Cholesterol - High - Density Lipoprotein (HDL_L).

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

Total Cholesterol

The laboratory method used to measure total cholesterol is an enzymatic assay. In this enzymatic assay, esterified cholesterol is converted to cholesterol by cholesterol esterase. The resulting cholesterol is then acted upon by cholesterol oxidase to produce cholest-4-en-3-one and hydrogen peroxide. The hydrogen peroxide then reacts with 4-aminophenazone in the presence of peroxidase to produce a colored product that is measured at 505 nm (secondary wavelength = 700 nm). The final step is known as the Trinder reaction. This method is a single reagent, endpoint reaction that is specific for cholesterol.

Cholesterol, a steroid molecule with a hydroxyl group in the C3 position, is synthesized in many types of tissue, but mainly in the liver and intestinal wall. About 75 percent of cholesterol is newly synthesized, with the remainder originating from dietary intake. Cholesterol measurement is performed to screen for atherosclerotic risk and in the diagnosis and treatment of disorders involving elevated cholesterol as well as lipid and lipoprotein metabolic disorders.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Serum total cholesterol levels were directly measured. Please see below the Data Processing and Editing section for more details. For laboratory methods used for LDL-C and triglycerides, and HDL-C, please refer to the accompanying documentation for the respective dataset.

There were no changes to the lab method or lab site for this component in the NHANES August 2021-2023 cycle. However, the lab equipment used changed from the Cobas 6000 Analyzer to the Cobas 8000 during the cycle.

Laboratory Method Files

[Total Cholesterol \(Frozen\) Laboratory Procedure Manual](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the University of Minnesota, Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manuals \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to University of Minnesota for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split specimens collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One derived variable was created in this data file. The variable was created using the following formula:

LBDTCSI

The total cholesterol in mg/dL (LBXTC) was converted to mmol/L (LBDTCSI) by multiplying by 0.02586.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

The laboratory data file can be linked to other NHANES data file using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for this analyte in the data set.

The lower limit of detection (LLOD, in mg/dL) for total cholesterol:

Variable Name	Analyte Description	LLOD
LBXTC	Serum Total cholesterol (Frozen)	4

No Correction Need for Total Cholesterol Results for NHANES August 2021 - August 2023

A method validation (bridging) study was performed to compare results from an instrument change in the August 2021-August 2023 cycle. The Cobas 6000 Chemistry Analyzer was upgraded to the Cobas 8000 Chemistry Analyzer during the cycle. Randomly selected serum samples (n=178) from previous NHANES were measured using both instruments and the results were used to conduct the analysis. On average, serum total cholesterol values measured from the Cobas 6000 were 0.56% higher than values from the Cobas 8000 ($p = 0.0001$). Data from the bridging study indicated the correlation coefficient (r) between the measurements was 0.998. Regression analyses were performed using Analyse-it, v4.30.4. The weighted Deming regression equation did not show a significant slope or intercept (95% confidence interval [CI] for slope included 1 and for intercept included 0 [mg/L]):

$\text{Cobas 8000} = -0.985 + 1 * (\text{Cobas 6000})$; 95% CI of slope (0.9909 to 1.009) and intercept (-2.536 to 0.5662).

Therefore, the NHANES August 2021-August 2023 total cholesterol values did not have to be adjusted.

References

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<https://www.cdc.gov/nchs/data/databriefs/db328-h.pdf>
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent Sequence Number

Variable Name:	SEQN
SAS Label:	Respondent Sequence Number
English Text:	Respondent Sequence Number
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBXTC - Total Cholesterol (mg/dL)

Variable Name: LBXTC

SAS Label: Total Cholesterol (mg/dL)

English Text: Total Cholesterol (mg/dL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
62 to 438	Range of Values	6890	6890	
.	Missing	1178	8068	

LBDTCSI - Total Cholesterol (mmol/L)

Variable Name: LBDTCSI

SAS Label: Total Cholesterol (mmol/L)

English Text: Total Cholesterol (mmol/L)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.6 to 11.33	Range of Values	6890	6890	
.	Missing	1178	8068	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Transferrin Receptor (TFR_L)

Data File: TFR_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Soluble transferrin receptor (sTfR) is a measure of iron deficiency and is particularly useful in persons with inflammation, infection, or chronic disease, where ferritin levels do not correlate with true iron levels. Low storage of iron can lead to iron deficiency anemia. High levels of iron storage, also called iron overload, occurs when excess iron is accumulated in the body, primarily the liver.

The objectives of this component are: 1) to provide data for monitoring secular trends in measures of nutritional status in the U.S. population; 2) to evaluate the effects of people's habits and behaviors, such as physical activity and the use of alcohol, tobacco, and dietary supplements on nutritional status; and 3) to evaluate the effect of changes in nutrition and public health policies, including welfare reform legislation, food fortification policy, and child nutrition programs on the nutritional status of the U.S. population. Data will be used for research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion.

Eligible Sample

Examined participants aged 1 to 5 years and females aged 12-49 years were eligible.

Description of Laboratory Methodology

The method principle for measurement of sTfR is a particle enhanced immunoturbidimetric assay that uses Roche kits on the Cobas® c501 clinical analyzer. Latex particles coated with anti-sTfR antibodies react with the antigen in the sample to form an antigen/antibody complex. Following agglutination, the precipitate is determined photometrically.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[Tina-quant Soluble Transferrin Receptor](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta,

GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et, al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable LBDTFRSI was created using the following:

LBDTFRSI: The transferrin receptor value in mg/L (LBXTFR) was converted to nmol/L (LBDTFRSI) by multiplying LBXTFR by 11.8.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES Aug. 2021–Aug. 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDTFRLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. For analytes with analytic results below the lower limit of detection (ex., LBDTFRLC =1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 ($LLOD/\sqrt{2}$). The other variable prefixed LBX (ex., LBXTFR, provides the analytic result for that analyte.

The lower limit of detection (LLOD in mg/L) for LBXTFR:

Variable Name	SAS Label	LLOD
LBXTFR	Transferrin receptor	0.5 mg/L

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 5 YEARS
Target:	Females only 12 YEARS - 49 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 MEC Weight
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6213.3002375 to 253478.77765	Range of Values	2051	2051	
0	No blood sample provided	513	2564	
.	Missing	0	2564	

LBXTFR - Transferrin receptor (mg/L)

Variable Name: LBXTFR
SAS Label: Transferrin receptor (mg/L)
English Text: Transferrin receptor (mg/L)
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.41 to 29.4	Range of Values	1949	1949	
.	Missing	615	2564	

LBDTFRSI - Transferrin receptor (nmol/L)

Variable Name: LBDTFRSI
SAS Label: Transferrin receptor (nmol/L)
English Text: Transferrin receptor (nmol/L)
Target: Both males and females 1 YEARS - 5 YEARS
Target: Females only 12 YEARS - 49 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
16.6 to 347	Range of Values	1949	1949	
.	Missing	615	2564	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Cholesterol - Low-Density Lipoproteins (LDL) & Triglycerides (TRIGLY_L)

Data File: TRIGLY_L.xpt

First Published: September 2025

Last Revised: NA

Component Description

Heart disease is the leading cause of death in the United States (Murphy, et. al., 2024). Blood lipid levels are fundamental measures included in NHANES that can be used for cardiovascular risk assessment. The goals of NHANES blood lipid measurements include: 1) monitoring the prevalence and trends in major cardiovascular conditions and overall risk factors in the U.S.; 2) evaluating prevention and treatment programs targeting cardiovascular disease in the U.S.; and 3) monitoring the status of hyperlipidemia.

In 2018, new Blood Cholesterol Guidelines were released, by the American College of Cardiology and American Heart Association Task Force on Clinical Practice Guidelines, which aim to reduce the risk of atherosclerotic cardiovascular disease through cholesterol management (Grundy, et. al., 2018). The blood lipids measurements in NHANES include total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoproteins cholesterol (LDL-C), and triglycerides. The present file provides data on LDL-C and triglycerides. Data on total cholesterol are provided in the Cholesterol - Total (TCHOL_L) file, and HDL-C data are provided in Cholesterol - High - Density Lipoprotein (HDL_L).

Eligible Sample

Participants aged 12 years and older who were examined in the morning sessions were eligible.

Description of Laboratory Methodology

This method is based on the work by Wahlefeld using a lipoprotein lipase from microorganisms for the rapid and complete hydrolysis of triglycerides to glycerol followed by oxidation to dihydroxyacetone phosphate and hydrogen peroxide. The hydrogen peroxide produced then reacts with 4-aminophenazone and 4-chlorophenol under the catalytic action of peroxidase to form a red dyestuff (Trinder endpoint reaction). The color intensity of the red dyestuff formed is directly proportional to the triglyceride concentration and can be measured photometrically.

There was a change in the lab method in the NHANES August 2021-August 2023 cycle for triglycerides measurement. The glycerol blanked assay used in previous cycles were phased out by the manufacturer. In addition, the lab equipment used for the measurement was updated from the Cobas 6000 Analyzer to Cobas 8000. Please refer to the Analytic Notes section for additional information.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Serum LDL-C levels were calculated from directly measured values of total cholesterol, triglycerides, and HDL-C. Please see below the Data Processing and Editing section for more

details. For laboratory methods used for total cholesterol and HDL-C, please refer to the accompanying documentation: [TCHOL_L](#) and [HDL_L](#).

Laboratory Method Files

[Triglycerides](#) (September 2025)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to University of Minnesota, Advanced Research Diagnostics Laboratory (ARDL), Minneapolis, MN for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to University of Minnesota for testing.

The NHANES quality assurance and quality control (QA/QC) meet the 1988 Clinical Laboratory Improvement Amendment mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Seven derived variables were created in this data file: one for triglycerides, and six for LDL-C. The formulas used to derive these variables are provided below.

Triglycerides - International System of Units (LBDTRSI)

The triglyceride values in mg/dL (**LBXTLG**) were converted to mmol/L (**LBDTRSI**) by multiplying by 0.01129.

Calculated LDL-C LEVELS

Serum LDL-C levels were derived for participants examined in the morning session only. LDL-C is calculated from directly measured values of total cholesterol (LBXTC), triglycerides (LBXTLG), and HDL-C (LBDHDD). Three equations are available for calculating LDL-C: the Friedewald equation; the Martin-Hopkins equation; and the NIH Equation 2.

The Friedewald equation has been the standard equation used for the calculation of LDL-C for clinical use for decades. The Friedewald equation estimates LDL-C concentration using directly measured total cholesterol, triglycerides, and HDL-C results, and a fixed factor of 5 to estimate the triglyceride to very low-density lipoprotein cholesterol (VLDL-C) ratio and is based on a direct LDL-C by beta quantification reference method from 448 samples (Friedewald, 1972). NHANES has used the Friedewald equation to calculate LDL-C since the release of the Third National Health and Nutrition Examination Survey (NHANES III) (1988-1994 data) in 1996. Studies over the years have shown that the equation underestimates LDL-C at lower levels, specifically those with LDL-C levels less than 70 mg/dL and/or high triglyceride levels, potentially leading to an undertreatment with lipid-lowering medications (Sathiyakumar, et. al., 2020). The Friedewald equation is not valid for triglyceride results greater than 400 mg/dL.

The Martin-Hopkins equation for calculating LDL-C was recommended in the 2018 Blood Cholesterol Guidelines, specifically those with LDL-C levels less than 70 mg/dL. It is based on a Vertical Auto Profile direct ultracentrifugation method from 1.35 million patients (Martin, et. al., 2013). The Martin-Hopkins equation applies an adjustable factor on an individual participant's median triglyceride to VLDL-C ratio based on non-HDL cholesterol (NONHDL, calculated by subtracting HDL-C from total cholesterol [NON-HDL=LBXTC-LBDHDD]) and triglyceride concentrations (Martin, et. al., 2013). To establish the adjustable factors 10, 20, 30, 60, 90, 120, 150, 180, 200, 300, 360, 400, 720, 800, 1000, and 2000 cell two-dimensional strata tables were developed (Martin, et. al., 2013). NHANES employs the 360-cell strata table to generate adjustable factors for the triglyceride to VLDL-C ratio for calculating LDL-C. Like the Friedewald equation, the Martin-Hopkins equation is not valid for triglyceride results greater than 400 mg/dL.

In January 2020, the NIH Equation 2 was released (Sampson, et. al., 2020). Like the Friedewald equation, it is based on a direct LDL-C by beta quantification reference method, though from a much larger sample consisting of 8,656 patients (18,715 lipid samples) (Sampson, et. al., 2020). Of the three equations for calculating LDL-C, the NIH Equation 2 has the best accuracy at triglyceride levels greater than or equal to 400 mg/dL. NIH Equation 2 has not been as scrutinized as the Friedewald and Martin-Hopkins equations and requires additional external validation, as it was released in 2020. However, it has been implemented by one major U.S. laboratory with an expansive nationwide network. The NIH Equation 2 is not valid for triglyceride results greater than 800 mg/dL.

All three equations are being released in this dataset. All calculated LDL-C data were converted into International System of Units (SI units) for each equation.

Friedewald Equation (LBDLDL):

$$\text{LBDLDL [LDL-C (Friedewald)]} = \text{LBXTC} - \text{LBDHDD} - \text{LBXTLG}^a/5$$

^aWhere triglyceride levels are less than 400 mg/dL.

Friedewald Equation - SI units (LBDLDLSI):

The LDL-C (Friedewald) in mg/dL (**LBDLDL**) was converted to mmol/L (**LBDLDLSI**) by multiplying by 0.02586.

$$\text{LBDLDLSI} = .02586 * \text{LBDLDL}$$

Martin-Hopkins Equation (LBDLDLM):

$$\text{LBDLDLM [LDL-C (Martin-Hopkins)]} = \text{LBXTC-LBDHDD-LBXTLG}^a / \text{Adjustable Factor}^b$$

^aWhere triglyceride levels are less than 400 mg/dL

^bWhere the Adjustable factor, an estimate of the triglyceride to VLDL cholesterol ratio, depends upon the values of non-HDL-C and triglycerides.

Martin-Hopkins Equation - SI units (LBDLDMSI).

The LDL-C (Martin-Hopkins) in mg/dL (**LBDLDLM**) was converted to mmol/L (**LBDLDMSI**) by multiplying by 0.02586.

$$\text{LBDLDMSI} = .02586 * \text{LBDLDLM}$$

NIH Equation 2 (LBDLDLN):

$$\text{LBDLDLN [LDL-C (NIH)]} =$$

$$\text{LBXTC}/0.948 - \text{LBDHDD}/0.971 - (\text{LBXTLG}/8.56 + \text{LBXTLG}^c * \text{NONHDL}/2140 - \text{LBXTLG}^2/16100) - 9.44$$

^cWhere triglyceride levels are less than 800 mg/dL.

$$\text{LDL} - \text{C} = \frac{\text{TC}}{.948} - \frac{\text{HDL} - \text{C}}{.971} - \left(\frac{\text{TG}}{8.56} + \frac{[\text{TG} * \text{non} - \text{HDL}]}{2140} - \frac{\text{TG}^2}{16100} \right) - 9.44$$

NIH Equation 2 - SI units (LBDLDNSI).

The LDL-C (NIH Equation 2) in mg/dL (**LBDLDLN**) was converted to mmol/L (**LBDLDNSI**) by multiplying by 0.02586.

$$\text{LBDLDNSI} = .02586 * \text{LBDLDLN}$$

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed.

Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Subsample Weights

Triglycerides were measured in a fasting subsample of participants 12 years and older. Special sample weights are required to analyze these data properly. Specific sample weights for this subsample are included in this data file and should be used when analyzing these data.

Participants included in the fasting subsample but did not provide a blood specimen (n=324) have an assigned sample weight value of "0" in their records. In addition, participants who provided a blood specimen but did not meet the 8 to less than 24 hours fasting criteria (n=311) have the sample weight value assigned as "0" (WTSAPRP =0) as well. However, these participants have data in at least one other fasting subsample tests (e.g., plasma fasting glucose). Therefore, they have a sample weight larger than "0," regardless of missing their triglyceride test results.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The August 2021-August 2023 [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

The laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Triglycerides (LBXTLG)

Serum levels were measured for participants that were examined in the morning session only. The distribution of serum triglycerides should only be estimated on participants aged 12 years and over who fasted at least 8 hours or more, but less than 24 hours. PHAFSTHR (total length of "food fast", hours) and PHAFSTMN (Total length of "food fast", minutes) can be found in the [Fasting Questionnaire File](#).

The TRIGLY_L data file contains laboratory test results for triglycerides (LBXTLG) using the reference analytic method. However, the NHANES Standard Biochemistry Profile (BIOPRO_L) also includes measurements of triglycerides (LBXSTR). The appropriate variable to use for the most accurate data analysis is **LBXTLG** from the TRIGLY_L data file.

Detection Limits

The detection limits were constant for this analyte in the data set. The variable prefixed LBX (ex., LBXTLG) provides the analytic result for that analyte.

The lower limit of detection (LLOD, in mg/dL) for triglycerides:

Variable Name	SAS Label	LLOD
LBXTLG	Serum Triglycerides	9

Triglyceride regression equations to compare 2017-March 2020 and August 2021-August 2023 data:

A method validation (bridging) studies were performed to compare results from an instrument change in the August 2021 – August 2023 cycle: the Cobas 6000 Chemistry Analyzer was upgraded to the Cobas 8000 Chemistry Analyzer. There was also a change in method as the manufacturer phased out glycerol blanked assay for frozen triglycerides. Randomly selected serum sample (n=89) from previous NHANES were measured using both instruments and methods and the results were used to conduct the analysis. Data from the bridging study indicated the correlation coefficient (r) between the measurements was 0.999. Regression analyses were performed using Analyse-it, v4.30.4. Given that the data showed proportional

differences in variability, a weighted Deming regression was chosen to describe the relation between the serum triglyceride results (mg/L) from the two methods as below:

Forward: $LBXTLG_{(New\ Instrument + New\ Triglyceride\ Method)} = 12.45 + 1.022 LBXTR_{(Old\ Instrument)}$; 95% CI of intercept (11.27 to 13.64) and slope (1.010 to 1.034)

Backward: $LBXTR_{(Old\ Instrument)} = -12.19 + 0.9785 LBXTLG_{(New\ Instrument + New\ Triglyceride\ Method)}$; 95% CI of intercept (-13.46 to -10.91) and slope (0.9669 to 0.9900)

These regression equations are provided for analytic use and should be applied according to the analytic aims and interest. The backward equation can be used to ensure comparability between the August 2021-August 2023 cycle and previous years that were measured using the glycerol blanked method (such as in 2017-March 2020). The forward equation may be used if the interest is ensuring comparability with data collected using non-glycerol blanked method in August 2021-August 2023 when comparing or combining the August 2021-August 2023 values with previous years of data. For more detailed information on the triglyceride data in the previous cycles, please refer to the documentations accompanying these datasets.

References

- Friedewald WT, Levy RI, Fredrickson DS. Estimation of the concentration of low-density lipoprotein cholesterol in plasma, without use of the preparative ultracentrifuge. Clin Chem. 1972;18(6):499–502.
- Grundy SM, Stone NJ, Bailey AL, Beam C, Birtcher KK, Blumenthal RS, Braun LT, Braun LT, de Ferranti S, Faiella-Tommasino J, Forman DE, Goldberg R, Heidenreich PA, Hlatky MA, Jones DW, Lloyd-Jones D, Lopez-Pajares N, Ndumele CE, Orringer CE, Peralta CA, Saseen JJ, Smith Jr SC, Sperling L, Virani SS, Yeboah J, 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: Executive Summary, Journal of the American College of Cardiology (2018), doi: <https://doi.org/10.1016/j.jacc.2018.11.002>.
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- Sampson M. Ling C, Qian Sun, Harb R, et al. A New Equation for Calculation of Low-Density Lipoprotein Cholesterol in Patients with Normolipidemia and/or Hypertriglyceridemia. JAMA Cardiol. February 26, 2020. E1-E8.
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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 12 YEARS - 150 YEARS

WTSAF2YR - Fasting Subsample 2 Year MEC Weight

Variable Name: WTSAF2YR
SAS Label: Fasting Subsample 2 Year MEC Weight
English Text: Fasting Subsample 2 Year MEC Weight
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11459.183689 to 561922.42956	Range of Values	3361	3361	
0	No Lab Result or Not Fasting for 8 to <24 hours	635	3996	
.	Missing	0	3996	

LBXTLG - Triglyceride (mg/dL)

Variable Name: LBXTLG

SAS Label: Triglyceride (mg/dL)

English Text: Triglyceride (mg/dL)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
19 to 1745	Range of Values	3517	3517	
.	Missing	479	3996	

LBDTRSI - Triglyceride (mmol/L)

Variable Name: LBDTRSI

SAS Label: Triglyceride (mmol/L)

English Text: Triglyceride (mmol/L)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.215 to 19.701	Range of Values	3517	3517	
.	Missing	479	3996	

LBDLDL - LDL-Cholesterol, Friedewald (mg/dL)

Variable Name: LBDLDL

SAS Label: LDL-Cholesterol, Friedewald (mg/dL)

English Text: LDL-Cholesterol, Friedewald equation (mg/dL). LBDLDL = (LBXTC - (LBDHDD + LBXTR/5), round to 0 decimal places) for LBXTR less than 400 mg/dL, and missing for LBXTR greater than 400 mg/dL. LBDHDD from public release file HDL_J

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3 to 314	Range of Values	3480	3480	
.	Missing	516	3996	

LBDLDLSI - LDL-Cholesterol, Friedewald (mmol/L)

Variable Name: LBDLDLSI
SAS Label: LDL-Cholesterol, Friedewald (mmol/L)
English Text: LDL-Cholesterol, Friedewald equation (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.078 to 8.12	Range of Values	3480	3480	
.	Missing	516	3996	

LBDLDLM - LDL-Cholesterol, Martin-Hopkins (mg/dL)**Variable Name:** LBDLDLM**SAS Label:** LDL-Cholesterol, Martin-Hopkins (mg/dL)**English Text:** LDL-Cholesterol, Martin-Hopkins equation (mg/dL). LBDLDLM = (LBXTC-(LBDHDD + LBXTR/Adjustable Factor), round to 0 decimal places) for LBXTR less than 400 mg/dL, and missing for LBXTR greater than 400 mg/dL. LBDHDD from public release file HDL_J**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
17 to 313	Range of Values	3480	3480	
.	Missing	516	3996	

LBDLDMSI - LDL-Cholesterol, Martin-Hopkins (mmol/L)

Variable Name: LBDLDMSI

SAS Label: LDL-Cholesterol, Martin-Hopkins (mmol/L)

English Text: LDL-Cholesterol, Martin-Hopkins equation (mmol/L)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.44 to 8.094	Range of Values	3480	3480	
.	Missing	516	3996	

LBDLDLN - LDL-Cholesterol, NIH equation 2 (mg/dL)

Variable Name: LBDLDLN

SAS Label: LDL-Cholesterol, NIH equation 2 (mg/dL)

English Text:
$$\text{LBDLDLN} = (\text{LBXTC}/0.948 - \text{LBDHDD}/0.971 - (\text{LBXTR}/8.56 + (\text{LBXTR} * (\text{LBXTC} - \text{LBDHDD}))/2140 - \text{LBXTR}^2/16100) - 9.44),$$
 round 0 decimal places) for LBXTR less than 800 mg/dL, and missing for LBXTR GE 800 mg/dL. ^2 stands for power=2, or squared. LBDHDD from public release file HDL_J

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
13 to 314	Range of Values	3513	3513	
.	Missing	483	3996	

LBDLDNSI - LDL-Cholesterol, NIH equation 2 (mmol/L)

Variable Name: LBDLDNSI
SAS Label: LDL-Cholesterol, NIH equation 2 (mmol/L)
English Text: LDL-Cholesterol, NIH equation 2 (mmol/L)
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.336 to 8.12	Range of Values	3513	3513	
.	Missing	483	3996	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Sex Steroid Hormone Panel - Serum (TST_L)

Data File: TST_L.xpt

First Published: October 2024

Last Revised: NA

Component Description

The sex steroid hormone panel – Serum (TST) consists of 17 α -hydroxyprogesterone, androstenedione, anti-Müllerian hormone, dehydroepiandrosterone sulfate (DHEAS), estradiol, estrone, estrone sulfate, follicle-stimulating hormone, luteinizing hormone, progesterone, sex hormone binding globulin, and total testosterone. This data will allow for analysis of the selected steroid hormones and related binding protein that can be used to assist in disease diagnosis, treatment, and prevention of diseases, such as Polycystic Ovary Syndrome (PCOS), androgen deficiency, certain cancers, and hormone imbalances. An estimated 5 to 7 million women in the United States (U.S.) suffer with the effects of PCOS; and PCOS can occur in girls as young as 11 years old. PCOS is the most common hormonal disorder among women of reproductive age and is the leading cause of infertility. Androgen deficiency, such as hypogonadism, is associated with a range of chronic diseases. The prevalence of symptomatic androgen deficiency in men between 30 and 79 years of age is estimated to be 5.6% (Araujo et. al., 2007). Androgen deficiency in men and excess in women and the associated chronic diseases are a public health concern. Estradiol is the key biomarker for assessing reproductive function in females, including amenorrhea, infertility, and menopausal status. Estradiol levels decline greatly with age, and this decrease is associated with increased risk for cardiovascular disease, cognitive impairment, and bone fractures in older women. Estrogen hormone therapy, or use of estradiol as a supplement, raises health concerns related to estradiol concentration in blood, such as elevated levels in postmenopausal women increasing the risk of breast cancer.

17 α -hydroxyprogesterone

17 α -hydroxyprogesterone (17-OHP) is a steroid hormone that is primarily produced in the adrenal glands, as well as in the ovaries, testes, and placenta. It is derived from progesterone via 17 α -hydroxylase and is a chemical intermediate in the biosynthesis of several other steroids, including cortisol. Measurement of 17-OHP is useful in the diagnosis of congenital adrenal hyperplasia (CAH).

Androstenedione

Androstenedione is a steroid hormone that is produced in the adrenal glands and the gonads. It is a precursor of androgen and estrogen sex hormones, such as testosterone and estrone. Androstenedione is synthesized by the conversion of dehydroepiandrosterone. Elevated androstenedione levels may cause symptoms of hyperandrogenism in females. Measurement of androstenedione is useful in the diagnosis of congenital adrenal hyperplasia, in conjunction with other androgenic precursors, such as 17 α -hydroxyprogesterone.

Anti-Müllerian hormone (AMH)

Anti-Müllerian hormone (AMH) or Müllerian-inhibiting substance, is a dimeric glycoprotein secreted by granulosa cells of the preantral and small antral ovarian follicles. AMH levels in the blood can provide information about ovarian function, menopausal status, and can aid in

assessments for developmental disorders in children. An increased level of AMH is often seen with PCOS.

Dehydroepiandrosterone sulfate (DHEAS)

DHEAS is produced in the adrenal cortex. It is a conjugated steroid converted by the sulfation of dehydroepiandrosterone (DHEA) at the 3 β position via hydroxysteroid sulfotransferase. While DHEAS itself is hormonally inactive, it can be converted to DHEA, which in turn can serve as a precursor to more active steroid hormones, such as testosterone or estradiol. Measurements of DHEAS provides information about the adrenal gland function and can help with the diagnosis of congenital adrenal hyperplasia and PCOS.

Estradiol

Estradiol is produced primarily in the ovary (follicle, corpus luteum), but small quantities are also formed in the testes and in the adrenal cortex, as well as in fat cells. During pregnancy, estradiol is mainly formed in the placenta. About 98% of estradiol is bound to transport proteins (SHBG and albumin). Estradiol secretion has two surges during the menstrual cycle. The determination of estradiol is utilized clinically in the elucidation of fertility disorders in the hypothalamus-pituitary-gonad axis, gynecomastia, estrogen-producing ovarian and testicular tumors and in hyperplasia of the adrenal cortex. Further clinical indications are the monitoring of fertility therapy and determining the time of ovulation within the framework of in vitro fertilization (IVF).

Estrone

Estrone is mainly produced in the gonads through the aromatization of androstenedione but can also be reversibly converted from estradiol via 17 β -hydroxysteroid dehydrogenase. Serum estrone is primarily bound to albumin and partially bound to sex hormone-binding globulin (SHBG), with a small portion freely circulating. Measurement of estrone levels by itself or together with estradiol can provide information about potential metabolic disorders, tumors, or developmental disorders, such as early or delayed puberty.

Estrone sulfate

Estrone sulfate is the most abundant circulating estrogen in women and is thought to serve as a long-term reservoir for estrone and estradiol. Thus, estrone sulfate serves as a precursor and intermediate to the major estrogens, estrone and estradiol. It is produced from estrone by estrogen sulfotransferases. While it is mainly biological inactive, it maintains an equilibrium of more active estrogens, such as estrone. Estrone can then be converted to the more reactive estradiol by 17 β -hydroxysteroid dehydrogenase. The measurement of estrone sulfate can help with monitoring treatments for certain cancers.

Follicle Stimulating Hormone (FSH)

Follicle-stimulating hormone (FSH) is a glycoprotein with two polypeptide units that is produced by the anterior pituitary gland. FSH, often together with luteinizing hormone and/or other hormones, can provide information about ovarian function, as well as help with assessing disease, such as PCOS or disorders of the pituitary or the hypothalamus.

Luteinizing Hormone (LH)

Luteinizing hormone (LH) is a heterodimeric glycoprotein that consists of one alpha and one beta subunit. LH is produced mainly in gonadotropic cells in the anterior pituitary gland. LH often together with follicle stimulating hormone and other steroids, can provide information about ovarian function and can help with assessing disorders of the pituitary and hypothalamus.

Progesterone

Progesterone is a steroid hormone that belongs to the progestogens class of steroid hormones. It is produced in the adrenal glands, corpus luteum, and placenta. Progesterone can be used to track ovulation, monitor the health of a pregnancy, and can help with the diagnosis congenital adrenal hyperplasia.

Sex hormone-binding globulin (SHBG)

Sex hormone-binding globulin (SHBG) is the blood transport protein for androgens and estrogens. SHBG has a high binding affinity to dihydrotestosterone (DHT), medium affinity to testosterone and estradiol, and only a low affinity to estrone, dehydroepiandrosterone (DHEA), androstenedione, and estriol. Its synthesis and secretion are regulated by estrogen (Burger, et. al., 2002; Davis, et. al., 2001). SHBG serum concentrations depend on the extent, duration, and the kind of estrogen applied, and how regulation takes place. In the serum, SHBG mainly takes over the transportation of steroids and the reduction/regulation of the effect of androgen (Rosner, et. al., 1999; Burger, et. al., 2002). Decreased SHBG serum levels are associated with conditions where elevated androgen levels are present or where the effect of androgen on its target organs is excessive. This explains the gender-related differences seen between men and women, especially during puberty.

Measurement of SHBG can be an important indicator of an excessive/chronic androgenic action where androgen levels are normal, but where clinical symptoms would seem to indicate androgen in excess. SHBG is a useful supplementary parameter in the determination of androgen where a relatively high concentration of free androgen (e.g., testosterone) is suspected (Pugeat, et. al., 1996).

Elevated SHBG levels can be seen in elderly men, and are often found in patients with hyperthyroidism, cirrhosis of the liver, and some polymorphisms in the SHBG gene (Bhasin, et. al., 2018). SHBG levels also increase when oral contraceptives, estrogen or antiepileptic drugs are taken. Pregnant women have markedly higher SHBG serum concentrations due to their increased estrogen production. Decreased SHBG concentrations are often seen with hypothyroidism, PCOS, obesity, hirsutism, elevated androgen levels, alopecia, acromegaly, and some polymorphisms in the SHBG gene.

Testosterone

Testosterone is the most important androgenic steroid that has an anabolic effect in humans. It is synthesized in the testes of the male, and in much smaller amounts, in the ovary of the female, and in the adrenal gland of both female and male. In human males, testosterone plays a key role in the development of male reproductive tissues, such as the testis and prostate, as well as promoting secondary sexual characteristics, such as increased muscle and bone mass, and the growth of body hair. The secretion of testosterone is regulated by LH and is subject to negative feedback via the pituitary and hypothalamus. Most of the circulating testosterone is bound to carrier proteins (SHBG and albumin). In women, increased production of testosterone can cause hirsutism and virilization (depending on the increase). The determination of testosterone in the female is helpful in the evaluation of congenital adrenal hyperplasia, PCOS, and when an ovarian tumor, adrenal tumor, adrenal hyperplasia, or ovarian insufficiency is suspected. Testosterone is determined in men when reduced testosterone production is suspected, e.g., in hypogonadism, estrogen therapy, chromosome aberrations (as in the Klinefelter's syndrome) and liver cirrhosis.

Eligible Sample

All examined participants aged 3 years and older were eligible for 17 α -hydroxyprogesterone, androstenedione, dehydroepiandrosterone, estradiol, estrone, estrone sulfate, progesterone, sex hormone binding globulin, and total testosterone.

All examined participants aged 6 years and older were eligible for anti-Müllerian hormone,

follicle-stimulating hormone, and luteinizing hormone.

Description of Laboratory Methodology

The following tests for the TST: 17 α -hydroxyprogesterone, androstenedione, DHEAS, estrone, estrone sulfate, estradiol, progesterone, and testosterone are performed via the isotope dilution liquid chromatography tandem mass spectrometry (ID-LC-MS/MS) method. ID-LC-MS/MS is performed on a triple quadrupole mass spectrometer using electrospray ionization in both positive and negative modes. The analytes are identified based on chromatographic retention time and on specific mass to charge ratio transitions using selected reaction monitoring (SRM). Isotopically labeled internal standards are used for each analyte.

AMH is based on the reaction of AMH with immuno-antibodies and chemi-luminescence measurements of the reaction products after two incubation periods. This assay consists of two incubation steps and a chemiluminescent measurement obtained with a photomultiplier tube.

LH is a test based on the reaction of LH with immuno-antibodies and chemi-luminescence measurements for the reaction products. This assay consists of two incubation steps and a chemiluminescent measurement obtained with a photomultiplier tube.

FSH is a test based on the reaction of FSH with immuno-antibodies and chemi-luminescence measurements for the reaction products. This assay consists of two incubation steps and a chemiluminescent measurement obtained with a photomultiplier tube.

SHBG is based on the reaction of SHBG with immuno-antibodies and chemi-luminescence measurements of the reaction products that occurs after two incubation periods and subjecting to a magnetic field. The microparticles are captured on an electrode, where a chemiluminescent reaction occurs and can be measured by a photomultiplier tube.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component during the NHANES August 2021–August 2023 cycle.

Laboratory Method Files

[Panel of Steroid Hormones](#) (October 2024)

[Anti-Müllerian Hormone](#) (October 2024)

[Sex Hormone-Binding Globulin](#) (October 2024)

[Follicle-Stimulating Hormone](#) (October 2024)

[Luteinizing Hormone](#) (October 2024)

Laboratory Quality Assurance and Monitoring

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual](#) (LPM). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in

the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Environmental Health Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Nine derived variables were created in this data file. The variables were created using the following formulas:

LBD17HSI

17 α -hydroxyprogesterone in ng/dL (LBX17H) was converted to nmol/L (LBD17HSI) by multiplying by 0.0303.

LBDANDSI

Androstenedione in ng/dL (LBXAND) was converted to nmol/L (LBDANDSI) by multiplying by 0.0349.

LBDAMHSI

Anti-Müllerian hormone in ng/mL (LBXAMH) was converted to pmol/L (LBDAMHSI) by multiplying by 7.14.

LBDDHESI

DHEAS in μ g/dL (LBXDHE) was converted to μ mol/L (LBDDHESI) by multiplying by 0.0271.

LBDESTSI

Estradiol in pg/mL (LBXEST) was converted to pmol/L (LBDESTSI) by multiplying by 3.67.

LBDESOSI

Estrone in ng/dL (LBXESO) was converted to pmol/L (LBDESOSI) by multiplying by 37.0.

LBDES1SI

Estrone sulfate in pg/mL (LBXES1) was converted to pmol/L (LBDES1SI) by multiplying by 2.73.

LBDPG4SI

Progesterone in ng/dL (LBXPG4) was converted to nmol/L by multiplying by 0.0318.

LBDTSTSI

Total testosterone in ng/dL (LBXTST) was converted to nmol/L (LBDTSTSI) by multiplying by 0.0347.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [August 2021—August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, the time of venipuncture, and the conditions precluding venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable named ended "LC" (ex., LBDTSTLC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXTST) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDTSTLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD) for steroid hormone panel:

Variable Name	SAS Label	LLOD
LBX17H	17a-hydroxyprogesterone (ng/dL)	0.41
LBXAND	Androstenedione (ng/dL)	0.82
LBXAMH	Anti-Müllerian hormone	0.03*
LBXDHE	Dehydroepiandrosterone sulfate (ug/dL)	0.22
LBXEST	Estradiol (pg/mL)	1.72
LBXESO	Estrone (ng/dL)	0.13
LBXES1	Estrone Sulfate (pg/mL)	2.04
LBXFSH	Follicle Stimulating Hormone (mIU/mL)	0.300
LBXLUH	Luteinizing Hormone (mIU/mL)	0.100
LBXPG4	Progesterone (ng/dL)	0.86
LBXSHBG	SHBG (nmol/L)	0.350
LBDTST	Testosterone, total (ng/dL)	0.57

*Values not reported below the Limit of Quantitation (LLOQ)

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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 3 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7532	7532	
0	No blood sample provided	961	8493	
.	Missing	0	8493	

LBX17H - 17 α -hydroxyprogesterone (ng/dL)**Variable Name:** LBX17H**SAS Label:** 17 α -hydroxyprogesterone (ng/dL)**English Text:** 17 α -hydroxyprogesterone (ng/dL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.29 to 1840	Range of Values	7120	7120	
.	Missing	1373	8493	

LBD17HSI - 17 α -hydroxyprogesterone (nmol/L)

Variable Name: LBD17HSI
SAS Label: 17 α -hydroxyprogesterone (nmol/L)
English Text: 17 α -hydroxyprogesterone (nmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.00879 to 55.8	Range of Values	7120	7120	
.	Missing	1373	8493	

LBD17HLC - 17 α -hydroxyprogesterone Comment Code

Variable Name: LBD17HLC
SAS Label: 17 α -hydroxyprogesterone Comment Code
English Text: 17 α -hydroxyprogesterone Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7112	7112	
1	Below lower detection limit	8	7120	
.	Missing	1373	8493	

LBXAND - Androstenedione (ng/dL)**Variable Name:** LBXAND**SAS Label:** Androstenedione (ng/dL)**English Text:** Androstenedione (ng/dL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.58 to 755	Range of Values	7190	7190	
.	Missing	1303	8493	

LBDANDSI - Androstenedione (nmol/L)

Variable Name: LBDANDSI
SAS Label: Androstenedione (nmol/L)
English Text: Androstenedione (nmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0202 to 26.3	Range of Values	7190	7190	
.	Missing	1303	8493	

LBDANDLC - Androstenedione Comment Code

Variable Name: LBDANDLC

SAS Label: Androstenedione Comment Code

English Text: Androstenedione Comment Code

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7186	7186	
1	Below lower detection limit	4	7190	
.	Missing	1303	8493	

LBXAMH - Anti-Mullerian hormone (ng/mL)**Variable Name:** LBXAMH**SAS Label:** Anti-Mullerian hormone (ng/mL)**English Text:** Anti-Mullerian hormone (ng/mL)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 45.92	Range of Values	6823	6823	
.	Missing	1670	8493	

LBDAMHSI - Anti-Mullerian hormone (pmol/L)

Variable Name: LBDAMHSI
SAS Label: Anti-Mullerian hormone (pmol/L)
English Text: Anti-Mullerian hormone (pmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.14 to 327.87	Range of Values	6823	6823	
.	Missing	1670	8493	

LBDAMHLC - Anti-Mullerian hormone Comment Code

Variable Name: LBDAMHLC
SAS Label: Anti-Mullerian hormone Comment Code
English Text: Anti-Mullerian hormone Comment Code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	4902	4902	
1	Below lower detection limit	1921	6823	
.	Missing	1670	8493	

LBXDHE - DHEAS ($\mu\text{g/dL}$)**Variable Name:** LBXDHE**SAS Label:** DHEAS ($\mu\text{g/dL}$)**English Text:** Dehydroepiandrosterone sulfate ($\mu\text{g/dL}$)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.156 to 601	Range of Values	7065	7065	
.	Missing	1428	8493	

LBDDHESI - DHEAS ($\mu\text{mol/L}$)

Variable Name: LBDDHESI
SAS Label: DHEAS ($\mu\text{mol/L}$)
English Text: Dehydroepiandrosterone sulfate ($\mu\text{mol/L}$)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.00423 to 16.3	Range of Values	7065	7065	
.	Missing	1428	8493	

LBDDHELC - DHEAS Comment Code

Variable Name: LBDDHELC
SAS Label: DHEAS Comment Code
English Text: Dehydroepiandrosterone sulfate Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7057	7057	
1	Below lower detection limit	8	7065	
.	Missing	1428	8493	

LBXEST - Estradiol (pg/mL)**Variable Name:** LBXEST**SAS Label:** Estradiol (pg/mL)**English Text:** Estradiol (pg/mL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.22 to 13200	Range of Values	7148	7148	
.	Missing	1345	8493	

LBDESTSI - Estradiol (pmol/L)

Variable Name: LBDESTSI
SAS Label: Estradiol (pmol/L)
English Text: Estradiol (pmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.48 to 48400	Range of Values	7148	7148	
.	Missing	1345	8493	

LBDESTLC - Estradiol Comment Code

Variable Name: LBDESTLC
SAS Label: Estradiol Comment Code
English Text: Estradiol Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6450	6450	
1	Below lower detection limit	698	7148	
.	Missing	1345	8493	

LBXESO - Estrone (ng/dL)

Variable Name: LBXESO**SAS Label:** Estrone (ng/dL)**English Text:** Estrone (ng/dL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0919 to 510	Range of Values	7159	7159	
.	Missing	1334	8493	

LBDESOSI - Estrone (pmol/L)

Variable Name: LBDESOSI
SAS Label: Estrone (pmol/L)
English Text: Estrone (pmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.4 to 18900	Range of Values	7159	7159	
.	Missing	1334	8493	

LBDESOLC - Estrone Comment Code

Variable Name: LBDESOLC
SAS Label: Estrone Comment Code
English Text: Estrone Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6993	6993	
1	Below lower detection limit	166	7159	
.	Missing	1334	8493	

LBXES1 - Estrone Sulfate (pg/mL)**Variable Name:** LBXES1**SAS Label:** Estrone Sulfate (pg/mL)**English Text:** Estrone Sulfate (pg/mL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.44 to 25400	Range of Values	7013	7013	
.	Missing	1480	8493	

LBDES1SI - Estrone Sulfate (pmol/L)

Variable Name: LBDES1SI
SAS Label: Estrone Sulfate (pmol/L)
English Text: Estrone Sulfate (pmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.93 to 69300	Range of Values	7013	7013	
.	Missing	1480	8493	

LBDES1LC - Estrone Sulfate Comment Code

Variable Name: LBDES1LC

SAS Label: Estrone Sulfate Comment Code

English Text: Estrone Sulfate Comment Code

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6966	6966	
1	Below lower detection limit	47	7013	
.	Missing	1480	8493	

LBXFSH - Follicle Stimulating Hormone (mIU/mL)

Variable Name: LBXFSH

SAS Label: Follicle Stimulating Hormone (mIU/mL)

English Text: Follicle Stimulating Hormone (mIU/mL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.21 to 194.7	Range of Values	7014	7014	
.	Missing	1479	8493	

LBDFSHLC - FSH Comment Code

Variable Name: LBDFSHLC

SAS Label: FSH Comment Code

English Text: Follicle Stimulating Hormone Comment Code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6919	6919	
1	Below lower detection limit	95	7014	
.	Missing	1479	8493	

LBXLUH - Luteinizing Hormone (mIU/mL)**Variable Name:** LBXLUH**SAS Label:** Luteinizing Hormone (mIU/mL)**English Text:** Luteinizing Hormone (mIU/mL)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 116.2	Range of Values	7014	7014	
.	Missing	1479	8493	

LBDLUHLC - Luteinizing Hormone Comment Code

Variable Name: LBDLUHLC
SAS Label: Luteinizing Hormone Comment Code
English Text: Luteinizing Hormone Comment Code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6624	6624	
1	Below lower detection limit	390	7014	
.	Missing	1479	8493	

LBXPG4 - Progesterone (ng/dL)**Variable Name:** LBXPG4**SAS Label:** Progesterone (ng/dL)**English Text:** Progesterone (ng/dL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.608 to 12800	Range of Values	7136	7136	
.	Missing	1357	8493	

LBDPG4SI - Progesterone (nmol/L)

Variable Name: LBDPG4SI
SAS Label: Progesterone (nmol/L)
English Text: Progesterone (nmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0193 to 407	Range of Values	7136	7136	
.	Missing	1357	8493	

LBDPG4LC - Progesterone Comment Code

Variable Name: LBDPG4LC
SAS Label: Progesterone Comment Code
English Text: Progesterone Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	6701	6701	
1	Below lower detection limit	435	7136	
.	Missing	1357	8493	

LBXSHBG - SHBG (nmol/L)

Variable Name: LBXSHBG
SAS Label: SHBG (nmol/L)
English Text: Sex Hormone Binding Globulin (SHBG, nmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.18 to 671.9	Range of Values	7191	7191	
.	Missing	1302	8493	

LBDSHGLC - SHBG Comment Code

Variable Name: LBDSHGLC
SAS Label: SHBG Comment Code
English Text: Sex Hormone Binding Globulin Comment Code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7191	7191	
1	Below lower detection limit	0	7191	
.	Missing	1302	8493	

LBXTST - Testosterone, total (ng/dL)**Variable Name:** LBXTST**SAS Label:** Testosterone, total (ng/dL)**English Text:** Testosterone, total (ng/dL)**Target:** Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.403 to 5180	Range of Values	7160	7160	
.	Missing	1333	8493	

LBDTSTSI - Testosterone, total (nmol/L)

Variable Name: LBDTSTSI
SAS Label: Testosterone, total (nmol/L)
English Text: Testosterone, total (nmol/L)
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.014 to 180	Range of Values	7160	7160	
.	Missing	1333	8493	

LBDTSTLC - Testosterone comment code

Variable Name: LBDTSTLC
SAS Label: Testosterone comment code
English Text: Testosterone comment code
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7062	7062	
1	Below lower detection limit	98	7160	
.	Missing	1333	8493	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Urine Pregnancy Test (UCPREG_L)

Data File: UCPREG_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

A urine pregnancy test was performed in the mobile examination center (MEC) on menstruating female survey participants 8 years and older. All positive test results excluded pregnant women from the DXA component at the MEC.

Eligible Sample

Examined female participants aged 12–59 years, and menstruating females aged 8–11 years were eligible. However, due to disclosure risks, only females 20–44 years of age have urine pregnancy results in this file.

Description of Laboratory Methodology

The Icon 25 hCG test kit (Beckman Coulter) is a rapid chromatographic immunoassay for the qualitative detection of human chorionic gonadotropin (hCG) in urine or serum to aid in the early detection of pregnancy. The test utilizes a combination of monoclonal and polyclonal antibodies to selectively detect elevated levels of hCG in urine or serum.

Refer to the Laboratory Methods Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES 2021–2023 cycle.

Laboratory Quality Assurance and Monitoring

Urine specimens were processed in the NHANES mobile examination centers.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#).

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures

and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

Refer to the August 2021-August 2023 Laboratory Data Overview for general information on NHANES laboratory data.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES August 2021- August 2023 Demographics File contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

Since this data is reported as qualitative data, the use of lower LLOD is not applicable.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Females only 20 YEARS - 44 YEARS

URXPREG - Urine Pregnancy Result

Variable Name: URXPREG
SAS Label: Urine Pregnancy Result
English Text: Urine Pregnancy Result
Target: Females only 20 YEARS - 44 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	41	41	
2	Negative	1078	1119	
3	Not Done	11	1130	
4	Invalid	0	1130	
.	Missing	4	1134	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Vitamin D (VID_L)

Data File: VID_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Vitamin D is functionally a hormone rather than a vitamin, and in conjunction with parathyroid hormone and calcitonin, it is one of the most important biological regulators of calcium metabolism. Vitamin D and its main metabolites may be categorized into two families of secosteroids: cholecalciferol (vitamin D3) and ergocalciferol (vitamin D2). Both vitamins D3 and D2 are enzymatically hydroxylated in the liver to 25-hydroxy forms and then further metabolized in the kidney to the bioactive 1,25-dihydroxy forms. Although 25-hydroxyvitamin D (25OHD) is not the bioactive form, it is the predominant circulating form of vitamin D, and thus, it is considered to be the most reliable index of vitamin D status (Olkowski, 2003; Saenger, 2006). Vitamin D3 is a naturally occurring form of vitamin D that is produced in the skin after 7-dehydrocholesterol is exposed to UV-B radiation. Commercially, vitamin D2 is produced by UV irradiation of plant-derived ergosterol. The two forms differ in the structures of their side chains, but they are metabolized identically. Good sources of vitamin D3 are fatty fish, while mushrooms provide a good source of vitamin D2. Both forms are used for fortification of a limited selection of foods including milk, juice, margarines, cheese and nutrition bars. Because these two parent compounds provide various contributions to vitamin D status, it is informative when both forms are measured separately (Olkowski, 2003; Saenger, 2006).

The measurement of 25OHD is becoming increasingly important in the management of patients with various disorders of calcium metabolism associated with rickets, osteomalacia, nutritional and renal osteodystrophy, hypoparathyroidism, and postmenopausal osteoporosis (Holick, 2005). These data will be used to estimate deficiencies and toxicities of specific nutrients in the population and subgroup, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to serum levels of nutrients. Data will be used in research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion.

Eligible Sample

Examined participants aged 1 year and older were eligible.

Description of Laboratory Methodology

The test principle for the CDC method utilizes high performance liquid chromatography-tandem mass spectrometry (HPLC-MS/MS) for the quantitative detection of 25-hydroxyvitamin D3 (25OHD3), 3-epi-25-hydroxyvitamin D3 (epi-25OHD3), and 25-hydroxyvitamin D2 (25OHD2) in human serum. The analytes are chromatographically separated generally on one of three pentafluorophenyl (PFP) columns. Mobile phase composition for optimized chromatography varies slightly for the three columns but is between 69% and 72% methanol in water. The composition of the solution added to the serum prior to extraction, the solution used for reconstitution, and the needle wash should match that used for the mobile phase.

Refer to the Laboratory Method Files section for a detailed description of the methods used.

There were no changes to the lab method, lab equipment, or lab site for this component during the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Vitamin D](#) (September 2024)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS periodically. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Science's QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The

specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information such as fasting status, the time of venipuncture, and the conditions precluding venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. The SAS variable name ending in LC (ex., LBXVD2LC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXVIDMS) provides the result for the analyte. For analytes with analytic results below the lower limit of detection (ex., LBXVIDLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt (2)).

The lower limits of detection (LLOD in nmol/L) for Vitamin D metabolites are:

Variable Name	Analyte Description	LLOD
LBXVD2MS	25-hydroxyvitamin D2	2.216 nmol/L
LBXVD3MS	25-hydroxyvitamin D3	0.439 nmol/L
LBXVE3MS	epi-25-hydroxyvitamin D3	0.755 nmol/L

Analysis of 25-Hydroxyvitamin D Data from NHANES III (1988-1994) and NHANES

2001-2006, with NHANES data collected since 2007

The LC-MS/MS-equivalent 25-Hydroxyvitamin D data are recommended for all analyses, but especially for analyzing secular trends for serum vitamin D involving NHANES III (1988-1994), NHANES 2001-2006 and NHANES data collected since 2007. Users are strongly encouraged to review the [Analytical Note for 25-Hydroxyvitamin D Data Analysis \(cdc.gov\)](#) prior to the analysis.

References

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- Olkowski AA, Aranda-Osorio G, and McKinnon J. Rapid HPLC method for measurement of vitamin D3 and 25(OH)D3 in blood plasma. *Int J Vitam Nutr Res*, 2003;73(1):15-18.
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- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 1 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 253478.77765	Range of Values	7626	7626	
0	No blood sample provided	1101	8727	
.	Missing	0	8727	

LBXVIDMS - 25OHD2+25OHD3 (nmol/L)

Variable Name: LBXVIDMS
SAS Label: 25OHD2+25OHD3 (nmol/L)
English Text: 25-hydroxyvitamin D2 + D3
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.97 to 424	Range of Values	7307	7307	
.	Missing	1420	8727	

LBDVIDLC - 25OHD2+25OHD3 comment code

Variable Name: LBDVIDLC
SAS Label: 25OHD2+25OHD3 comment code
English Text: 25-hydroxyvitamin D2 + D3 comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7307	7307	
1	Below lower detection limit	0	7307	
.	Missing	1420	8727	

LBXVD2MS - 25OHD2 (nmol/L)

Variable Name: LBXVD2MS
SAS Label: 25OHD2 (nmol/L)
English Text: 25-hydroxyvitamin D2 (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.57 to 239	Range of Values	7307	7307	
.	Missing	1420	8727	

LBDVD2LC - 25OHD2 comment code

Variable Name: LBDVD2LC
SAS Label: 25OHD2 comment code
English Text: 25-hydroxyvitamin D2 comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	1447	1447	
1	Below lower detection limit	5860	7307	
.	Missing	1420	8727	

LBXVD3MS - 25OHD3 (nmol/L)

Variable Name: LBXVD3MS
SAS Label: 25OHD3 (nmol/L)
English Text: 25-hydroxyvitamin D3 (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.1 to 422	Range of Values	7308	7308	
.	Missing	1419	8727	

LBDVD3LC - 25OHD3 comment code

Variable Name: LBDVD3LC
SAS Label: 25OHD3 comment code
English Text: 25-hydroxyvitamin D3 comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7308	7308	
1	Below lower detection limit	0	7308	
.	Missing	1419	8727	

LBXVE3MS - epi-25OHD3 (nmol/L)

Variable Name: LBXVE3MS
SAS Label: epi-25OHD3 (nmol/L)
English Text: epi-25-hydroxyvitamin D3 (nmol/L)
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.54 to 62.8	Range of Values	7238	7238	
.	Missing	1489	8727	

LBDVE3LC - epi-25OHD3 comment code

Variable Name: LBDVE3LC
SAS Label: epi-25OHD3 comment code
English Text: epi-25-hydroxyvitamin D3 comment code
Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7168	7168	
1	Below lower detection limit	70	7238	
.	Missing	1489	8727	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Volatile Organic Compounds and Trihalomethanes/MTBE - Blood (VOCWB_L)

Data File: VOCWB_L.xpt

First Published: September 2025

Last Revised: NA

Component Description

Volatile Organic Compounds and Trihalomethanes/MTBE (Whole Blood)

Volatile organic compounds (VOCs) are a large group of chemicals that have been used as solvents, degreasers, and cleaning agents in industry and consumer products. Many of the VOCs were found to contaminate ground water and drinking water sources. Because of human health concerns, these VOCs have been banned or restricted from most uses.

Halogenated solvents are VOCs consisting of a hydrocarbon chain or one hydrocarbon substituted with one or more chlorine or bromine atoms. Most of these chemicals are used as degreasers and solvents in various products, such as paint. In the past, 1,1,1-trichloroethane was used as a dry-cleaning agent, insect fumigant, and solvent in consumer products. Methylene chloride, tetrachloroethene, and trichloroethene are other VOCs that were widely used in the past.

Benzene, toluene, ethylbenzene, xylene, and styrene, collectively referred to as BTEXS, are components of tobacco smoke. Along with 2,5-dimethylfuran, these VOCs are usually detected in the blood of cigarette smokers at higher levels than in non-smokers. Chlorobenzene (monochlorobenzene) and the three dichlorobenzenes are halogenated aromatic hydrocarbons primarily used in industrial and chemical synthetic processes. Chlorobenzene has been used to produce DDT, phenol, and nitrobenzene. The dichlorobenzenes are also chemical intermediates in the synthesis of dyes, pesticides, and other industrial products. 1,4-Dichlorobenzene (para-dichlorobenzene) is used also as a moth repellent and as a deodorizer. Disinfection by-products (DBP), including bromodichloromethane, dibromochloromethane, bromoform, and chloroform are formed when chlorine interacts with natural organic materials found in water. Primary sources of DBPs are chlorinated drinking water and recreational water bodies, such as swimming pools.

The prevalence of disinfection by-products in drinking water supplies has raised concerns about possible adverse health effects from chronic exposure to these potentially carcinogenic compounds. Methyl-tert-butyl ether (MTBE) was used as an additive to replace lead in gasoline, but its use was banned after widespread ground water contamination was discovered.

Inhalation is the most common VOC route of exposure in the general population, including indoor sources such as paints, adhesives, cleaning solutions, and aerosolized insecticide sprays; industries producing these solvents; and contaminated waste disposal sites. Drinking water may contribute to exposure when underground drinking water supplies are contaminated. After they are absorbed in the body, VOCs are rapidly eliminated in exhaled breath, or may be rapidly metabolized and eliminated in the urine.

Eligible Sample

Examined participants aged 12 years and older from a one-half sample were eligible.

Description of Laboratory Methodology

An automated analytical method was developed using capillary gas chromatography (GC) and mass spectrometry (MS) with selected-ion monitoring (SIM) detection and isotope-dilution. This method quantifies levels of individual VOCs and Trihalomethanes (THMs) and methyl tert-butyl ether (MTBE) in whole blood to low-parts-per-trillion range. Because non-occupationally exposed individuals have blood VOC concentrations within this range, this method is applicable for determining these quantities and investigating cases of sustained or recent low-level exposure.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab method, lab equipment, or lab site for this component in the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Volatile Organic Compounds \(VOCs\) & Trihalomethanes/MTBE Laboratory Procedure Manual](#) (September 2025)

Laboratory Quality Assurance and Monitoring

Whole blood specimens were processed, stored, and shipped to Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual](#) (LPM). Vials are stored under appropriate refrigerated (2-8°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendment mandates. Detailed QA/QC instructions are discussed in the NHANES [LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any

special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Services' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed.

Analyst should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and analytic issues.

Volatile Toxicant Questionnaire

A volatile toxicant questionnaire (VTQ) was administered on the mobile examination center (MEC), by trained interviewers, using the Computer-Assisted Personal Interview (CAPI) system. The VTQ section includes data about the participant's home, activities, amount of time spent in various locations, and exposure to different chemicals over the past 48 hours. This questionnaire data can be used in conjunction with the VOC laboratory dataset and found in the Volatile Toxicant Data File in the [NHANES August 2021 – August 2023 Questionnaire Data](#) section.

Subsample Weights

Whole blood VOCs were measured in a one-half subsample of participants 12 years and older. Special sample weights are required to analyze these data properly. Specific sample weights for this subsample are included in this data file and should be used when analyzing these data.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES [August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBD2DFLC) indicates whether the result was below the limit of detection: "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBX2DF) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBD2DFLC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by square root of 2 (LLOD/sqrt [2]).

Lower Limit of Detection (LLOD, in ng/mL) for Whole Blood VOCs:

VARIABLE NAME	SAS LABEL	LLOD
LBX2DF	Blood 2,5-Dimethylfuran (ng/mL)	0.0110
LBXV06	Blood Hexane (ng/mL)	0.122
LBXV07N	Blood Heptane (ng/mL)	0.100
LBXV08N	Blood Octane (ng/mL)	0.100
LBXV1D	Blood 1,2-Dichlorobenzene (ng/mL)	0.0250
LBXV2A	Blood 1,2-Dichloroethane (ng/mL)	0.0100
LBXV3B	Blood 1,3-Dichlorobenzene (ng/mL)	0.0250
LBXV4C	Blood Tetrachloroethene (ng/mL)	0.0480
LBXVAPN	Blood a-pinene (ng/mL)	0.0200
LBXVBF	Blood Bromoform (ng/mL)	0.0080
LBXVBM	Blood Bromodichloromethane (ng/mL)	0.0060
LBXVBZ	Blood Benzene (ng/mL)	0.0240
LBXVBZN	Blood Benzonitrile (ng/mL)	0.150
LBXVC6	Blood Cyclohexane (ng/mL)	0.0200
LBXVCB	Blood Chlorobenzene (ng/mL)	0.0110
LBXVCF	Blood Chloroform (ng/mL)	0.0080
LBXVCM	Blood Dibromochloromethane (ng/mL)	0.0050
LBXVCT	Blood Carbon Tetrachloride (ng/mL)	0.0050
LBXVDB	Blood 1,4-Dichlorobenzene (ng/mL)	0.0400
LBXVDEE	Blood Diethyl Ether (ng/mL)	0.0400
LBXVEA	Blood Ethyl Acetate (ng/mL)	0.158
LBXVEB	Blood Ethylbenzene (ng/mL)	0.0240
LBXVEC	Blood Chloroethane (ng/mL)	0.0450
LBXVFN	Blood Furan (ng/mL)	0.0250
LBXVIBN	Blood Isobutyronitrile (ng/mL)	0.0400
LBXVIPB	Blood Isopropylbenzene (ng/mL)	0.0400
LBXVMC	Blood Methylene Chloride (ng/mL)	0.250
LBXVMCP	Blood Methylcyclopentane (ng/mL)	0.0200
LBXVME	Blood MTBE (ng/mL)	0.0100
LBXVMIK	Blood Methyl Isobutyl Ketone (ng/mL)	0.100
LBXVOX	Blood o-Xylene (ng/mL)	0.0240
LBXVST	Blood Styrene (ng/mL)	0.0300
LBXVTC	Blood Trichloroethene (ng/mL)	0.0120
LBXVTE	Blood 1,1,1-Trichloroethane (ng/mL)	0.0100
LBXVTHF	Blood Tetrahydrofuran (ng/mL)	0.125
LBXVTO	Blood Toluene (ng/mL)	0.0250
LBXVTP	Blood 1,2,3-Trichloropropane (ng/mL)	0.0400
LBXVXY	Blood m-/p-Xylene (ng/mL)	0.0340

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTSVOC2Y - VOC Subsample Weight

Variable Name: WTSVOC2Y

SAS Label: VOC Subsample Weight

English Text: VOC Subsample Weight

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11386.962716 to 543590.56928	Range of Values	2960	2960	
0	No Lab Result	658	3618	
.	Missing	0	3618	

LBX2DF - Blood 2,5-Dimethylfuran (ng/mL)**Variable Name:** LBX2DF**SAS Label:** Blood 2,5-Dimethylfuran (ng/mL)**English Text:** Blood 2,5-Dimethylfuran (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0078 to 0.857	Range of Values	2960	2960	
.	Missing	658	3618	

LBD2DFLC - Blood 2,5-Dimethylfuran Comment Code

Variable Name: LBD2DFLC

SAS Label: Blood 2,5-Dimethylfuran Comment Code

English Text: Blood 2,5-Dimethylfuran Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	411	411	
1	Below lower detection limit	2549	2960	
.	Missing	658	3618	

LBXV06 - Blood Hexane (ng/mL)**Variable Name:** LBXV06**SAS Label:** Blood Hexane (ng/mL)**English Text:** Blood Hexane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.086 to 14	Range of Values	2914	2914	
.	Missing	704	3618	

LBDV06LC - Blood Hexane Comment Code

Variable Name: LBDV06LC
SAS Label: Blood Hexane Comment Code
English Text: Blood Hexane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	4	4	
1	Below lower detection limit	2910	2914	
.	Missing	704	3618	

LBXV07N - Blood Heptane (ng/mL)

Variable Name: LBXV07N**SAS Label:** Blood Heptane (ng/mL)**English Text:** Blood Heptane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.071 to 0.598	Range of Values	2931	2931	
.	Missing	687	3618	

LBDV07LC - Blood Heptane Comment Code

Variable Name: LBDV07LC
SAS Label: Blood Heptane Comment Code
English Text: Blood Heptane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	11	11	
1	Below lower detection limit	2920	2931	
.	Missing	687	3618	

LBXV08N - Blood Octane (ng/mL)**Variable Name:** LBXV08N**SAS Label:** Blood Octane (ng/mL)**English Text:** Blood Octane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.071 to 0.295	Range of Values	2959	2959	
.	Missing	659	3618	

LBDV08LC - Blood Octane Comment Code

Variable Name: LBDV08LC
SAS Label: Blood Octane Comment Code
English Text: Blood Octane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	5	5	
1	Below lower detection limit	2954	2959	
.	Missing	659	3618	

LBXV1D - Blood 1,2-Dichlorobenzene (ng/mL)**Variable Name:** LBXV1D**SAS Label:** Blood 1,2-Dichlorobenzene (ng/mL)**English Text:** Blood 1,2-Dichlorobenzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0177 to 0.0177	Range of Values	2936	2936	
.	Missing	682	3618	

LBDV1DLC - Blood 1,2-Dichlorobenzene Comment Code

Variable Name: LBDV1DLC

SAS Label: Blood 1,2-Dichlorobenzene Comment Code

English Text: Blood 1,2-Dichlorobenzene Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	0	0	
1	Below lower detection limit	2936	2936	
.	Missing	682	3618	

LBXV2A - Blood 1,2-Dichloroethane (ng/mL)**Variable Name:** LBXV2A**SAS Label:** Blood 1,2-Dichloroethane (ng/mL)**English Text:** Blood 1,2-Dichloroethane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0071 to 0.141	Range of Values	2939	2939	
.	Missing	679	3618	

LBDV2ALC - Blood 1,2-Dichloroethane Comment Code

Variable Name: LBDV2ALC
SAS Label: Blood 1,2-Dichloroethane Comment Code
English Text: Blood 1,2-Dichloroethane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	52	52	
1	Below lower detection limit	2887	2939	
.	Missing	679	3618	

LBXV3B - Blood 1,3-Dichlorobenzene (ng/mL)**Variable Name:** LBXV3B**SAS Label:** Blood 1,3-Dichlorobenzene (ng/mL)**English Text:** Blood 1,3-Dichlorobenzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0177 to 0.0177	Range of Values	2960	2960	
.	Missing	658	3618	

LBDV3BLC - Blood 1,3-Dichlorobenzene Comment Code

Variable Name: LBDV3BLC
SAS Label: Blood 1,3-Dichlorobenzene Comment Code
English Text: Blood 1,3-Dichlorobenzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	0	0	
1	Below lower detection limit	2960	2960	
.	Missing	658	3618	

LBXV4C - Blood Tetrachloroethene (ng/mL)**Variable Name:** LBXV4C**SAS Label:** Blood Tetrachloroethene (ng/mL)**English Text:** Blood Tetrachloroethene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0339 to 3.63	Range of Values	2960	2960	
.	Missing	658	3618	

LBDV4CLC - Blood Tetrachloroethene Comment Code

Variable Name: LBDV4CLC
SAS Label: Blood Tetrachloroethene Comment Code
English Text: Blood Tetrachloroethene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	171	171	
1	Below lower detection limit	2789	2960	
.	Missing	658	3618	

LBXVAPN - Blood a-pinene (ng/mL)**Variable Name:** LBXVAPN**SAS Label:** Blood a-pinene (ng/mL)**English Text:** Blood a-pinene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0141 to 5.49	Range of Values	2934	2934	
.	Missing	684	3618	

LBDVAPLC - Blood a-pinene Comment Code

Variable Name: LBDVAPLC
SAS Label: Blood a-pinene Comment Code
English Text: Blood a-pinene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	2020	2020	
1	Below lower detection limit	914	2934	
.	Missing	684	3618	

LBXVBF - Blood Bromoform (ng/mL)**Variable Name:** LBXVBF**SAS Label:** Blood Bromoform (ng/mL)**English Text:** Blood Bromoform (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0057 to 3.48	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVBFLC - Blood Bromoform Comment Code

Variable Name: LBDVBFLC
SAS Label: Blood Bromoform Comment Code
English Text: Blood Bromoform Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	60	60	
1	Below lower detection limit	2900	2960	
.	Missing	658	3618	

LBXVBM - Blood Bromodichloromethane (ng/mL)**Variable Name:** LBXVBM**SAS Label:** Blood Bromodichloromethane (ng/mL)**English Text:** Blood Bromodichloromethane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0042 to 0.0441	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVBMLC - Blood Bromodichloromethane Comment Code

Variable Name: LBDVBMLC

SAS Label: Blood Bromodichloromethane Comment Code

English Text: Blood Bromodichloromethane Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	264	264	
1	Below lower detection limit	2696	2960	
.	Missing	658	3618	

LBXVBZ - Blood Benzene (ng/mL)**Variable Name:** LBXVBZ**SAS Label:** Blood Benzene (ng/mL)**English Text:** Blood Benzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.017 to 2.15	Range of Values	2934	2934	
.	Missing	684	3618	

LBDVBZLC - Blood Benzene Comment Code

Variable Name: LBDVBZLC
SAS Label: Blood Benzene Comment Code
English Text: Blood Benzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	709	709	
1	Below lower detection limit	2225	2934	
.	Missing	684	3618	

LBXVBZN - Blood Benzonitrile (ng/mL)**Variable Name:** LBXVBZN**SAS Label:** Blood Benzonitrile (ng/mL)**English Text:** Blood Benzonitrile (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.106 to 1.6	Range of Values	2957	2957	
.	Missing	661	3618	

LBDVZBLC - Blood Benzonitrile Comment Code

Variable Name: LBDVZBLC
SAS Label: Blood Benzonitrile Comment Code
English Text: Blood Benzonitrile Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	133	133	
1	Below lower detection limit	2824	2957	
.	Missing	661	3618	

LBXVC6 - Blood Cyclohexane (ng/mL)

Variable Name: LBXVC6

SAS Label: Blood Cyclohexane (ng/mL)

English Text: Blood Cyclohexane (ng/mL)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0141 to 3.49	Range of Values	2934	2934	
.	Missing	684	3618	

LBDVC6LC - Blood Cyclohexane Comment Code

Variable Name: LBDVC6LC
SAS Label: Blood Cyclohexane Comment Code
English Text: Blood Cyclohexane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	20	20	
1	Below lower detection limit	2914	2934	
.	Missing	684	3618	

LBXVCB - Blood Chlorobenzene (ng/mL)

Variable Name: LBXVCB

SAS Label: Blood Chlorobenzene (ng/mL)

English Text: Blood Chlorobenzene (ng/mL)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0078 to 0.158	Range of Values	2949	2949	
.	Missing	669	3618	

LBDVCBLC - Blood Chlorobenzene Comment Code

Variable Name: LBDVCBLC
SAS Label: Blood Chlorobenzene Comment Code
English Text: Blood Chlorobenzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	2	2	
1	Below lower detection limit	2947	2949	
.	Missing	669	3618	

LBXVCF - Blood Chloroform (ng/mL)**Variable Name:** LBXVCF**SAS Label:** Blood Chloroform (ng/mL)**English Text:** Blood Chloroform (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0057 to 0.685	Range of Values	2939	2939	
.	Missing	679	3618	

LBDVCFLC - Blood Chloroform Comment Code

Variable Name: LBDVCFLC
SAS Label: Blood Chloroform Comment Code
English Text: Blood Chloroform Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	1616	1616	
1	Below lower detection limit	1323	2939	
.	Missing	679	3618	

LBXVCM - Blood Dibromochloromethane (ng/mL)**Variable Name:** LBXVCM**SAS Label:** Blood Dibromochloromethane (ng/mL)**English Text:** Blood Dibromochloromethane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0035 to 0.035	Range of Values	2959	2959	
.	Missing	659	3618	

LBDVCMLC - Blood Dibromochloromethane Comment Code

Variable Name: LBDVCMLC
SAS Label: Blood Dibromochloromethane Comment Code
English Text: Blood Dibromochloromethane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	228	228	
1	Below lower detection limit	2731	2959	
.	Missing	659	3618	

LBXVCT - Blood Carbon Tetrachloride (ng/mL)

Variable Name: LBXVCT

SAS Label: Blood Carbon Tetrachloride (ng/mL)

English Text: Blood Carbon Tetrachloride (ng/mL)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0035 to 0.0524	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVCTL - Blood Carbon Tetrachloride Comment Code

Variable Name: LBDVCTL

SAS Label: Blood Carbon Tetrachloride Comment Code

English Text: Blood Carbon Tetrachloride Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	9	9	
1	Below lower detection limit	2951	2960	
.	Missing	658	3618	

LBXVDB - Blood 1,4-Dichlorobenzene (ng/mL)**Variable Name:** LBXVDB**SAS Label:** Blood 1,4-Dichlorobenzene (ng/mL)**English Text:** Blood 1,4-Dichlorobenzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0283 to 101	Range of Values	2942	2942	
.	Missing	676	3618	

LBDVDBLC - Blood 1,4-Dichlorobenzene Comment Code

Variable Name: LBDVDBLC
SAS Label: Blood 1,4-Dichlorobenzene Comment Code
English Text: Blood 1,4-Dichlorobenzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	845	845	
1	Below lower detection limit	2097	2942	
.	Missing	676	3618	

LBXVDEE - Blood Diethyl Ether (ng/mL)**Variable Name:** LBXVDEE**SAS Label:** Blood Diethyl Ether (ng/mL)**English Text:** Blood Diethyl Ether (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0283 to 0.508	Range of Values	2914	2914	
.	Missing	704	3618	

LBDVEELC - Blood Diethyl Ether Comment Code

Variable Name: LBDVEELC
SAS Label: Blood Diethyl Ether Comment Code
English Text: Blood Diethyl Ether Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	3	3	
1	Below lower detection limit	2911	2914	
.	Missing	704	3618	

LBXVEA - Blood Ethyl Acetate (ng/mL)**Variable Name:** LBXVEA**SAS Label:** Blood Ethyl Acetate (ng/mL)**English Text:** Blood Ethyl Acetate (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.112 to 43.3	Range of Values	2797	2797	
.	Missing	821	3618	

LBDVEALC - Blood Ethyl Acetate Comment Code

Variable Name: LBDVEALC
SAS Label: Blood Ethyl Acetate Comment Code
English Text: Blood Ethyl Acetate Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	45	45	
1	Below lower detection limit	2752	2797	
.	Missing	821	3618	

LBXVEB - Blood Ethylbenzene (ng/mL)**Variable Name:** LBXVEB**SAS Label:** Blood Ethylbenzene (ng/mL)**English Text:** Blood Ethylbenzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.017 to 4.92	Range of Values	2936	2936	
.	Missing	682	3618	

LBDVEBLC - Blood Ethylbenzene Comment Code

Variable Name: LBDVEBLC
SAS Label: Blood Ethylbenzene Comment Code
English Text: Blood Ethylbenzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	543	543	
1	Below lower detection limit	2393	2936	
.	Missing	682	3618	

LBXVEC - Blood Chloroethane (ng/mL)**Variable Name:** LBXVEC**SAS Label:** Blood Chloroethane (ng/mL)**English Text:** Blood Chloroethane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0318 to 1.81	Range of Values	2890	2890	
.	Missing	728	3618	

LBDVECLC - Blood Chloroethane Comment Code

Variable Name: LBDVECLC
SAS Label: Blood Chloroethane Comment Code
English Text: Blood Chloroethane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	8	8	
1	Below lower detection limit	2882	2890	
.	Missing	728	3618	

LBXVFN - Blood Furan (ng/mL)**Variable Name:** LBXVFN**SAS Label:** Blood Furan (ng/mL)**English Text:** Blood Furan (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0177 to 0.35	Range of Values	2937	2937	
.	Missing	681	3618	

LBDVFNLC - Blood Furan Comment Code

Variable Name: LBDVFNLC
SAS Label: Blood Furan Comment Code
English Text: Blood Furan Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	386	386	
1	Below lower detection limit	2551	2937	
.	Missing	681	3618	

LBXVIBN - Blood Isobutyronitrile (ng/mL)**Variable Name:** LBXVIBN**SAS Label:** Blood Isobutyronitrile (ng/mL)**English Text:** Blood Isobutyronitrile (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0283 to 0.665	Range of Values	2918	2918	
.	Missing	700	3618	

LBDVIBLC - Blood Isobutyronitrile Comment Code

Variable Name: LBDVIBLC
SAS Label: Blood Isobutyronitrile Comment Code
English Text: Blood Isobutyronitrile Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	204	204	
1	Below lower detection limit	2714	2918	
.	Missing	700	3618	

LBXVIPB - Blood Isopropylbenzene (ng/mL)**Variable Name:** LBXVIPB**SAS Label:** Blood Isopropylbenzene (ng/mL)**English Text:** Blood Isopropylbenzene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0283 to 0.205	Range of Values	2959	2959	
.	Missing	659	3618	

LBDVIPLC - Blood Isopropylbenzene Comment Code

Variable Name: LBDVIPLC
SAS Label: Blood Isopropylbenzene Comment Code
English Text: Blood Isopropylbenzene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	2	2	
1	Below lower detection limit	2957	2959	
.	Missing	659	3618	

LBXVMC - Blood Methylene Chloride (ng/mL)**Variable Name:** LBXVMC**SAS Label:** Blood Methylene Chloride (ng/mL)**English Text:** Blood Methylene Chloride (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.177 to 15.7	Range of Values	2937	2937	
.	Missing	681	3618	

LBDVMCLC - Blood Methylene Chloride Comment Code

Variable Name: LBDVMCLC
SAS Label: Blood Methylene Chloride Comment Code
English Text: Blood Methylene Chloride Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7	7	
1	Below lower detection limit	2930	2937	
.	Missing	681	3618	

LBXVMCP - Blood Methylcyclopentane (ng/mL)**Variable Name:** LBXVMCP**SAS Label:** Blood Methylcyclopentane (ng/mL)**English Text:** Blood Methylcyclopentane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0141 to 15.6	Range of Values	2914	2914	
.	Missing	704	3618	

LBDVMPLC - Blood Methylcyclopentane Comment Code

Variable Name: LBDVMPLC
SAS Label: Blood Methylcyclopentane Comment Code
English Text: Blood Methylcyclopentane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	22	22	
1	Below lower detection limit	2892	2914	
.	Missing	704	3618	

LBXVME - Blood MTBE (ng/mL)**Variable Name:** LBXVME**SAS Label:** Blood MTBE (ng/mL)**English Text:** Blood Methyl-tert-butyl ether (MTBE) (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0071 to 0.0631	Range of Values	2944	2944	
.	Missing	674	3618	

LBDVMELC - Blood MTBE Comment Code

Variable Name: LBDVMELC

SAS Label: Blood MTBE Comment Code

English Text: Blood Methyl-tert-butyl ether (MTBE) Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	20	20	
1	Below lower detection limit	2924	2944	
.	Missing	674	3618	

LBXVMIK - Blood Methyl Isobutyl Ketone (ng/mL)**Variable Name:** LBXVMIK**SAS Label:** Blood Methyl Isobutyl Ketone (ng/mL)**English Text:** Blood Methyl Isobutyl Ketone (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.071 to 0.238	Range of Values	2942	2942	
.	Missing	676	3618	

LBDVMKLC - Blood Methyl Isobutyl Ketone Comt Code

Variable Name: LBDVMKLC
SAS Label: Blood Methyl Isobutyl Ketone Comt Code
English Text: Blood Methyl Isobutyl Ketone Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	21	21	
1	Below lower detection limit	2921	2942	
.	Missing	676	3618	

LBXVOX - Blood o-Xylene (ng/mL)**Variable Name:** LBXVOX**SAS Label:** Blood o-Xylene (ng/mL)**English Text:** Blood o-Xylene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.017 to 5.96	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVOXLC - Blood o-Xylene Comment Code

Variable Name: LBDVOXLC
SAS Label: Blood o-Xylene Comment Code
English Text: Blood o-Xylene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	548	548	
1	Below lower detection limit	2412	2960	
.	Missing	658	3618	

LBXVST - Blood Styrene (ng/mL)**Variable Name:** LBXVST**SAS Label:** Blood Styrene (ng/mL)**English Text:** Blood Styrene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0212 to 9.16	Range of Values	2933	2933	
.	Missing	685	3618	

LBDVSTLC - Blood Styrene Comment Code

Variable Name: LBDVSTLC
SAS Label: Blood Styrene Comment Code
English Text: Blood Styrene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	1388	1388	
1	Below lower detection limit	1545	2933	
.	Missing	685	3618	

LBXVTC - Blood Trichloroethene (ng/mL)**Variable Name:** LBXVTC**SAS Label:** Blood Trichloroethene (ng/mL)**English Text:** Blood Trichloroethene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0085 to 0.5	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVTCLC - Blood Trichloroethene Comment Code

Variable Name: LBDVTCLC
SAS Label: Blood Trichloroethene Comment Code
English Text: Blood Trichloroethene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	15	15	
1	Below lower detection limit	2945	2960	
.	Missing	658	3618	

LBXVTE - Blood 1,1,1-Trichloroethane (ng/mL)**Variable Name:** LBXVTE**SAS Label:** Blood 1,1,1-Trichloroethane (ng/mL)**English Text:** Blood 1,1,1-Trichloroethane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0071 to 0.122	Range of Values	2958	2958	
.	Missing	660	3618	

LBDVTELC - Blood 1,1,1-Trichloroethane Comment Code

Variable Name: LBDVTELC
SAS Label: Blood 1,1,1-Trichloroethane Comment Code
English Text: Blood 1,1,1-Trichloroethane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	17	17	
1	Below lower detection limit	2941	2958	
.	Missing	660	3618	

LBXVTHF - Blood Tetrahydrofuran (ng/mL)**Variable Name:** LBXVTHF**SAS Label:** Blood Tetrahydrofuran (ng/mL)**English Text:** Blood Tetrahydrofuran (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.088 to 19.7	Range of Values	2940	2940	
.	Missing	678	3618	

LBDVHTLC - Blood Tetrahydrofuran Comment Code

Variable Name: LBDVHTLC
SAS Label: Blood Tetrahydrofuran Comment Code
English Text: Blood Tetrahydrofuran Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	5	5	
1	Below lower detection limit	2935	2940	
.	Missing	678	3618	

LBXVTO - Blood Toluene (ng/mL)**Variable Name:** LBXVTO**SAS Label:** Blood Toluene (ng/mL)**English Text:** Blood Toluene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0177 to 16.7	Range of Values	2924	2924	
.	Missing	694	3618	

LBDVTOLC - Blood Toluene Comment Code

Variable Name: LBDVTOLC

SAS Label: Blood Toluene Comment Code

English Text: Blood Toluene Comment Code

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	2425	2425	
1	Below lower detection limit	499	2924	
.	Missing	694	3618	

LBXVTP - Blood 1,2,3-Trichloropropane (ng/mL)

Variable Name: LBXVTP**SAS Label:** Blood 1,2,3-Trichloropropane (ng/mL)**English Text:** Blood 1,2,3-Trichloropropane (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0283 to 0.0283	Range of Values	2960	2960	
.	Missing	658	3618	

LBDVTPLC - Blood 1,2,3-Trichloropropane Comt Code

Variable Name: LBDVTPLC
SAS Label: Blood 1,2,3-Trichloropropane Comt Code
English Text: Blood 1,2,3-Trichloropropane Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	0	0	
1	Below lower detection limit	2960	2960	
.	Missing	658	3618	

LBXVXY - Blood m-/p-Xylene (ng/mL)**Variable Name:** LBXVXY**SAS Label:** Blood m-/p-Xylene (ng/mL)**English Text:** Blood m-/p-Xylene (ng/mL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.024 to 12.6	Range of Values	2956	2956	
.	Missing	662	3618	

LBDVXYLC - Blood m-/p-Xylene Comment Code

Variable Name: LBDVXYLC
SAS Label: Blood m-/p-Xylene Comment Code
English Text: Blood m-/p-Xylene Comment Code
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	981	981	
1	Below lower detection limit	1975	2956	
.	Missing	662	3618	